KENNESAW STATE
UNIVERSITY

CS 4277
Object Detection

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Computer Science

College of Computing and Software Engineering

Content

- From single object to multiple objects
- Region proposal and R-CNN
- Fast R-CNN and Faster R-CNN

From single object to multiple objects

Object Detection

Classification

CAT

No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection

DOG, DOG, CAT

Instance Segmentation

DOG, DOG, CAT

Multiple Object

Object Detection

Classification

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No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SKY

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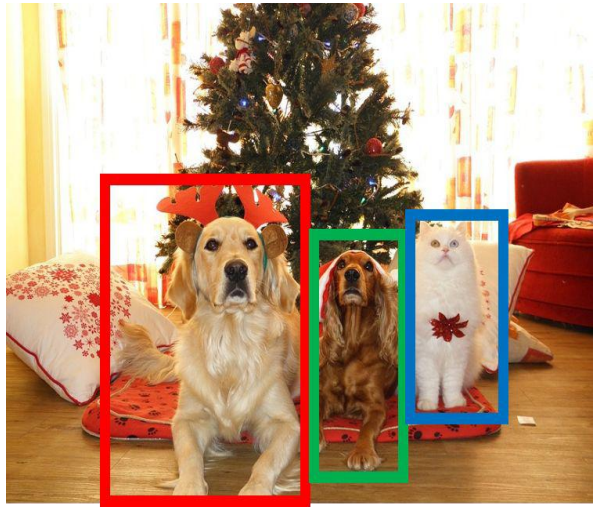
Object Detection



DOG, DOG, CAT

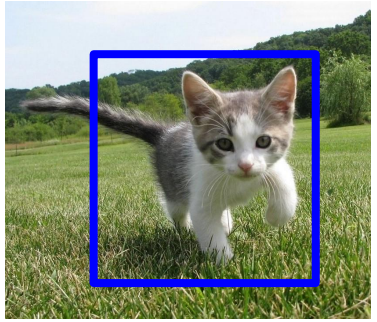
Instance Segmentation

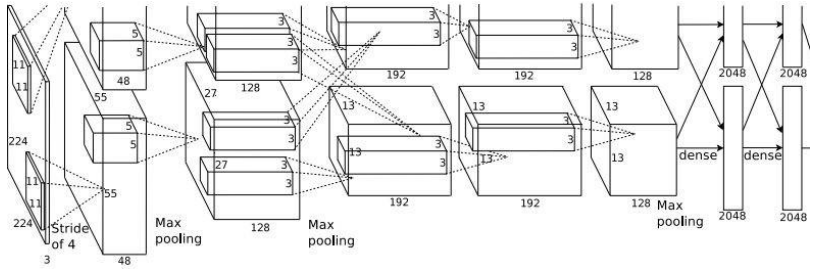
DOG, DOG, CAT

Multiple Object

Object Detection: Single Object

(Classification + Localization)





Vector: 4096

Fully Connected: 4096 to 1000

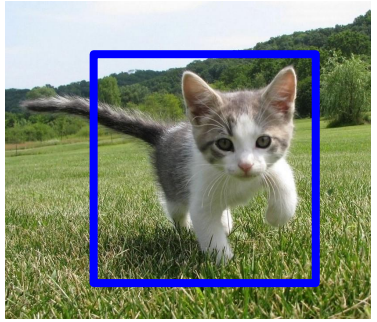
Class Scores
Cat: 0.9
Dog: 0.05
Car: 0.01
...

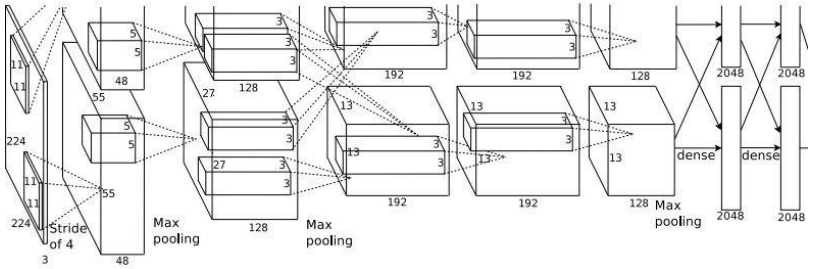
Fully Connected: 4096 to 4

Box Coordinates
(x, y, w, h)

Object Detection: Single Object

(Classification + Localization)





Vector: 4096

Fully Connected: 4096 to 1000

Class Scores
Cat: 0.9
Dog: 0.05
Car: 0.01
...

Softmax Loss

Correct label: Cat

Fully Connected: 4096 to 4

Box Coordinates (x, y, w, h)

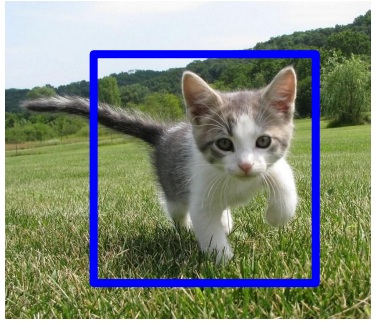
L2 Loss

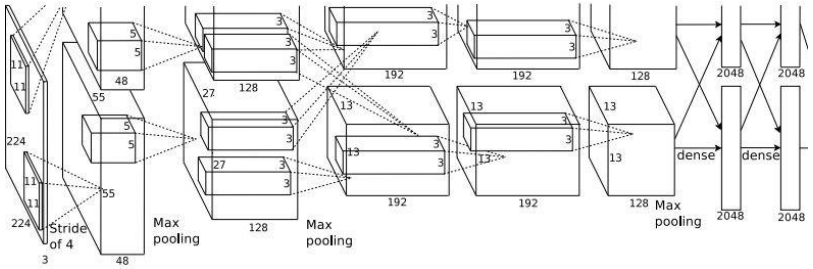
Correct box: (x', y', w', h')

Treat localization as a regression problem!

Object Detection: Single Object

(Classification + Localization)





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Class Scores
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...

Softmax Loss

Correct label: Cat

Fully Connected: 4096 to 4

Box Coordinates (x, y, w, h)

L2 Loss

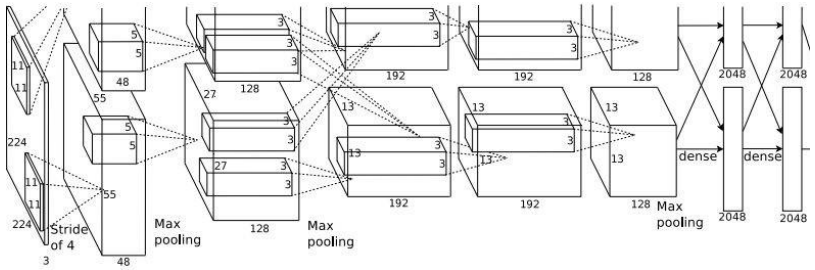
Correct box: (x', y', w', h')

Multitask Loss

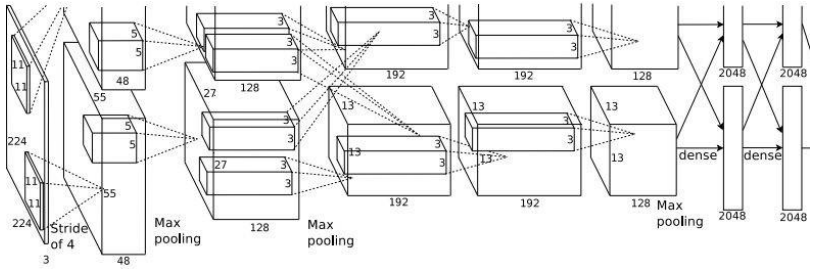
+ → **Loss**

Treat localization as a regression problem!

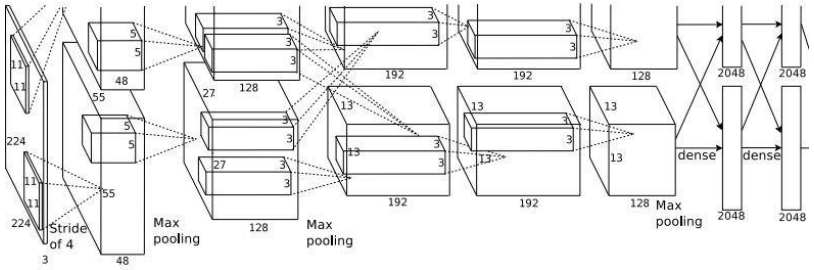
Object Detection: Multiple Objects



CAT: (x, y, w, h)



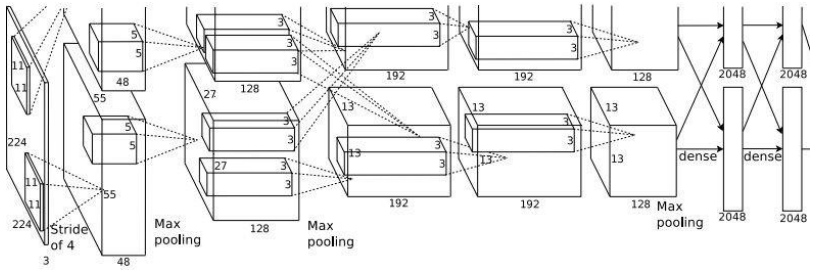
DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)



DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
….

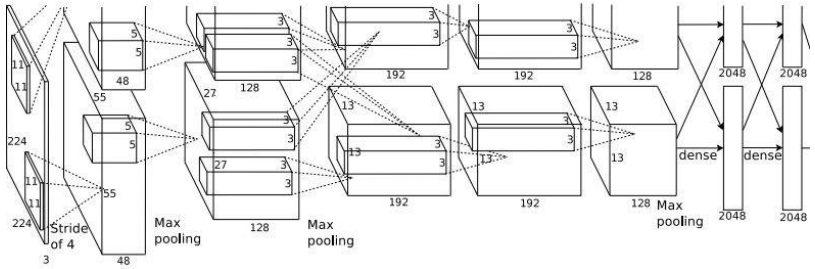
Object Detection: Multiple Objects

Each image needs a different number of outputs!



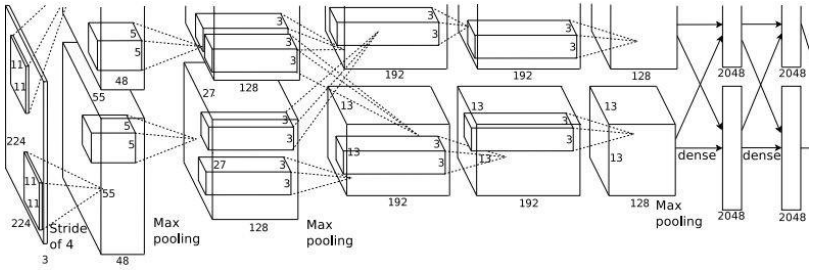
CAT: (x, y, w, h)

4 numbers



DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)

12 numbers

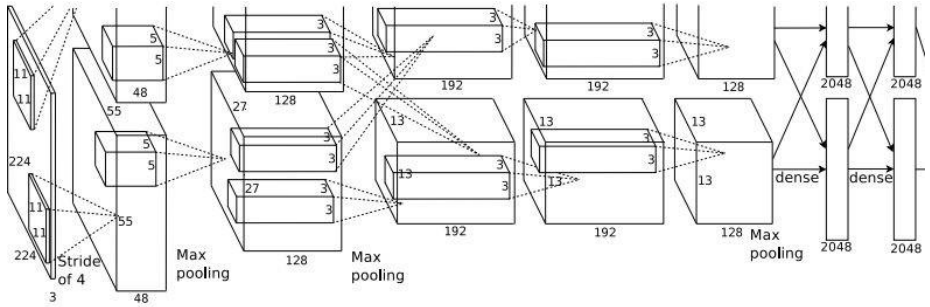


DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
….

Many numbers!

Object Detection: Multiple Objects

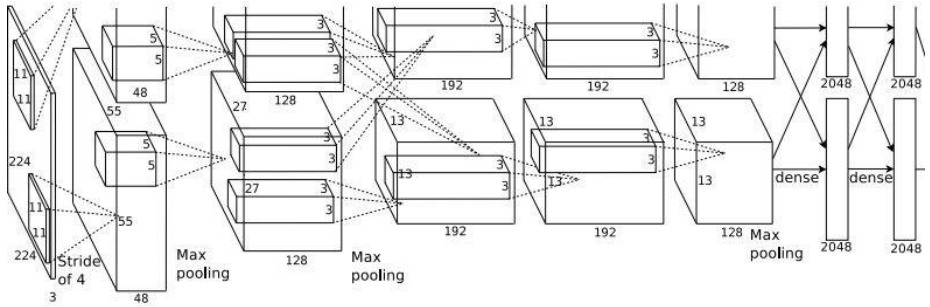
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? NO
Cat? NO
Background? YES

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



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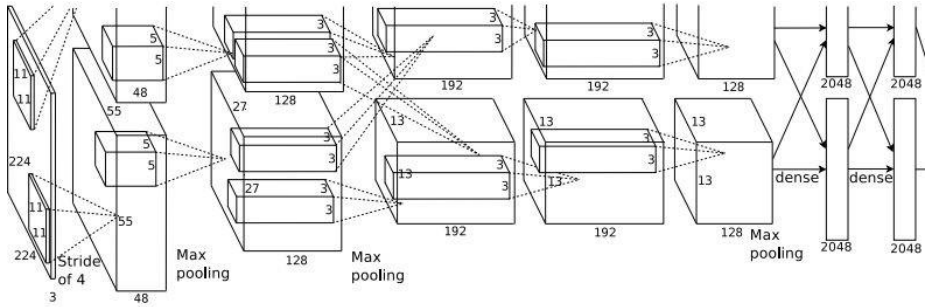

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Cat? NO
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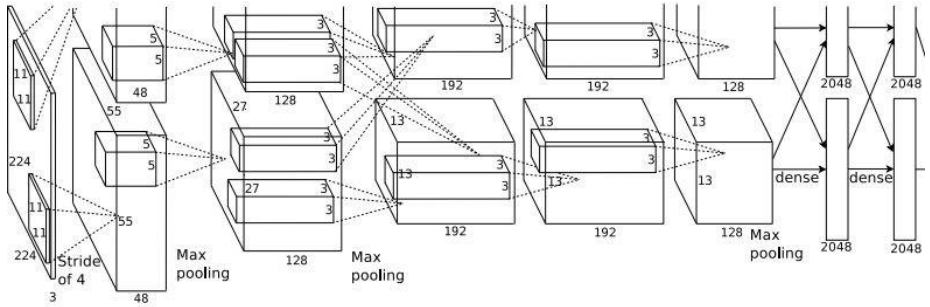
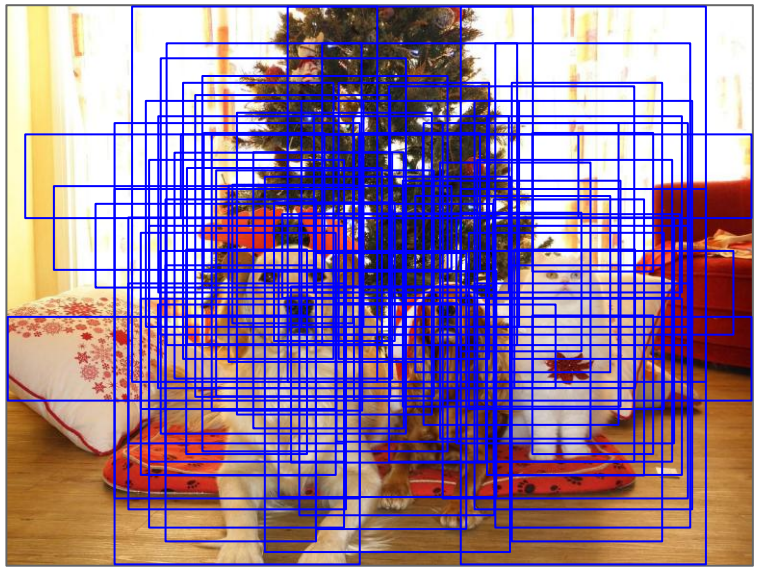


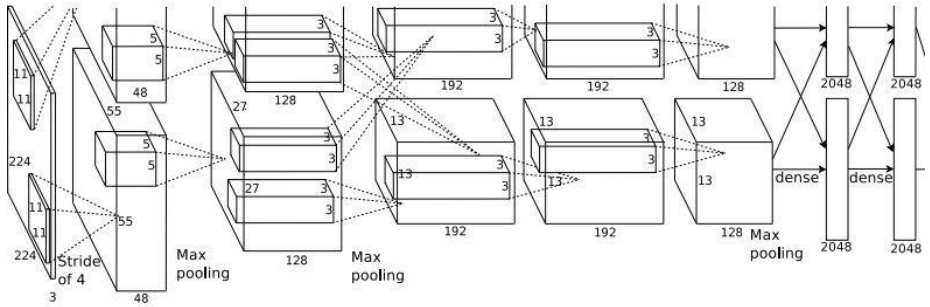
Dog? NO
Cat? YES
Background? NO

Q: What's the problem with this approach?

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background





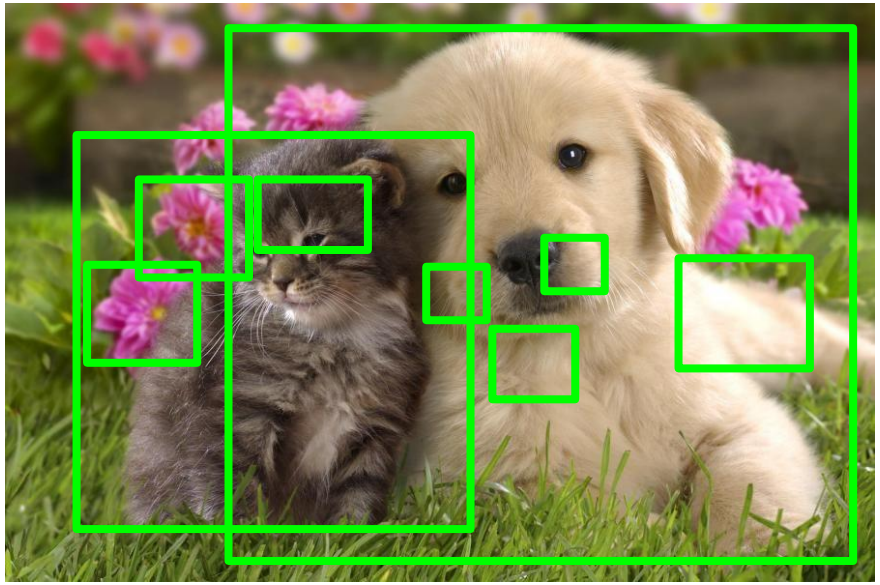
- Dog? NO Cat? YES
- Background? NO

- Problem: Need to apply CNN to huge number of locations, scales, and aspect ratios, very computationally expensive!

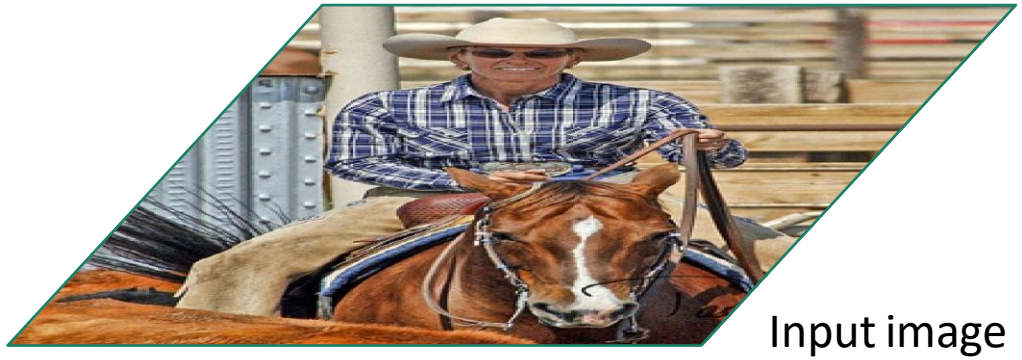
Region proposal and R-CNN

Region Proposals: Selective Search

- Find “blobby” image regions that are likely to contain objects
- Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU

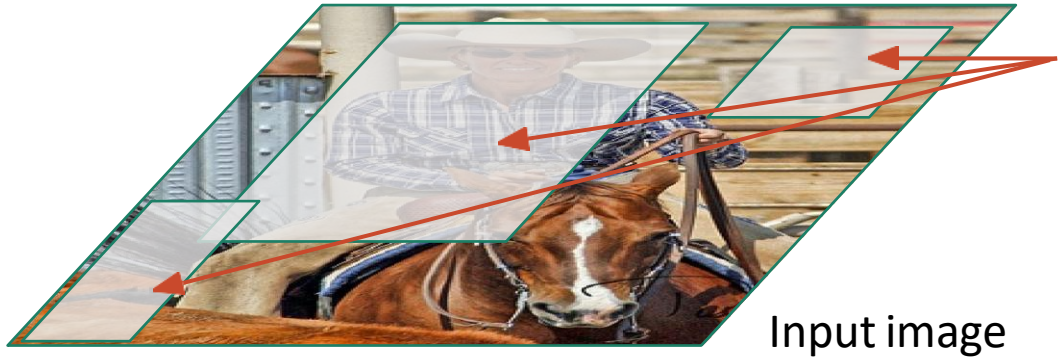


R-CNN



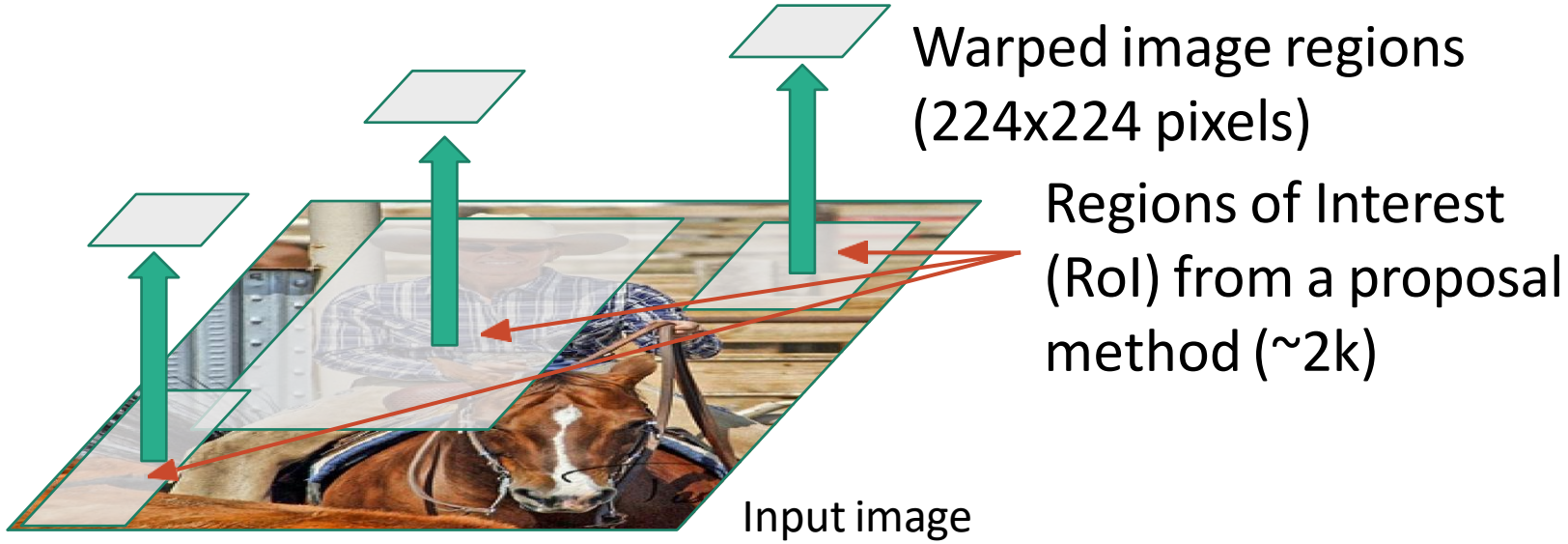
Input image

R-CNN

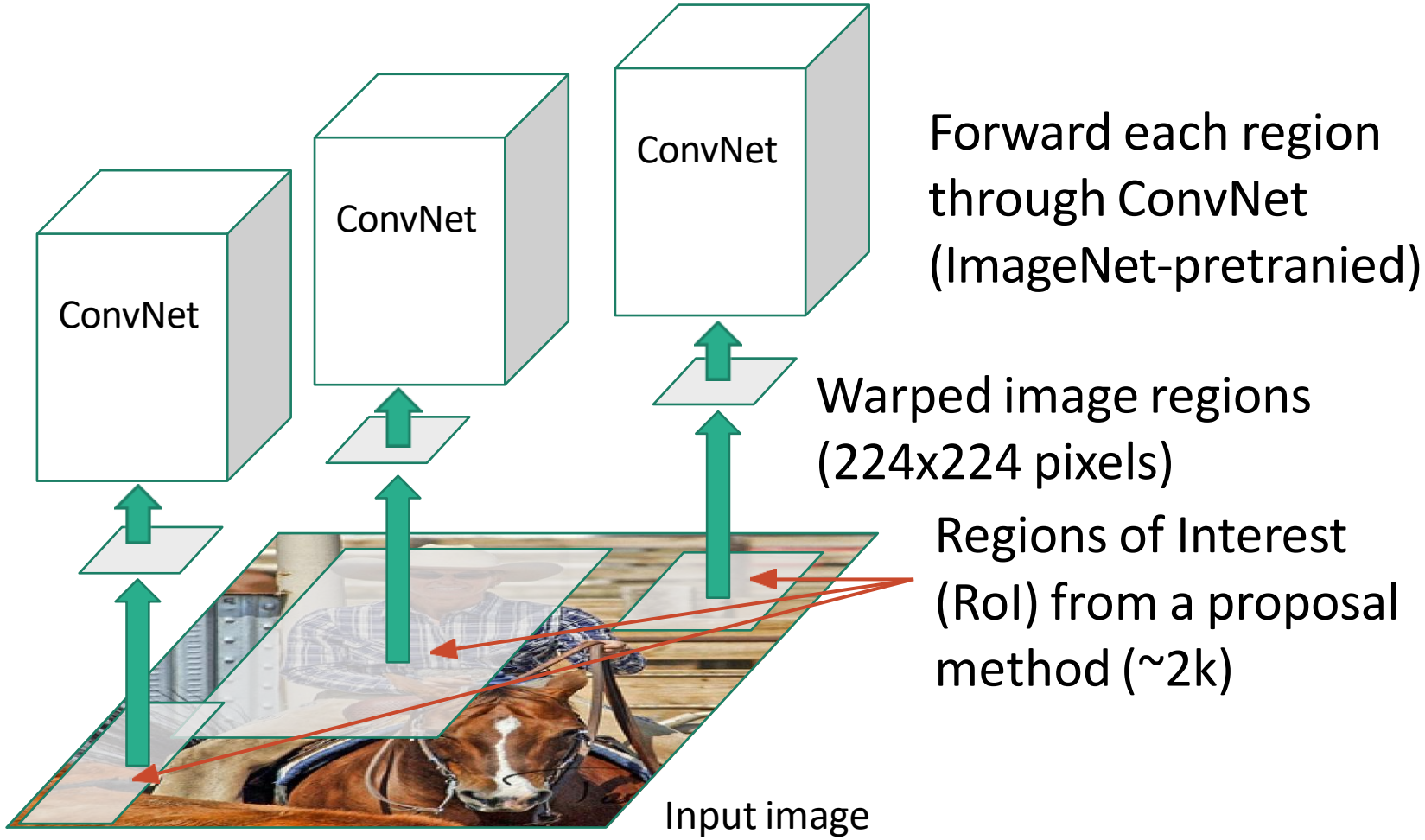


Regions of Interest (RoI) from a proposal method (~2k)

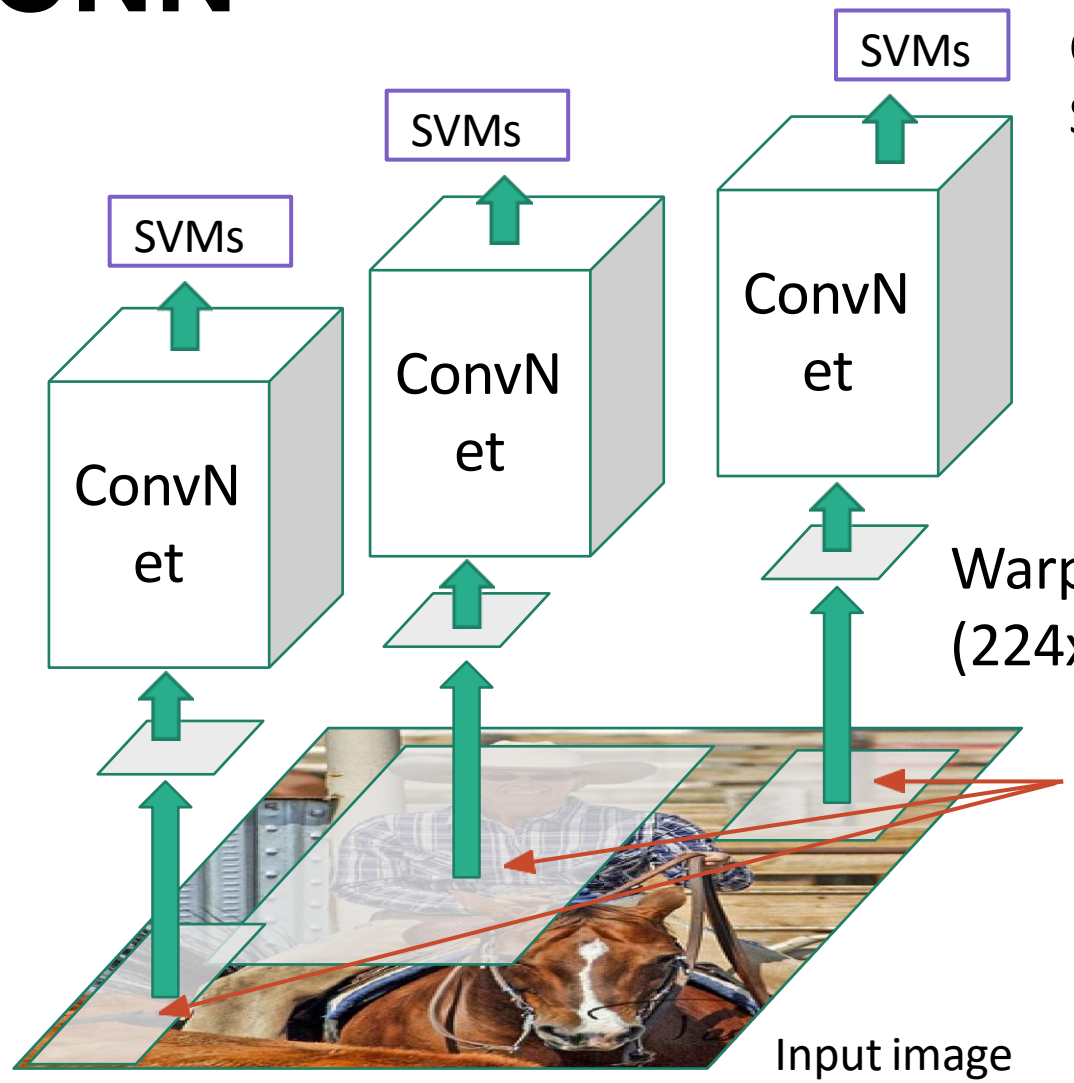
R-CNN


Warped image regions
(224x224 pixels)
Regions of Interest
(RoI) from a proposal
method (~2k)
Input image

R-CNN


ConvNet
ConvNet
ConvNet
Forward each region through ConvNet (ImageNet-pretranied)
Warped image regions (224x224 pixels)
Regions of Interest (RoI) from a proposal method (~2k)
Input image

R-CNN



Classify regions with SVMs

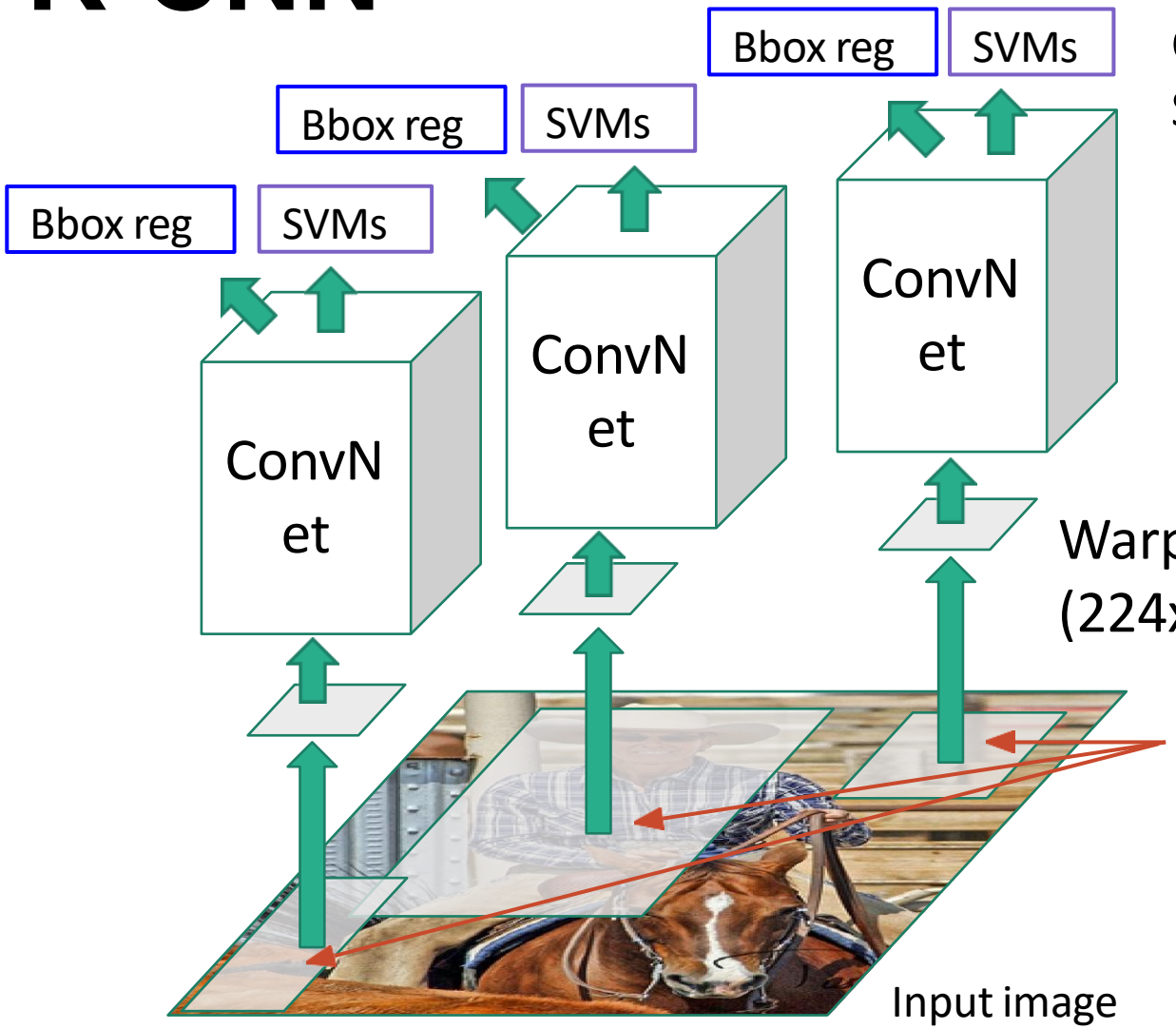
Forward each region through ConvNet (ImageNet-pretranied)

Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

R-CNN

Predict “corrections” to the RoI: 4 numbers: (dx, dy, dw, dh)



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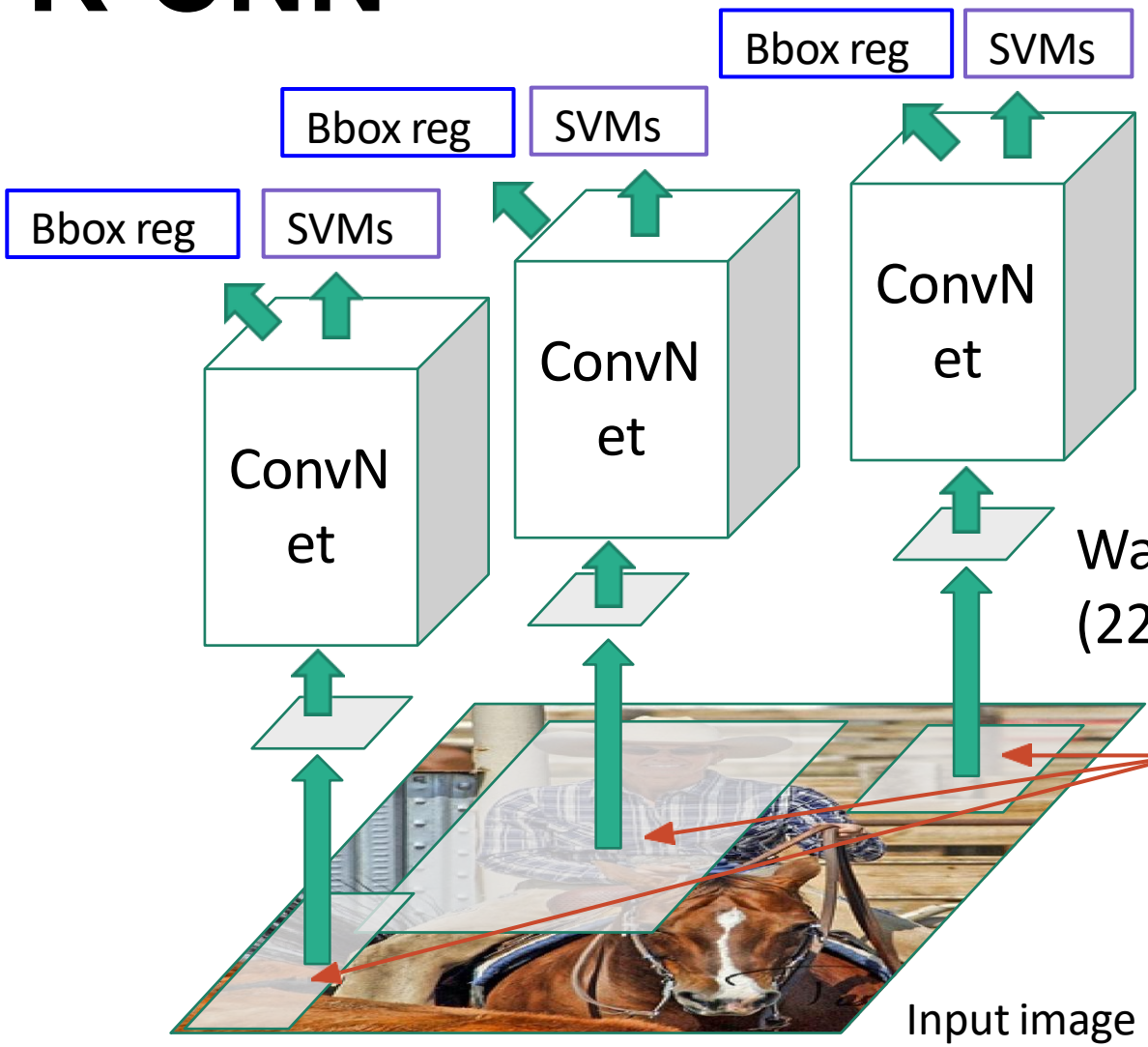


Classify regions with SVMs

Forward each region through ConvNet

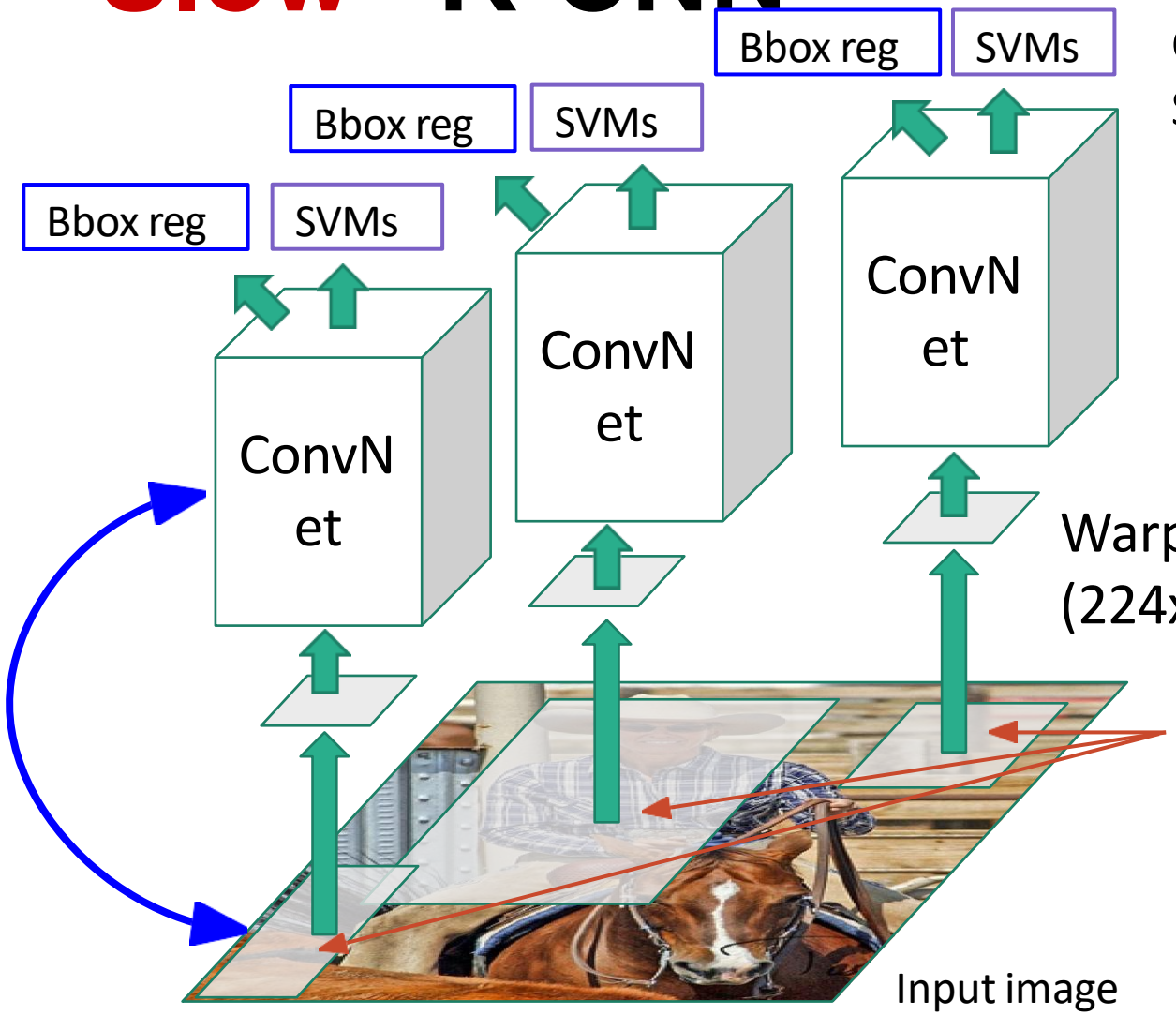
Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

Problem: Very slow! Need to do ~2k independent forward passes for each image!

"Slow" R-CNN

Predict "corrections" to the RoI: 4 numbers: (dx, dy, dw, dh)



Classify regions with SVMs

Forward each region through ConvNet

Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

Input image

Problem: Very slow! Need to do ~2k independent forward passes for each image!

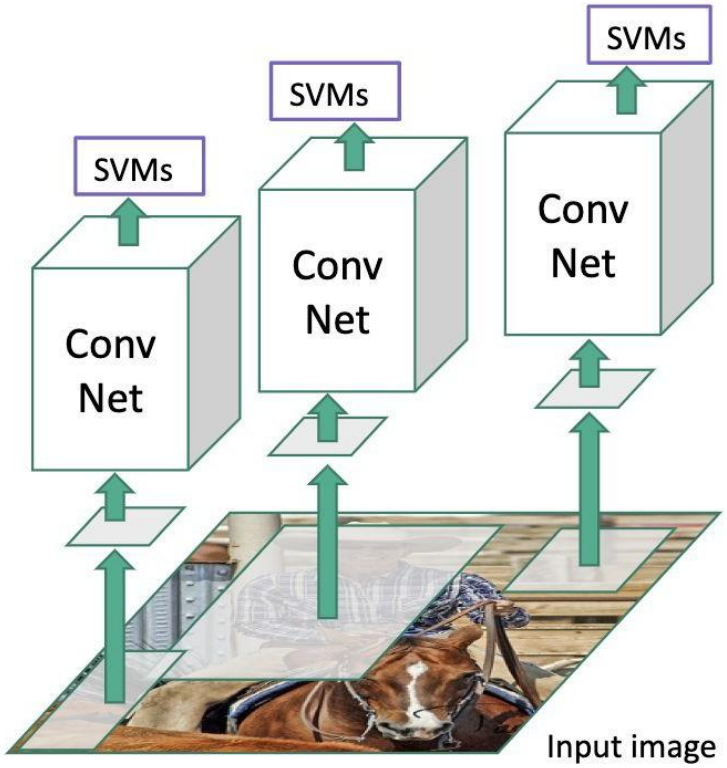
Idea: Pass the image through convnet before cropping! Crop the conv feature instead!

Fast R-CNN and Faster R-CNN

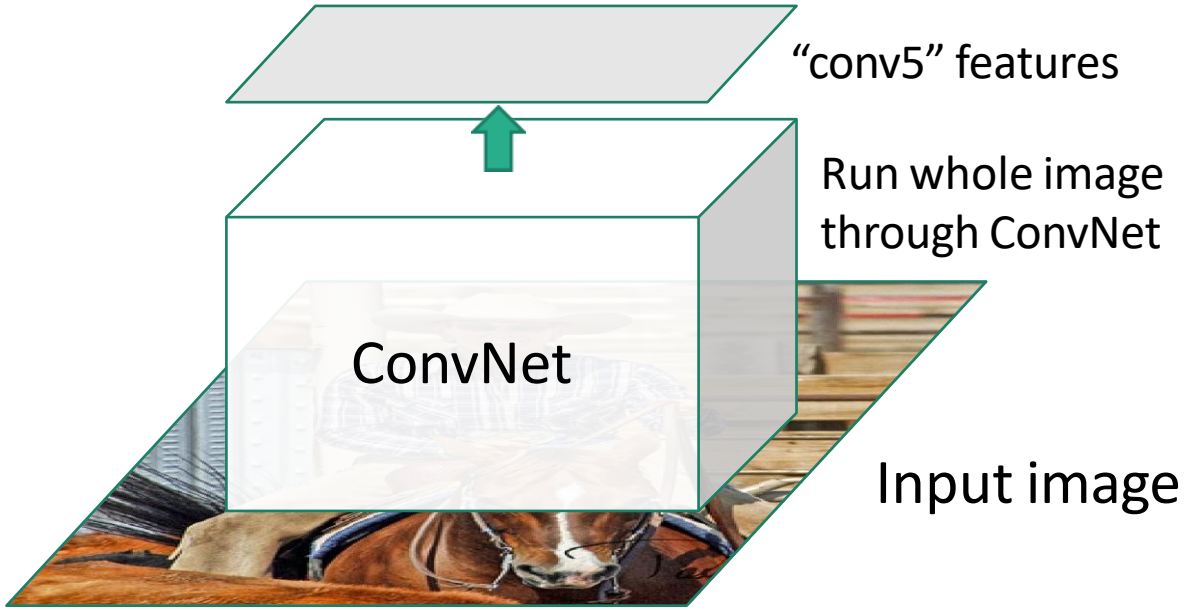
Fast R-CNN

Input image

“Slow” R-CNN



Fast R-CNN



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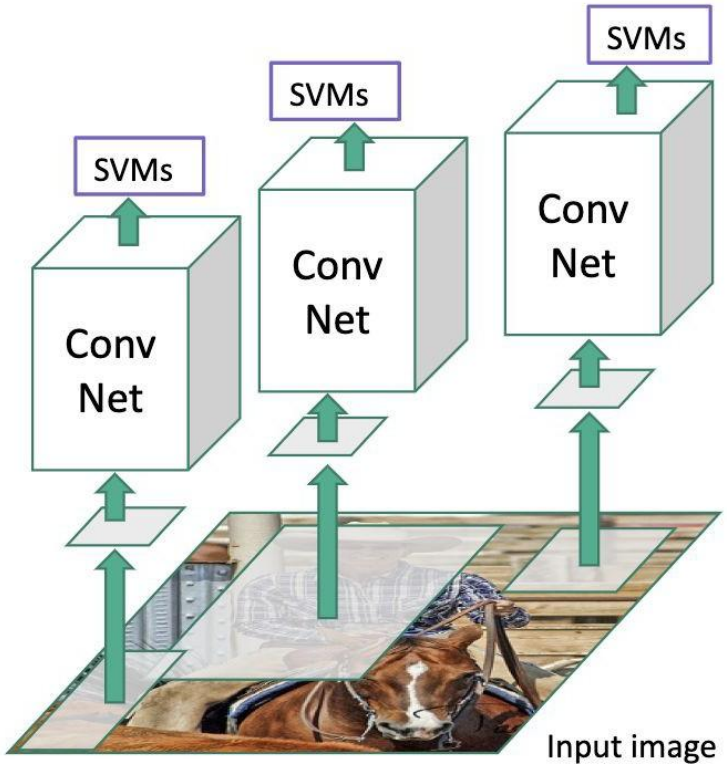
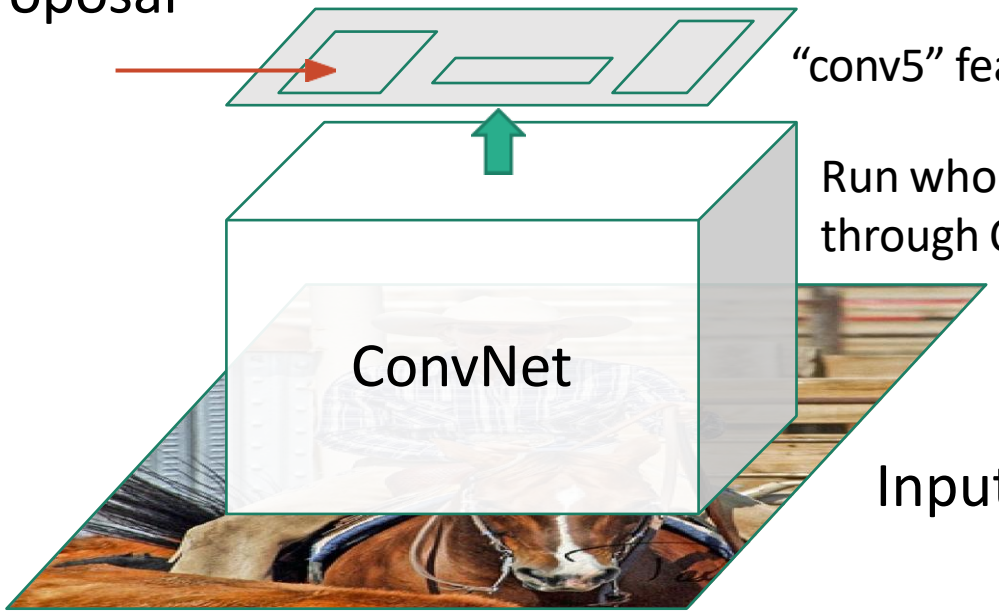


Fast R-CNN

Regions of Interest (RoIs) from a proposal method



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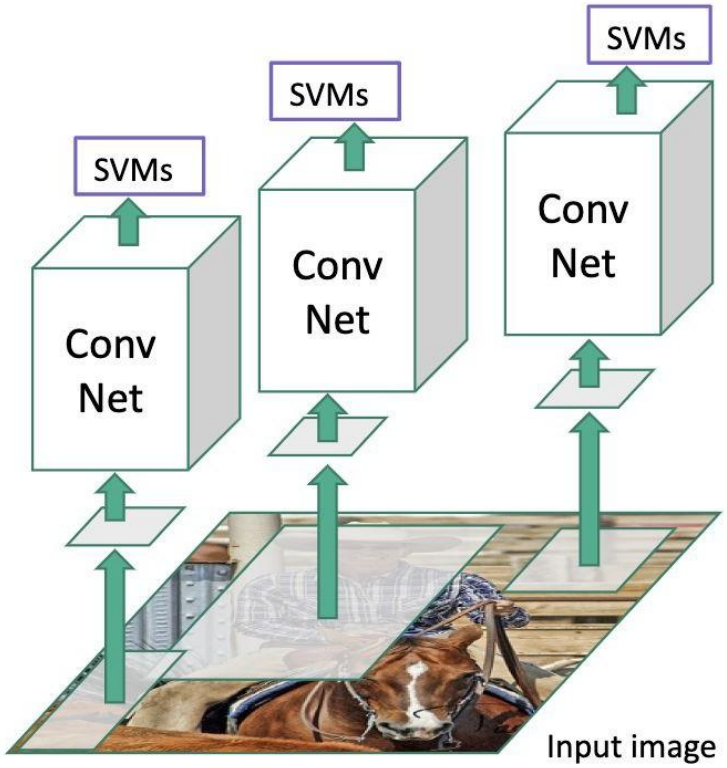

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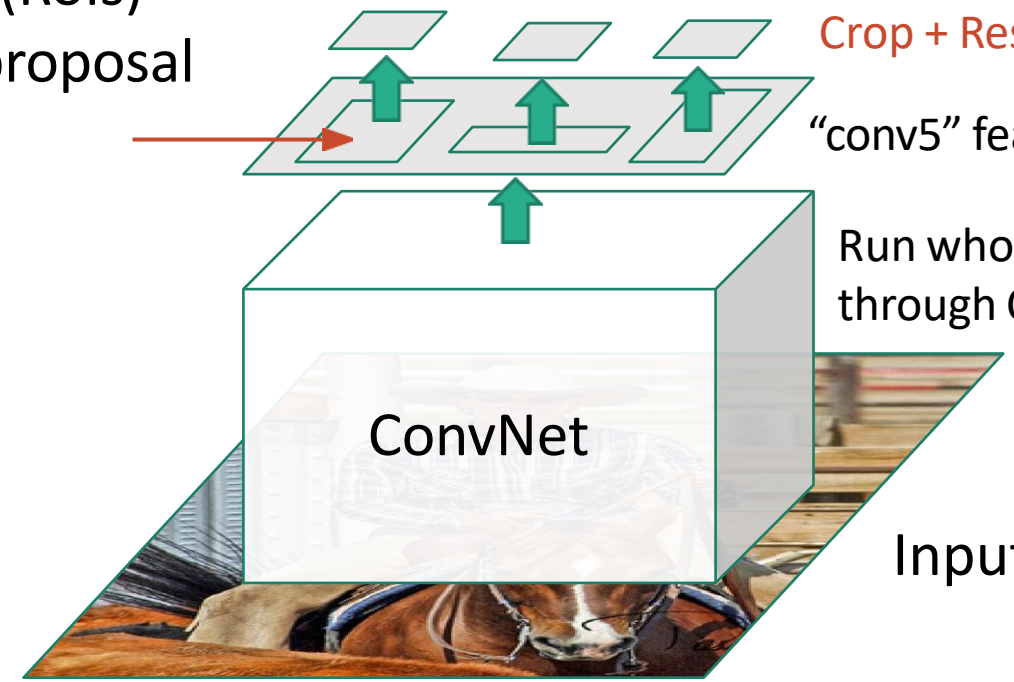

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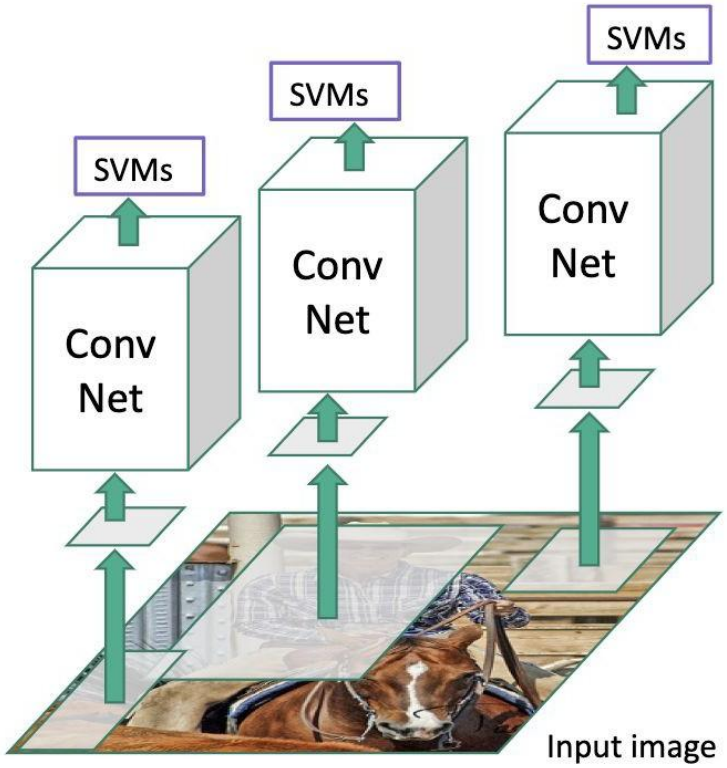

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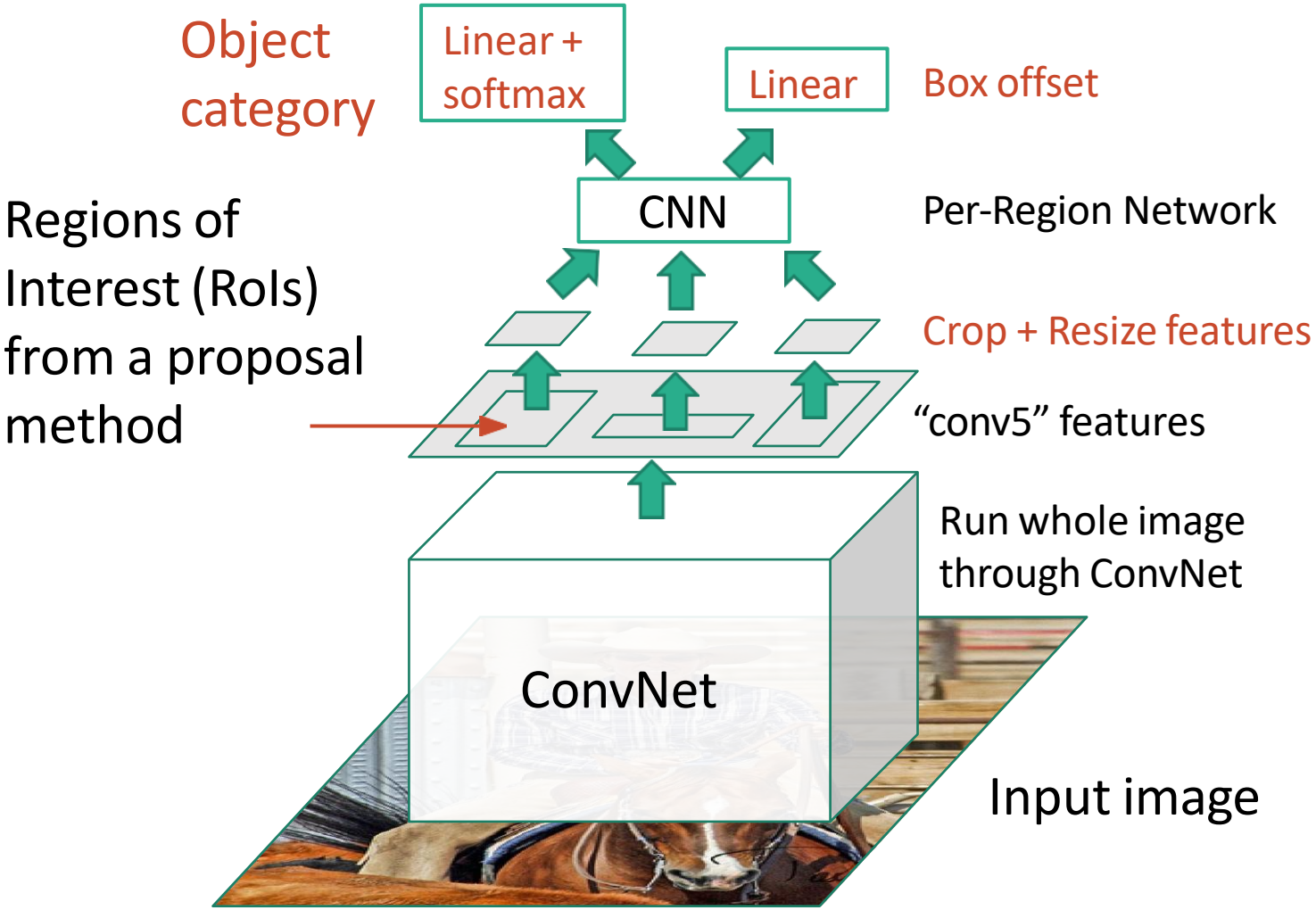

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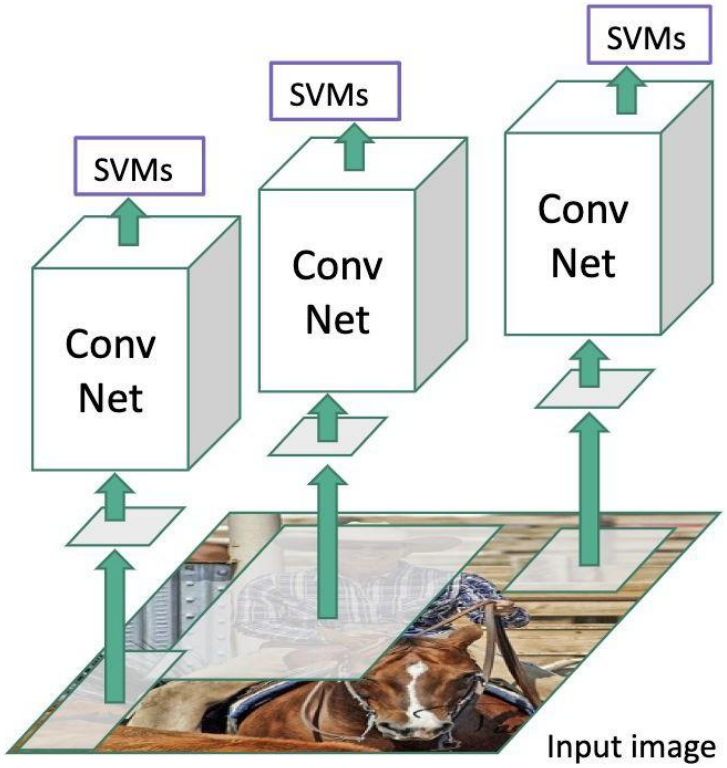

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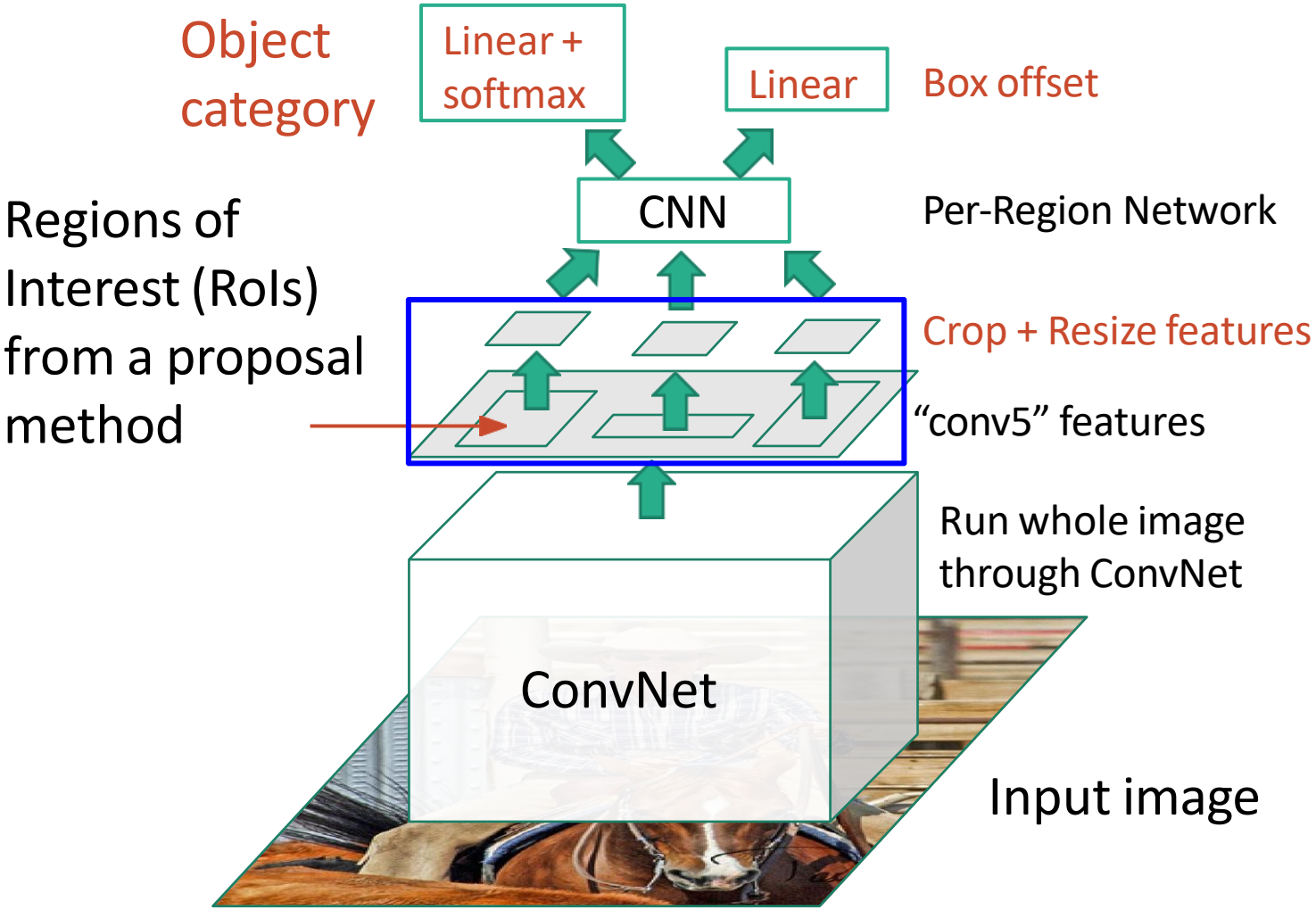

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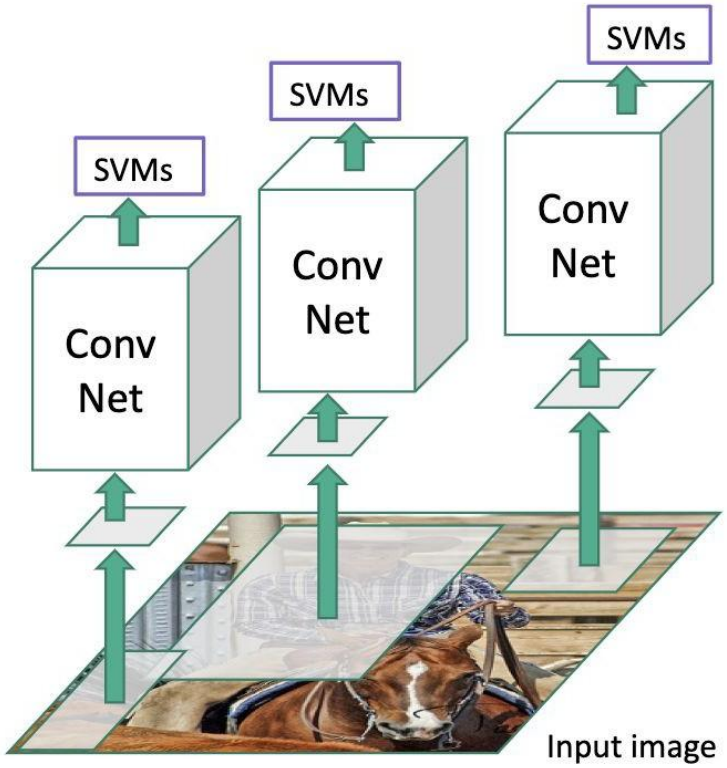
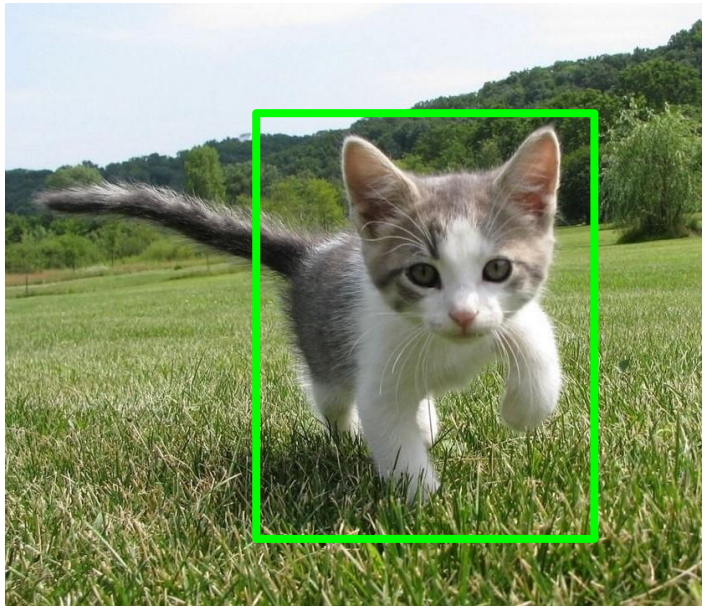


Cropping Features: RoI Pool



Input Image
(e.g. 3 x 640 x 480)

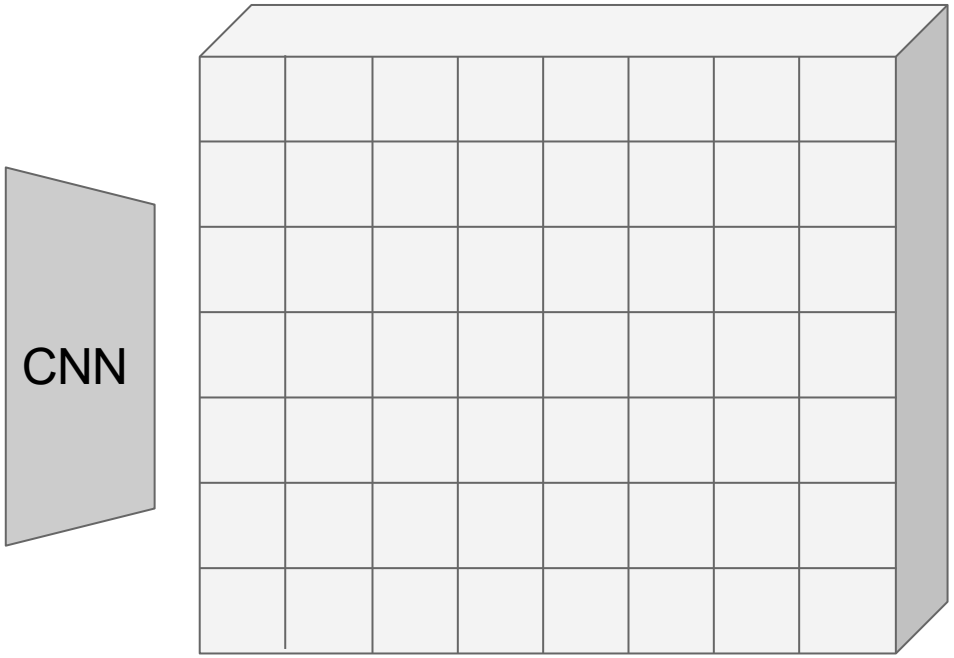
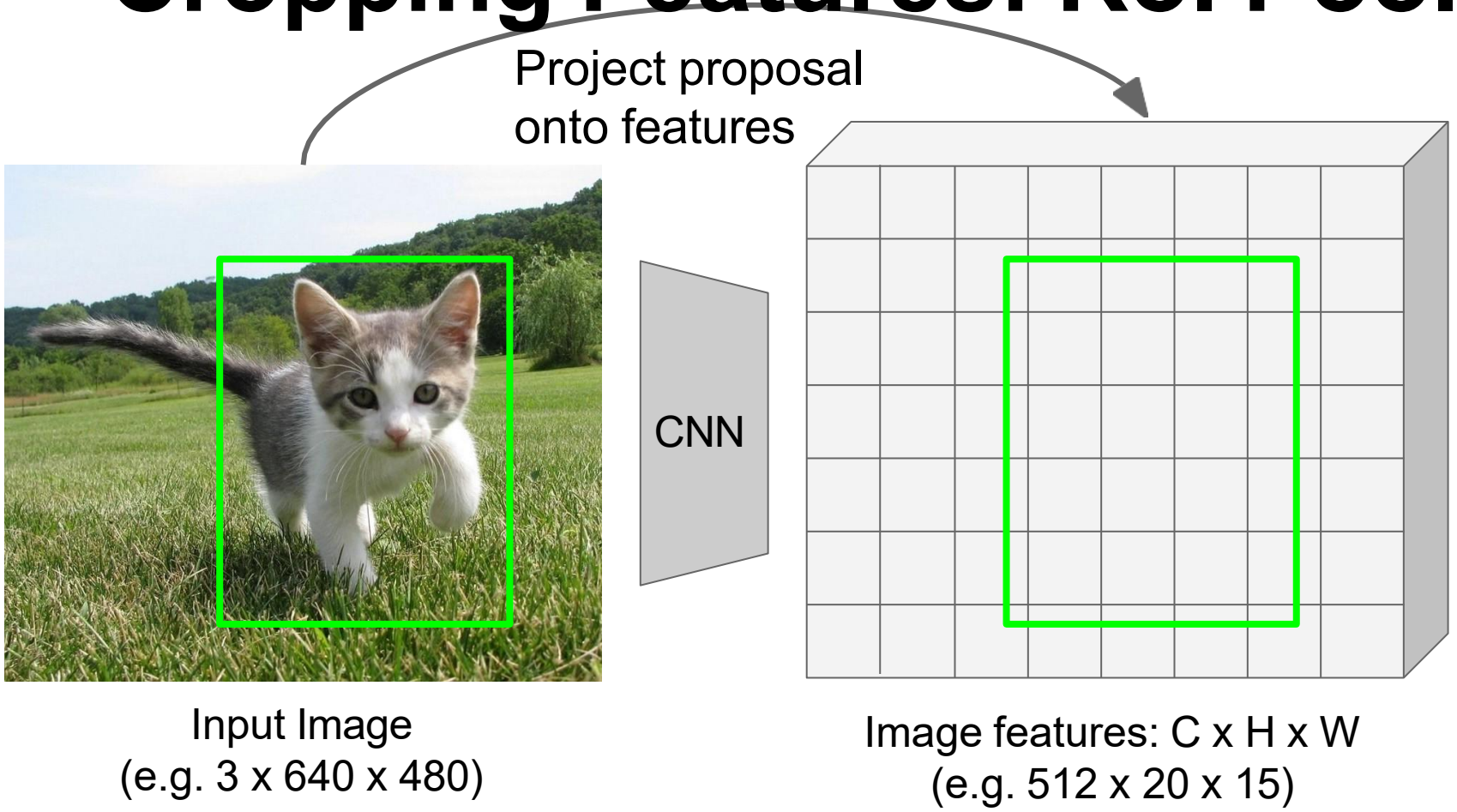


Image features: C x H x W
(e.g. 512 x 20 x 15)

Cropping Features: RoI Pool



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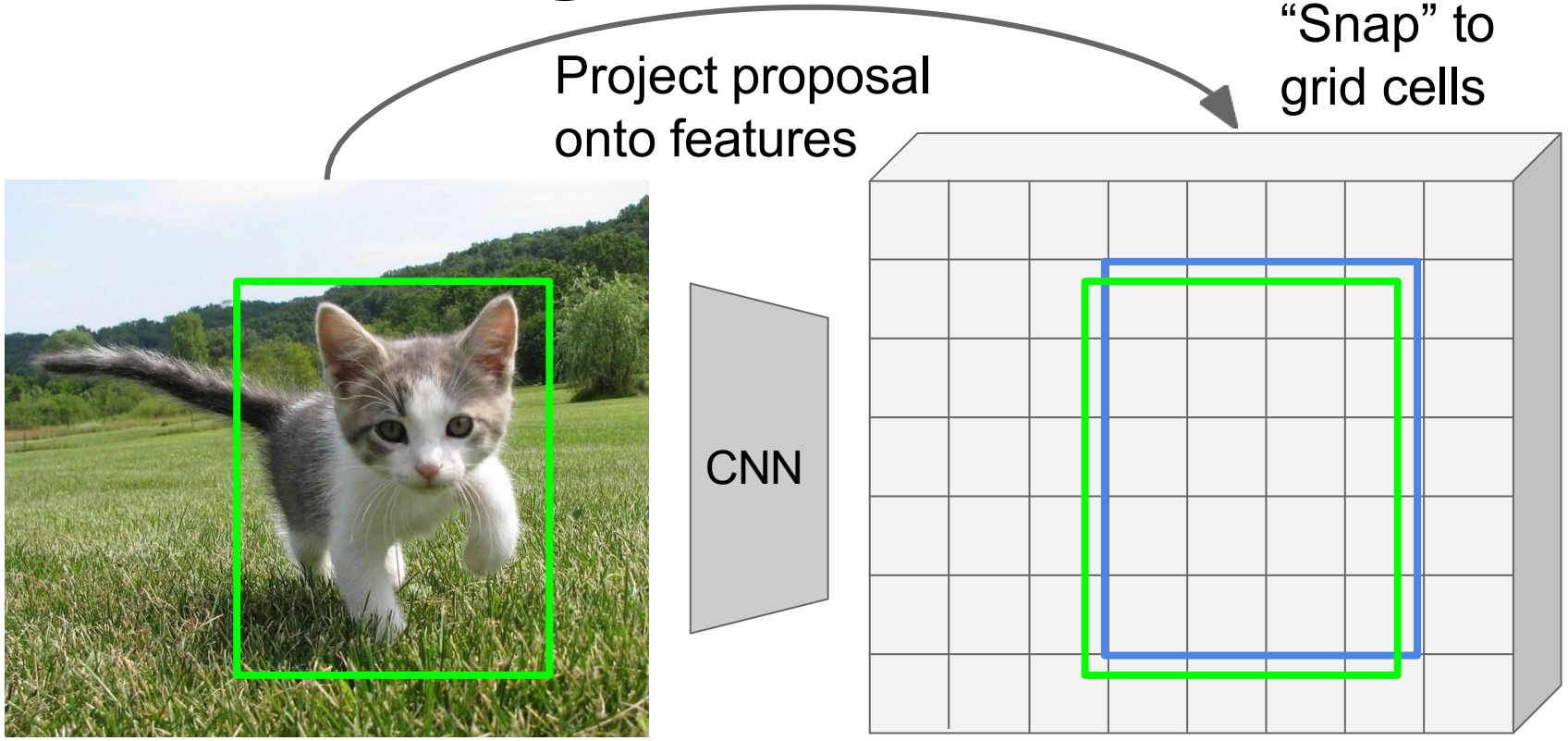

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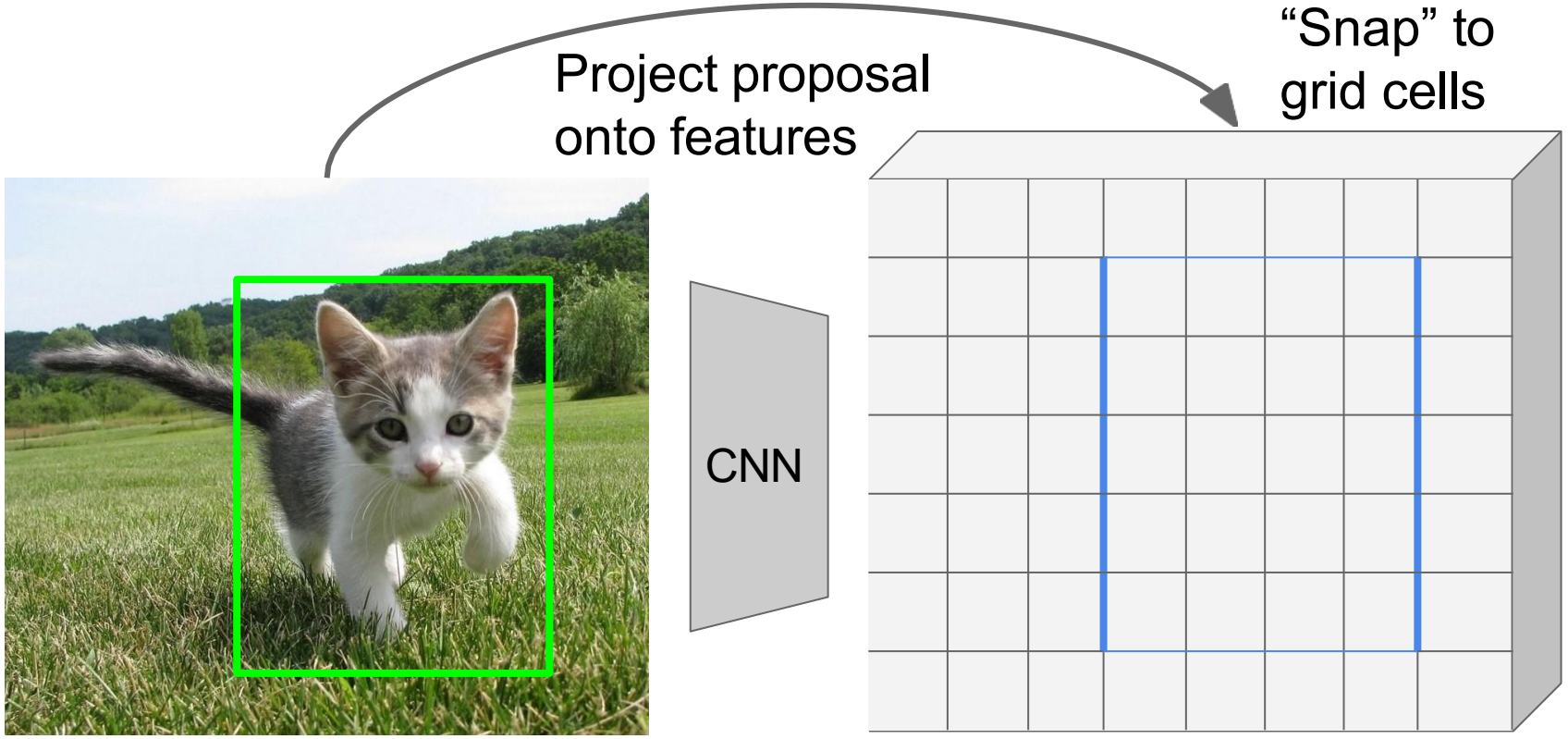
Image features: C x H x W
(e.g. 512 x 20 x 15)

Cropping Features: RoI Pool

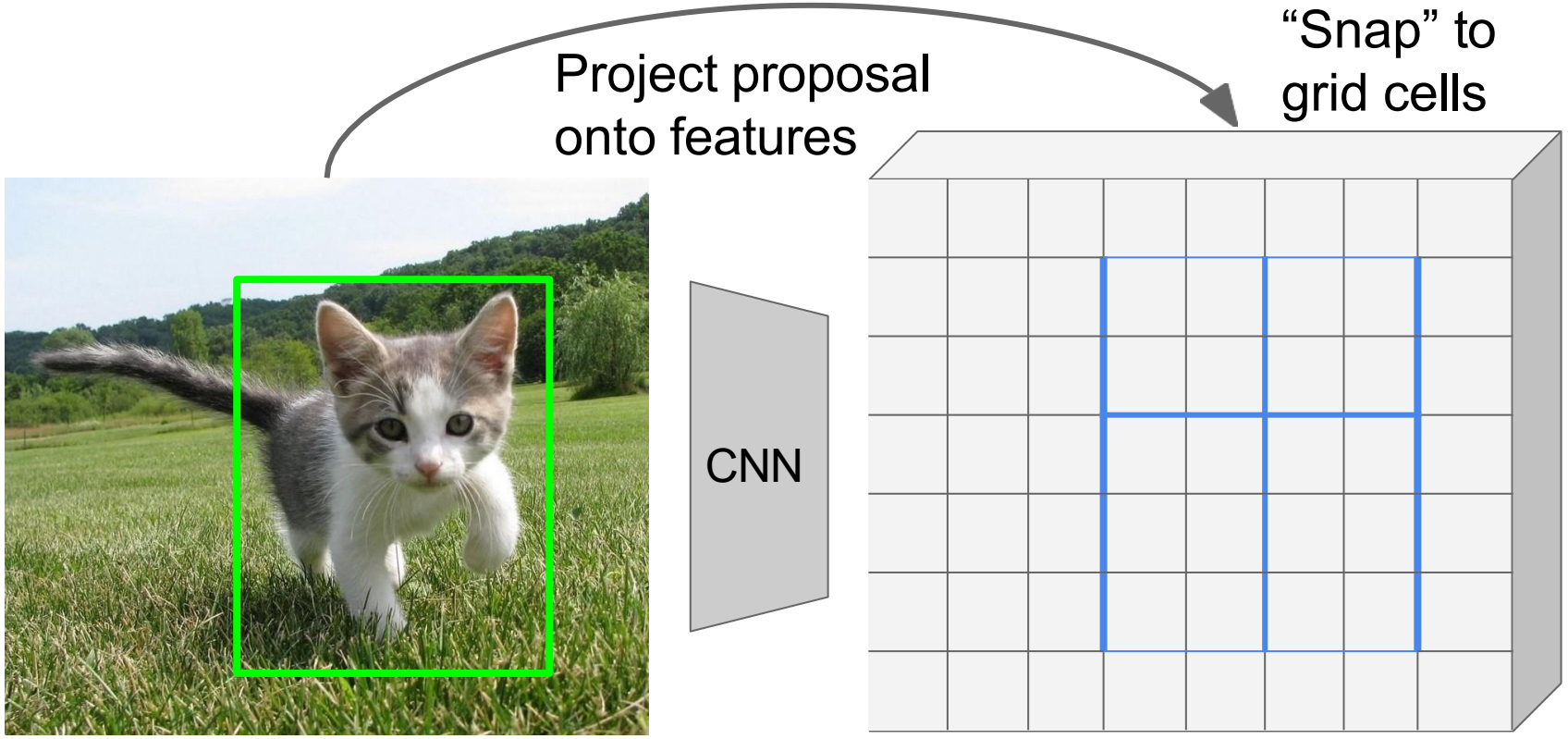


Input Image
(e.g. 3 x 640 x 480)

Image features: C x H x W
(e.g. 512 x 20 x 15)

Q: how do we resize the 512 x 5 x 4 region to, e.g., a 512 x 2 x 2 tensor?.

Cropping Features: RoI Pool



Input Image
(e.g. 3 x 640 x 480)

Image features: C x H x W
(e.g. 512 x 20 x 15)

Divide into 2x2 grid of (roughly) equal subregions

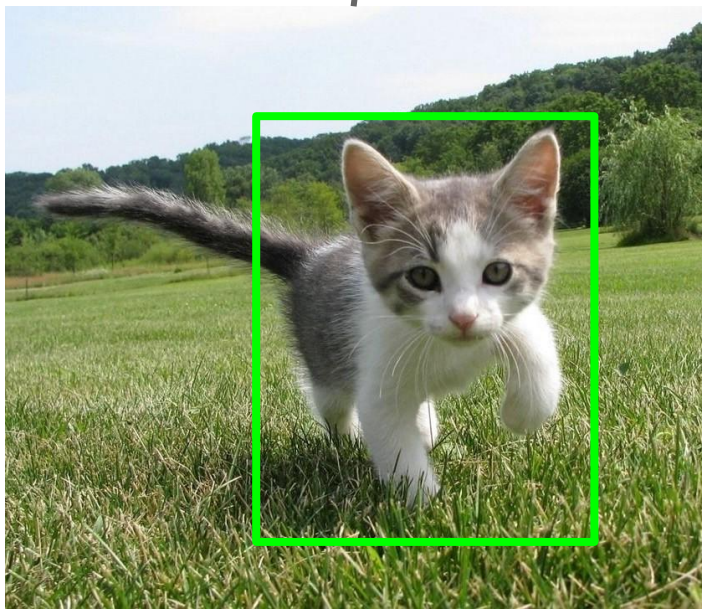
Q: how do we resize the 512 x 5 x 4 region to, e.g., a 512 x 2 x 2 tensor?.

Cropping Features: RoI Pool

Divide into 2x2 grid of (roughly) equal subregions

"Snap" to grid cells

Project proposal onto features



Input Image
(e.g. 3 x 640 x 480)

CNN

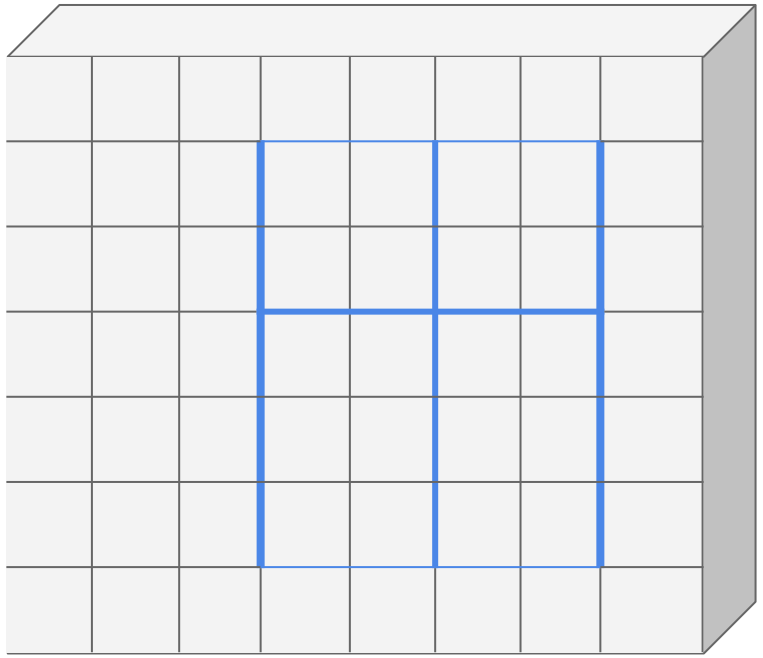
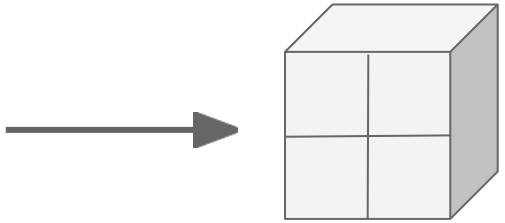


Image features: C x H x W
(e.g. 512 x 20 x 15)

Max-pool within each subregion



Region features
(here 512 x 2 x 2;
In practice e.g 512 x 7 x 7)

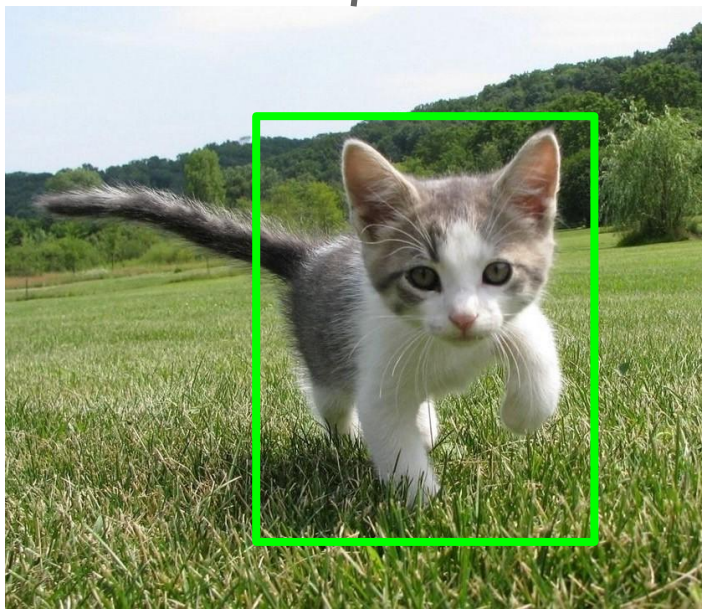
Region features always the same size even if input regions have different sizes!

Cropping Features: RoI Pool

Divide into 2x2 grid of (roughly) equal subregions

"Snap" to grid cells

Project proposal onto features



Input Image
(e.g. 3 x 640 x 480)

CNN

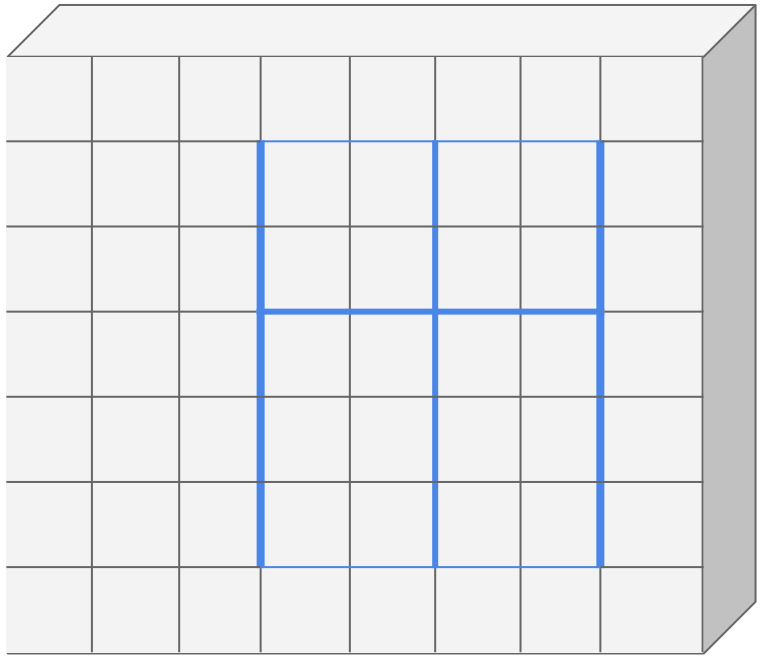
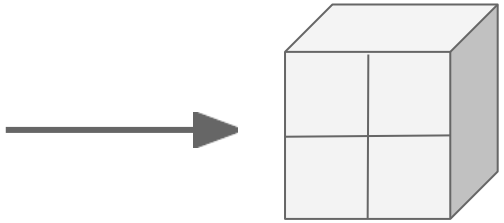


Image features: C x H x W
(e.g. 512 x 20 x 15)

Max-pool within each subregion

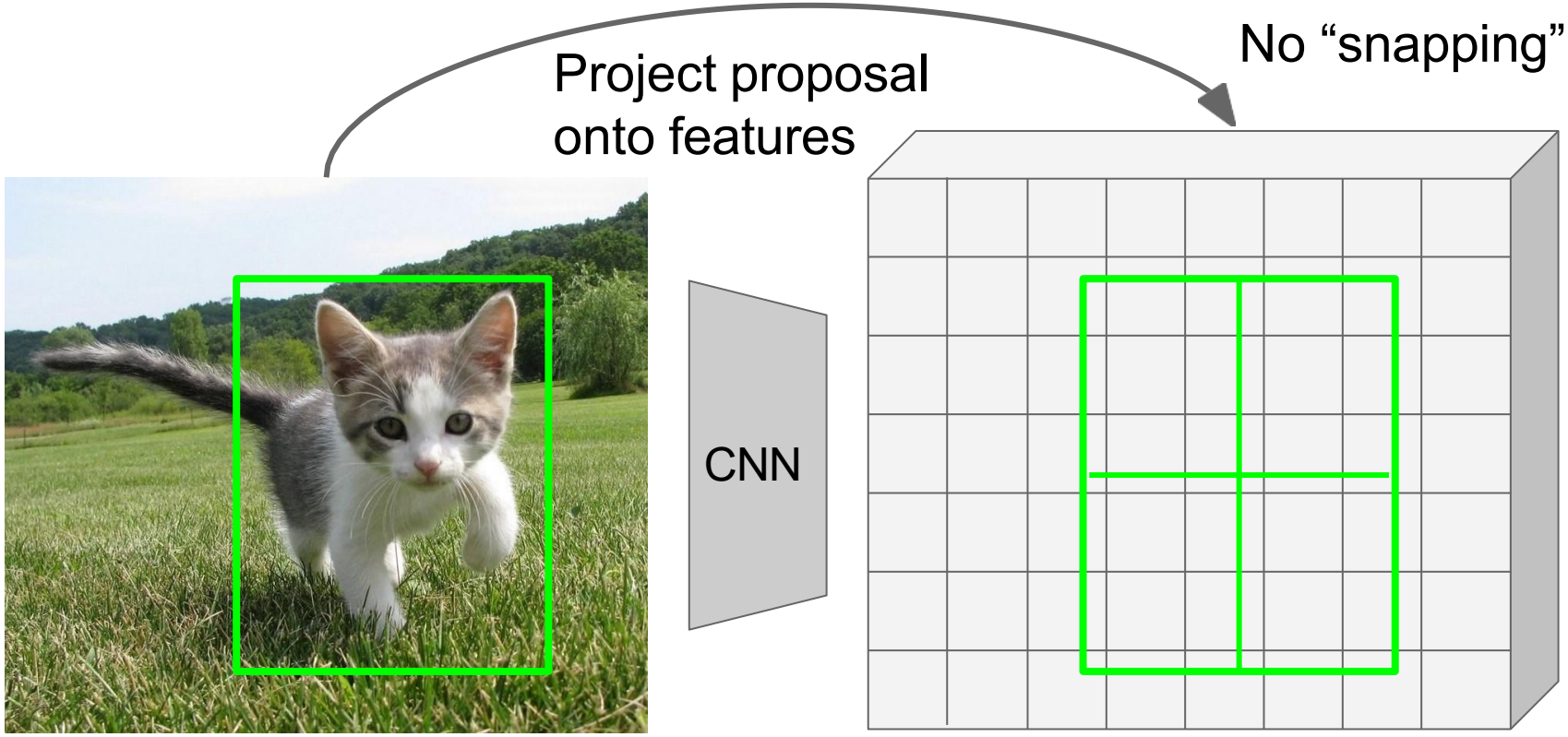


Region features
(here 512 x 2 x 2;
In practice e.g 512 x 7 x 7)

Region features always the same size even if input regions have different sizes!

Problem: Region features slightly misaligned

Cropping Features: RoI Align

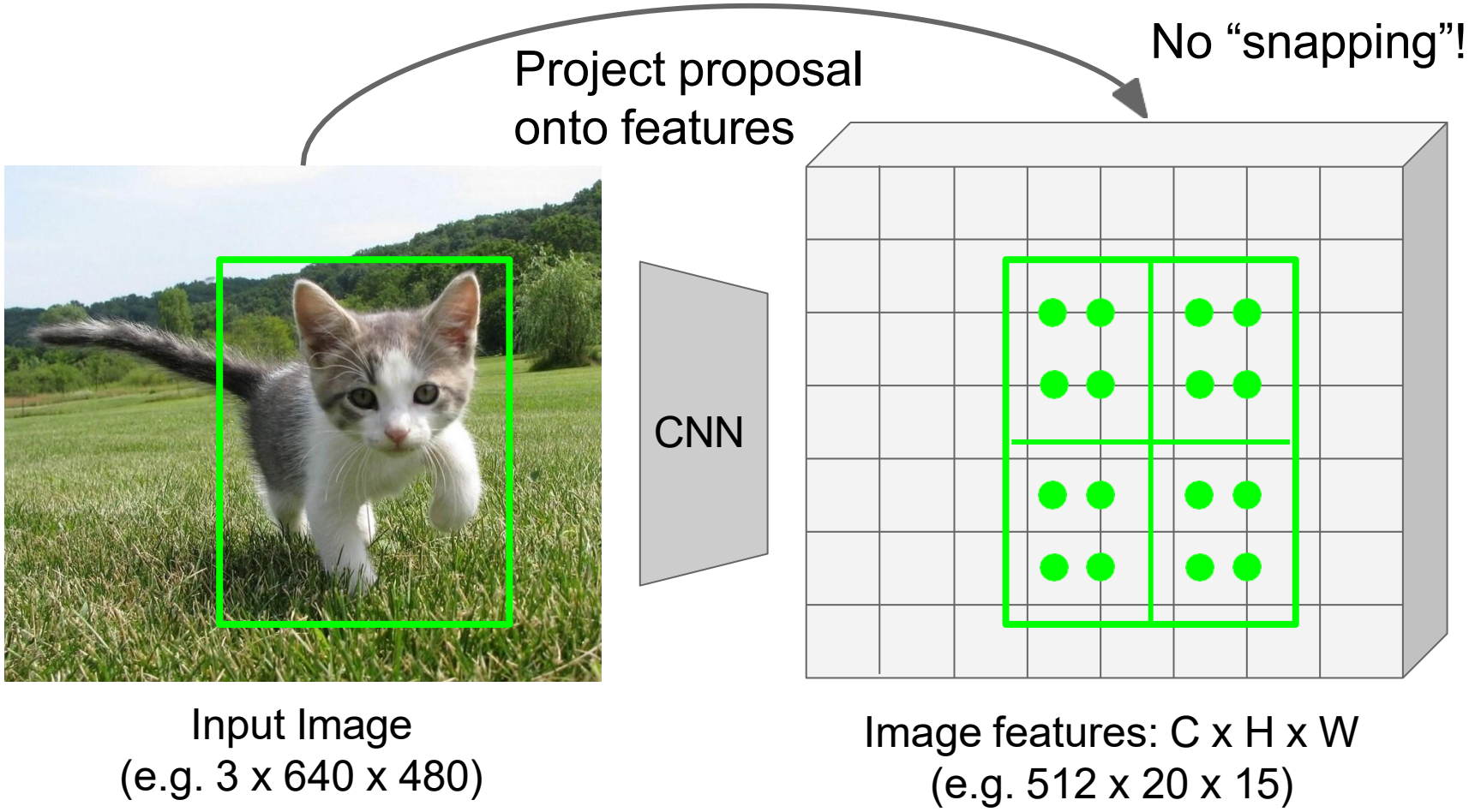


Input Image
(e.g. 3 x 640 x 480)

Image features: C x H x W
(e.g. 512 x 20 x 15)

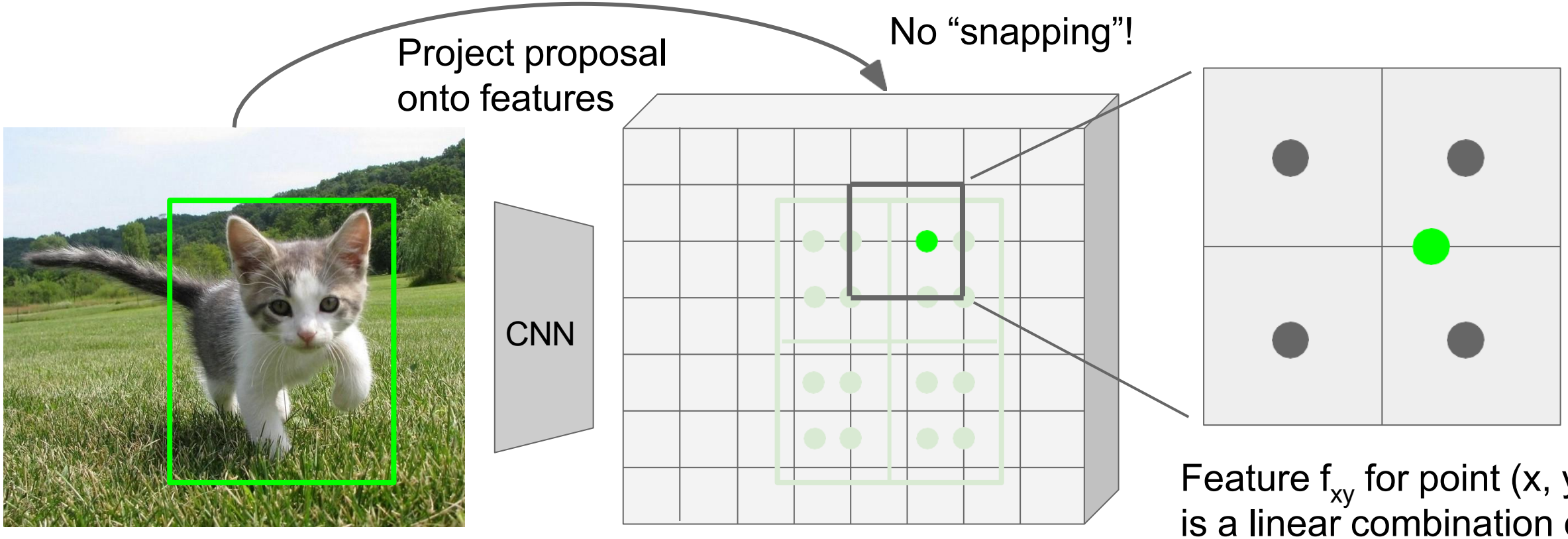
Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation



Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation



Input Image
(e.g. 3 x 640 x 480)

Image features: C x H x W
(e.g. 512 x 20 x 15)

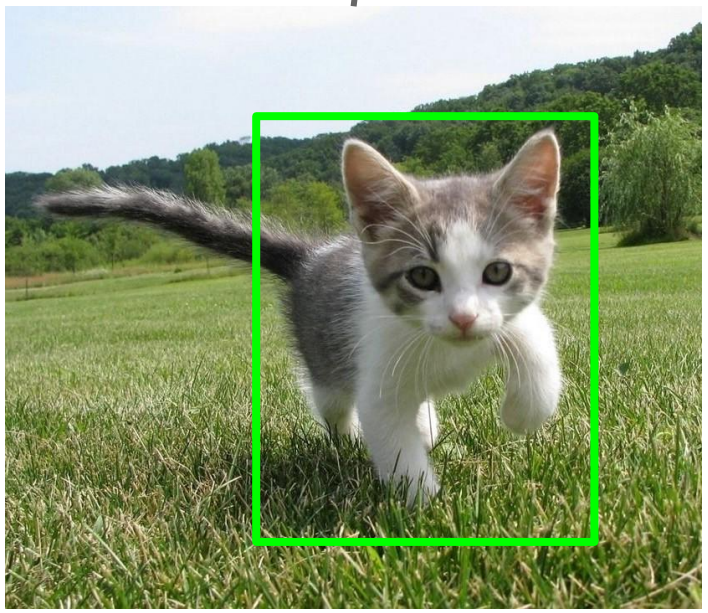
Feature f_{xy} for point (x, y) is a linear combination of features at its four neighboring grid cells:

Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation

No "snapping"!

Project proposal onto features



Input Image
(e.g. 3 x 640 x 480)

CNN

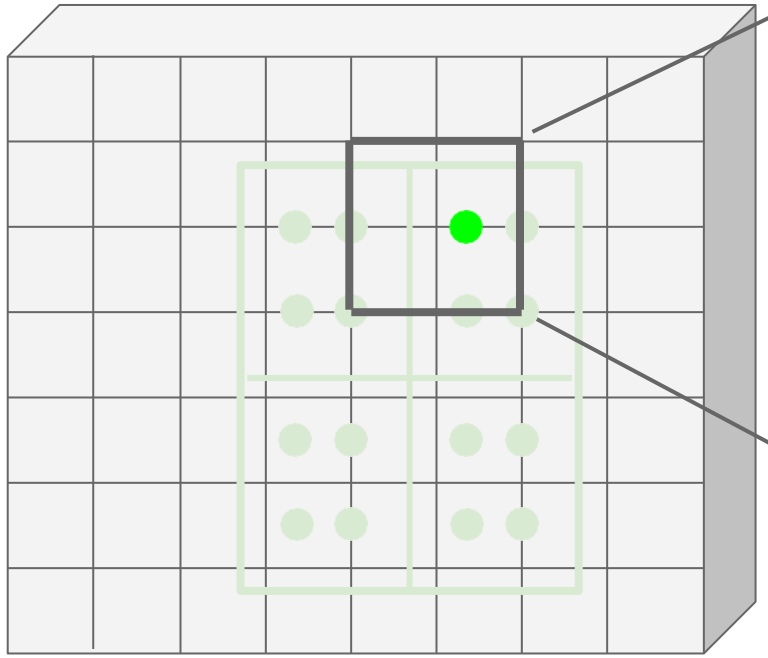
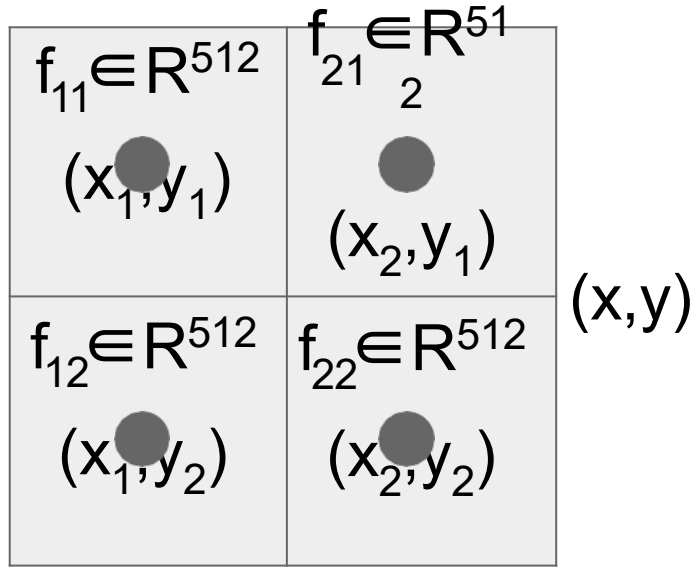


Image features: C x H x W
(e.g. 512 x 20 x 15)

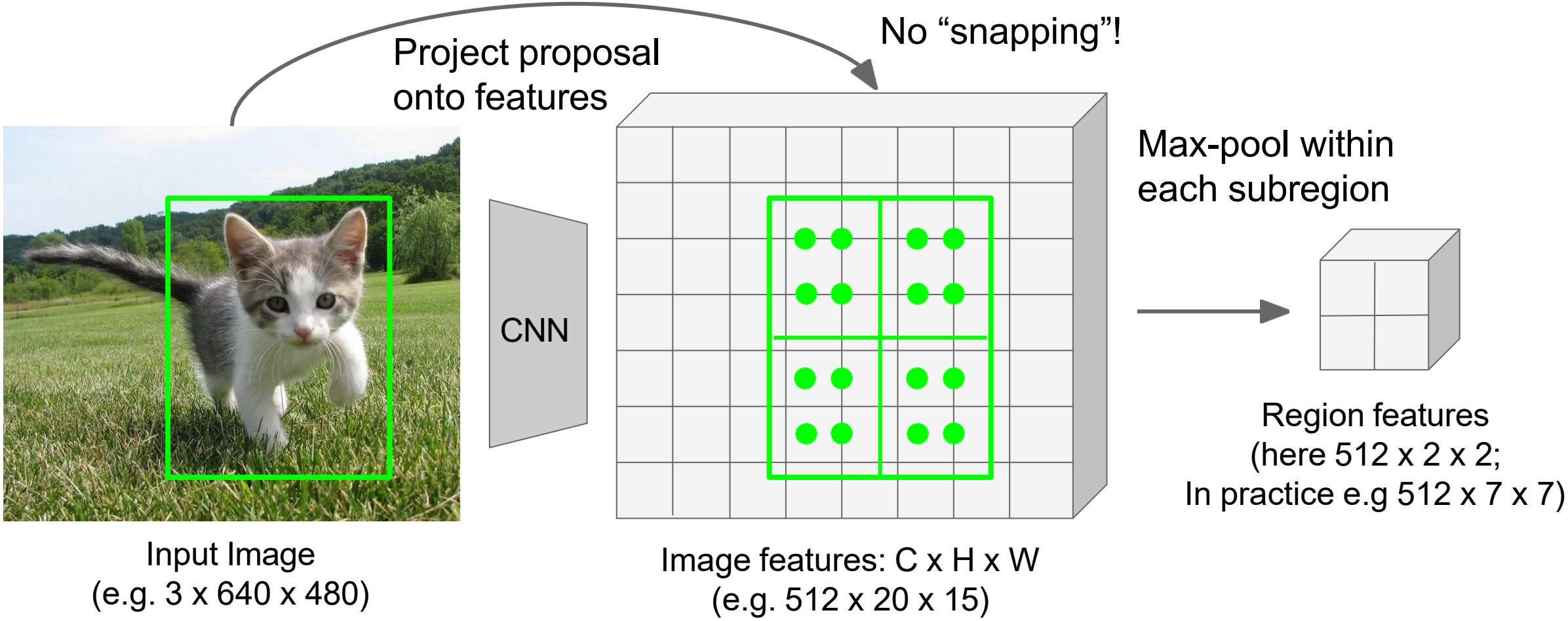


Feature f_{xy} for point (x, y) is a linear combination of features at its four neighboring grid cells:

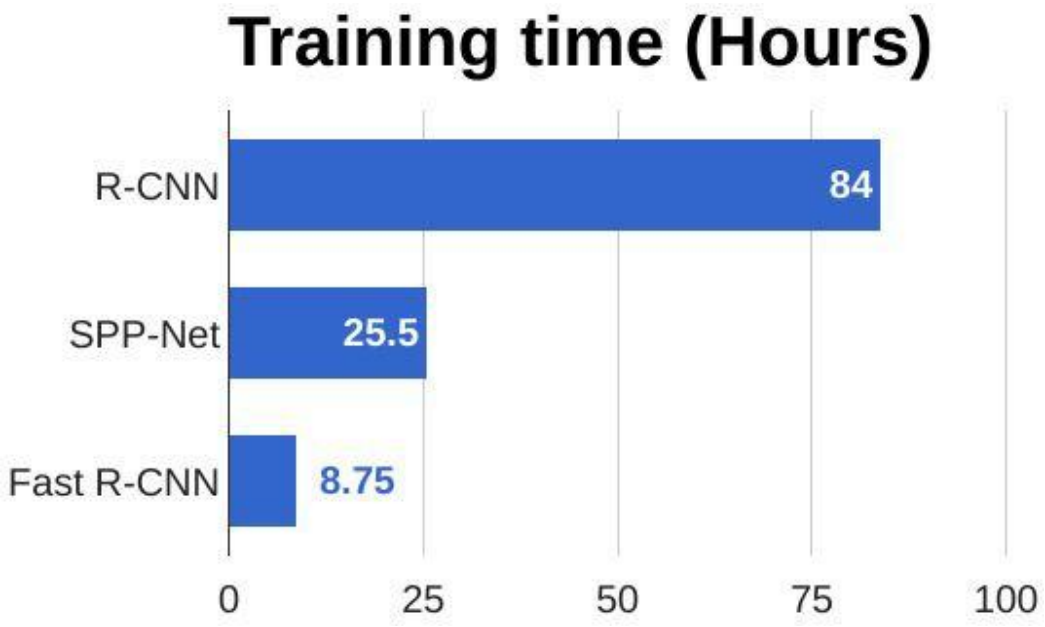
$$f_{xy} = \sum_{i,j=1}^{2} f_{i,j} \max(0, 1 - |x - x_i|) \max(0, 1 - |y - y_j|)$$

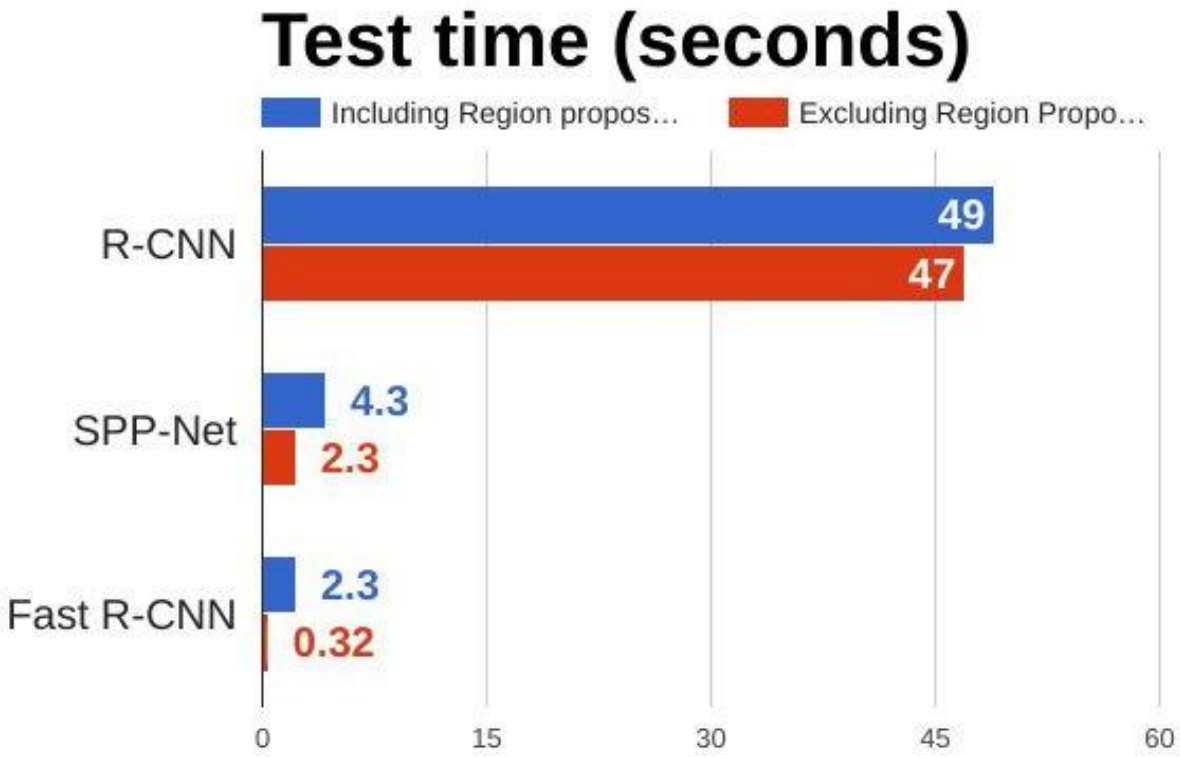
Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation

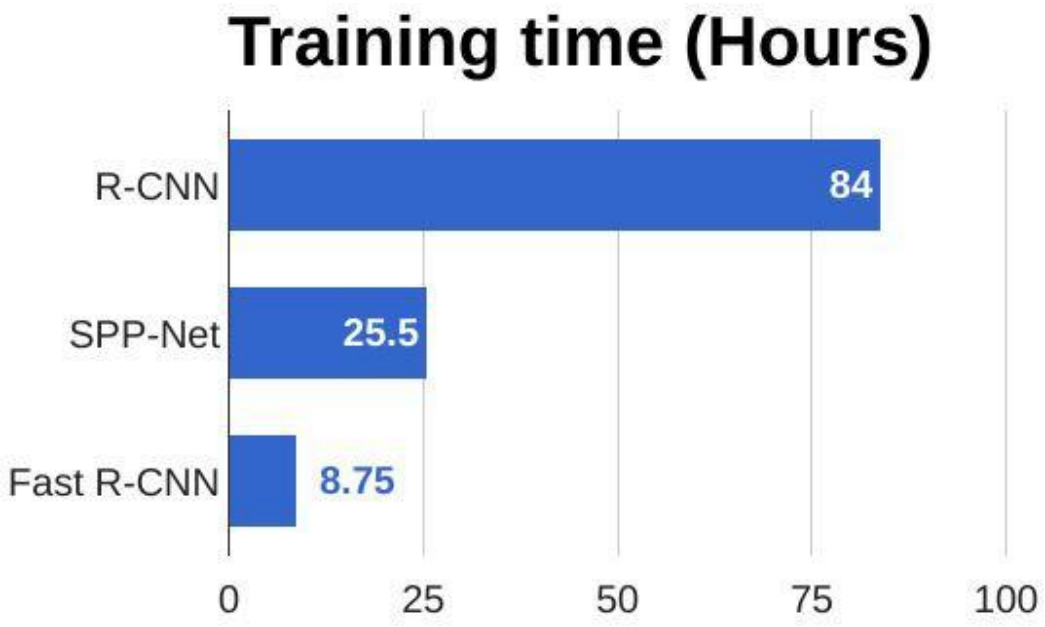


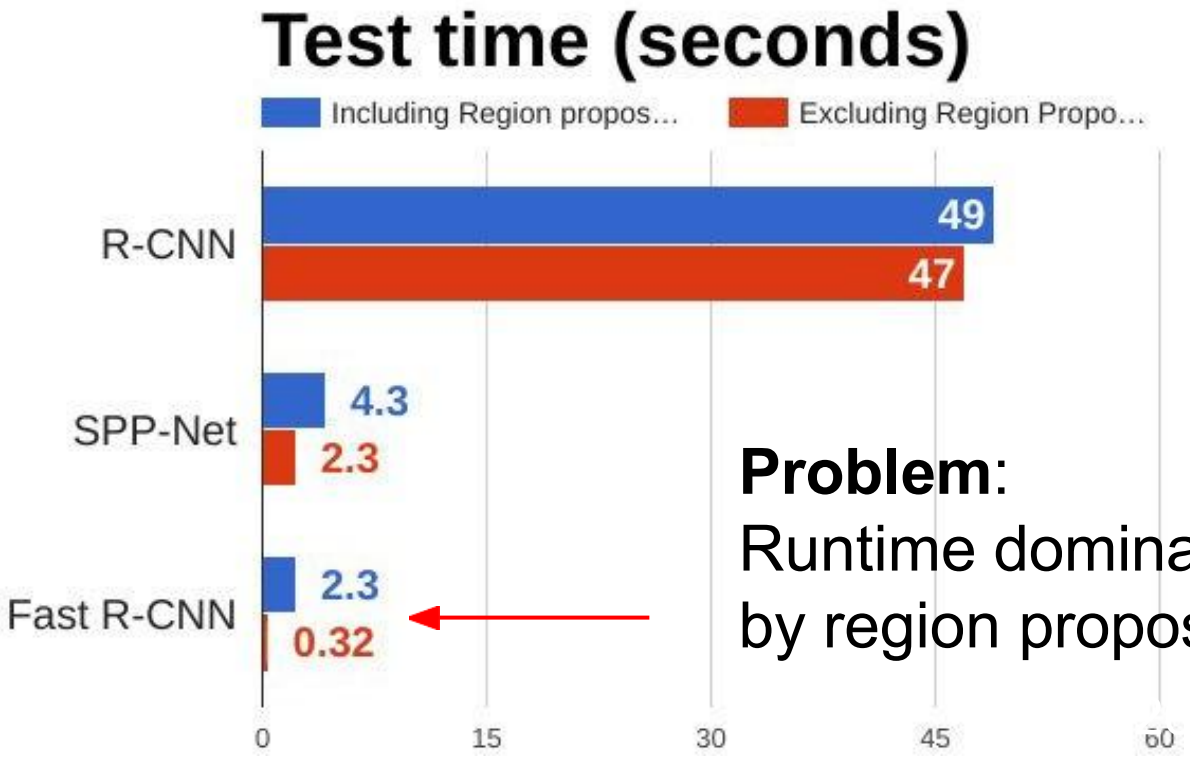
R-CNN vs Fast R-CNN





R-CNN vs Fast R-CNN





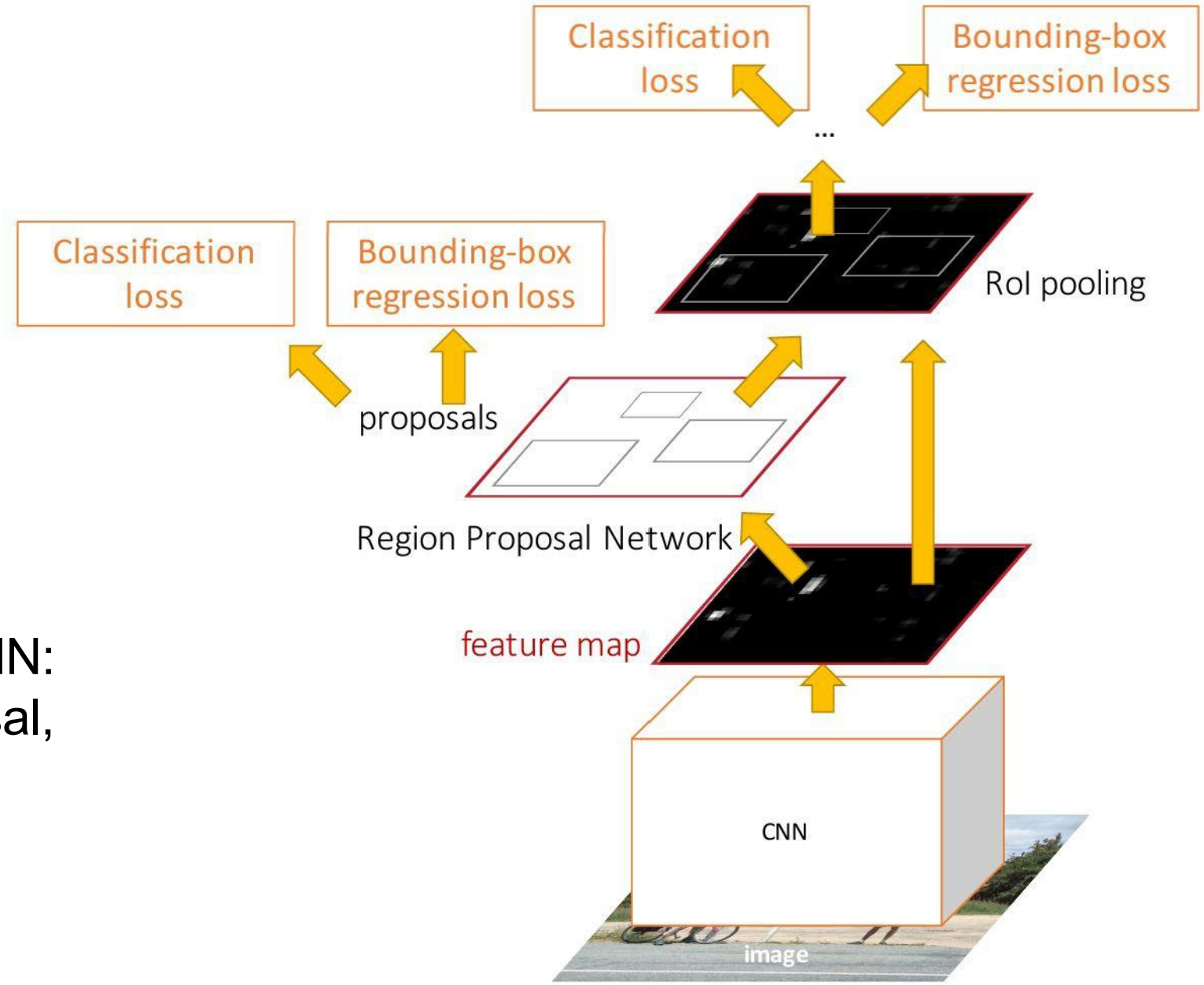
Problem:
Runtime dominated by region proposals!

Faster R-CNN:

Make CNN do proposals!

Insert **Region Proposal Network (RPN)** to predict proposals from features

Otherwise same as Fast R-CNN: Crop features for each proposal, classify each one



Region Proposal Network

Input Image
(e.g. 3 x 640 x 480)

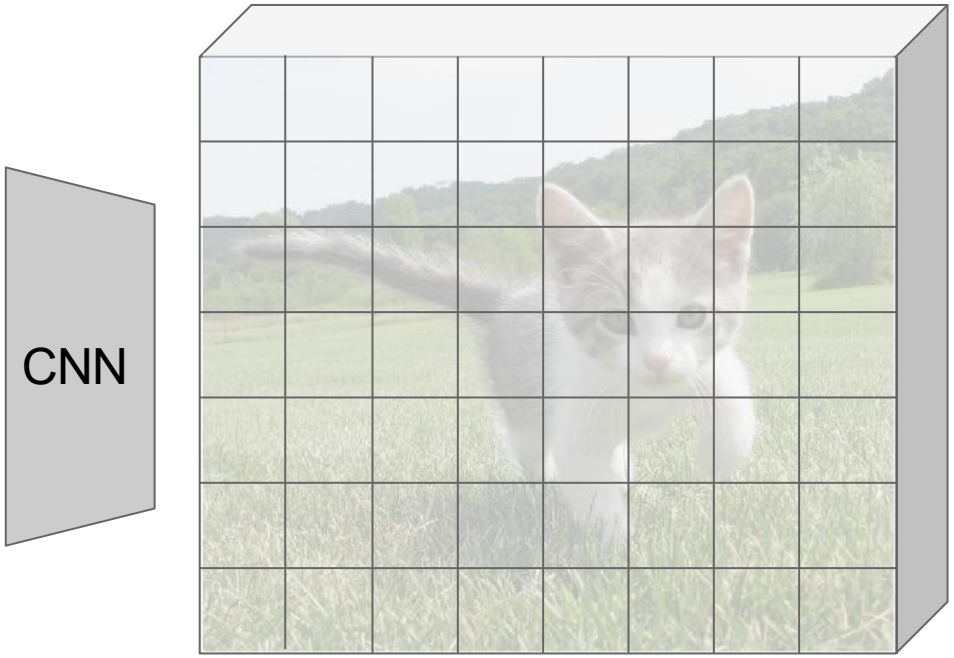


Image features
(e.g. 512 x 20 x 15)

Region Proposal Network

Imagine an **anchor box** of fixed size at each point in the feature map

Input Image
(e.g. 3 x 640 x 480)

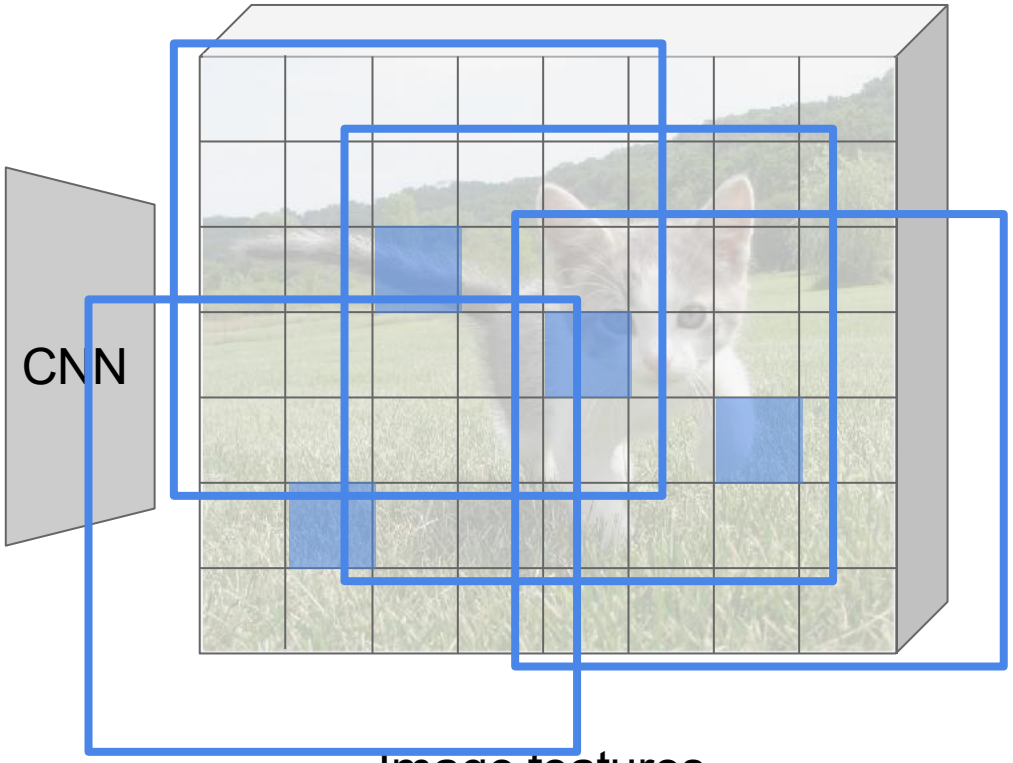


Image features
(e.g. 512 x 20 x 15)

Region Proposal Network

Imagine an **anchor box** of fixed size at each point in the feature map

Input Image
(e.g. 3 x 640 x 480)

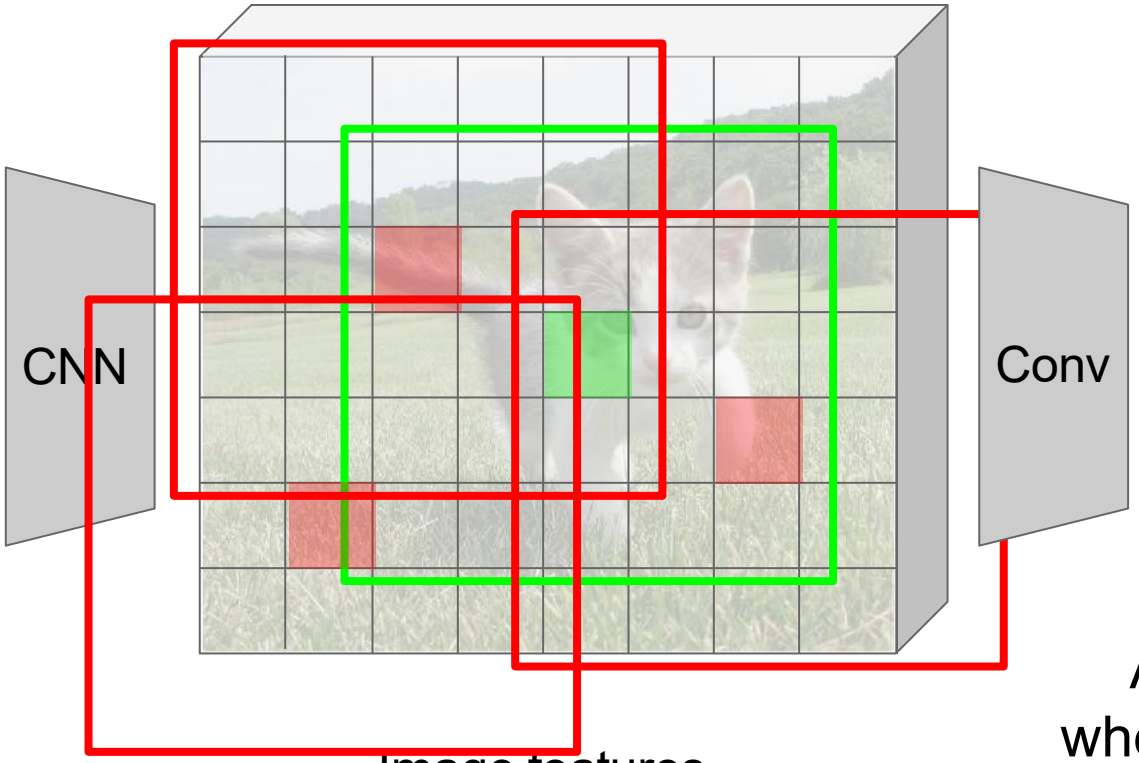


Image features
(e.g. 512 x 20 x 15)

Anchor is an object?
1 x 20 x 15

At each point, predict whether the corresponding anchor contains an object (binary classification)

Region Proposal Network

Imagine an **anchor box** of fixed size at each point in the feature map

Input Image
(e.g. 3 x 640 x 480)

CNN

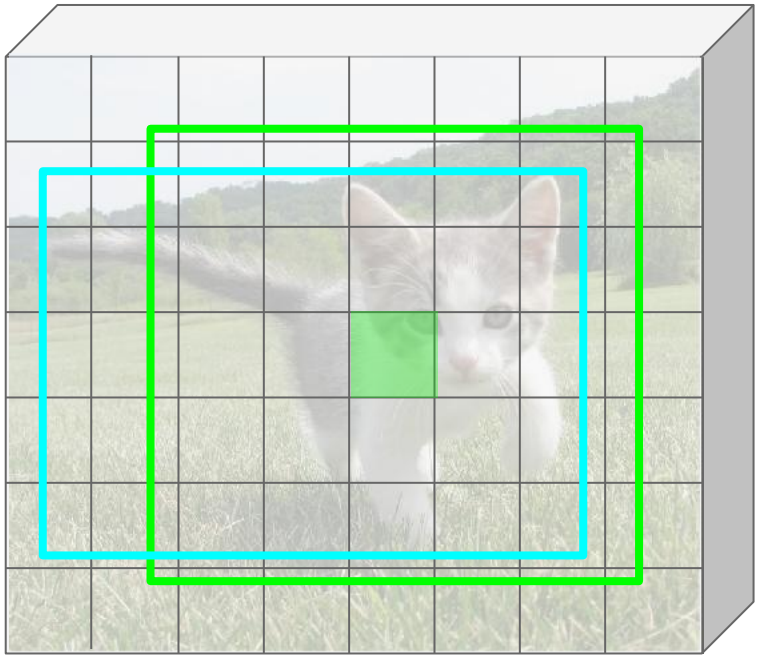


Image features
(e.g. 512 x 20 x 15)

Conv

Anchor is an object?
1 x 20 x 15

Box corrections
4 x 20 x 15

For positive boxes, also predict a corrections from the anchor to the ground-truth box (regress 4 numbers per pixel)

Region Proposal Network

In practice use K different anchor boxes of different size / scale at each point

Input Image
(e.g. 3 x 640 x 480)

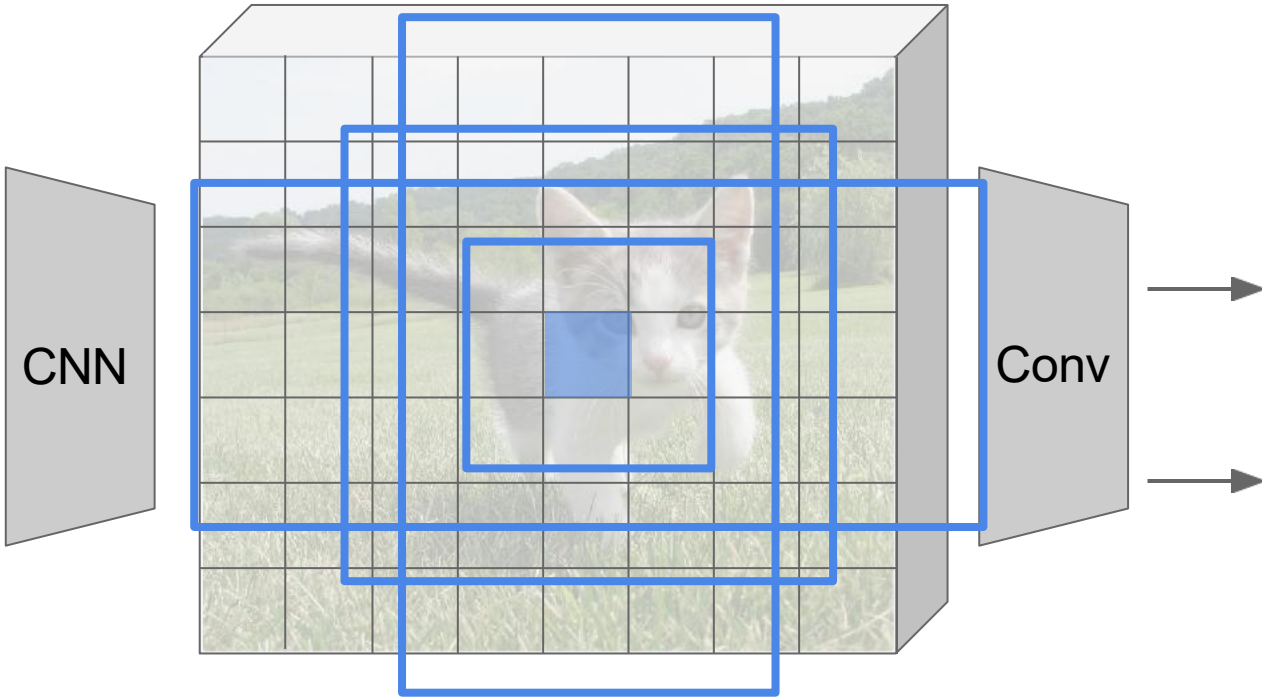


Image features
(e.g. 512 x 20 x 15)

Anchor is an object?
K x 20 x 15

Box transforms
4K x 20 x 15

Region Proposal Network

In practice use K different anchor boxes of different size / scale at each point

Input Image
(e.g. 3 x 640 x 480)

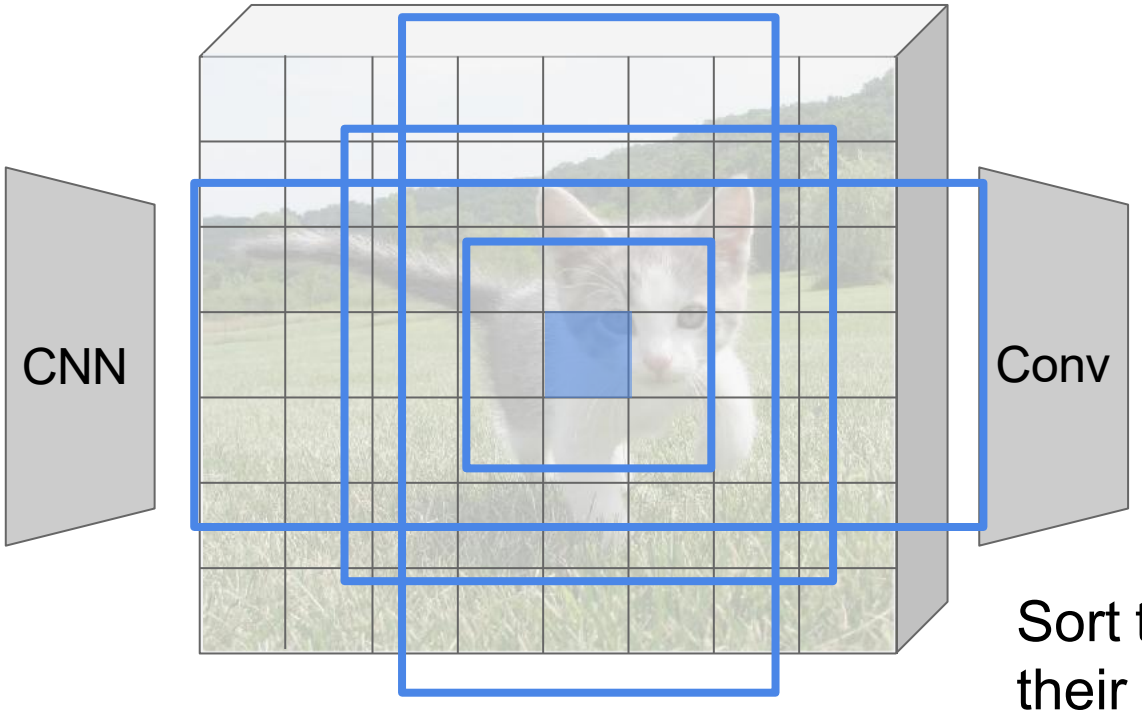


Image features
(e.g. 512 x 20 x 15)

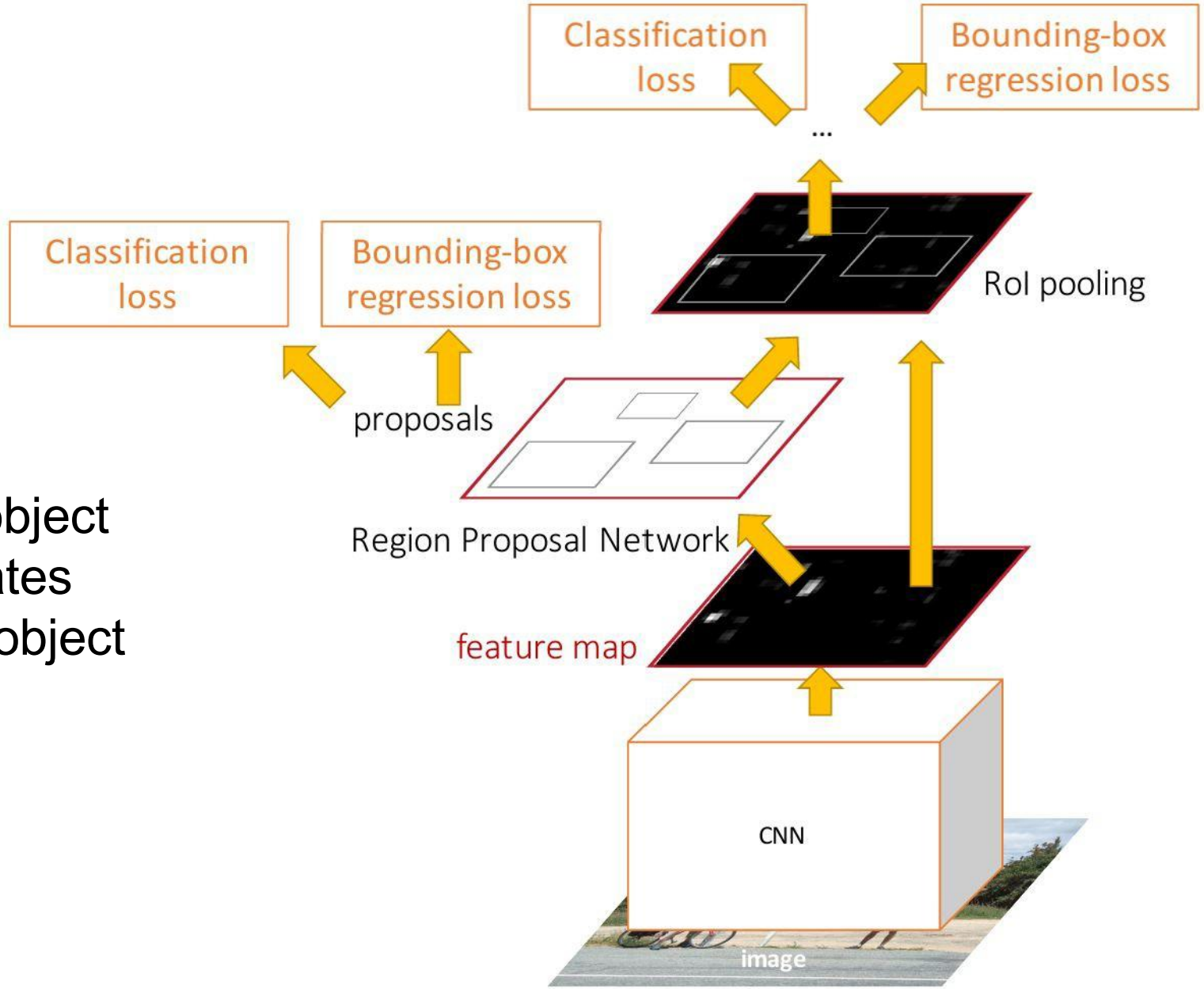
Anchor is an object?
K x 20 x 15

Box transforms
4K x 20 x 15

Sort the K*20*15 boxes by their “objectness” score, take top ~300 as our proposals

Faster R-CNN:

Make CNN do proposals!

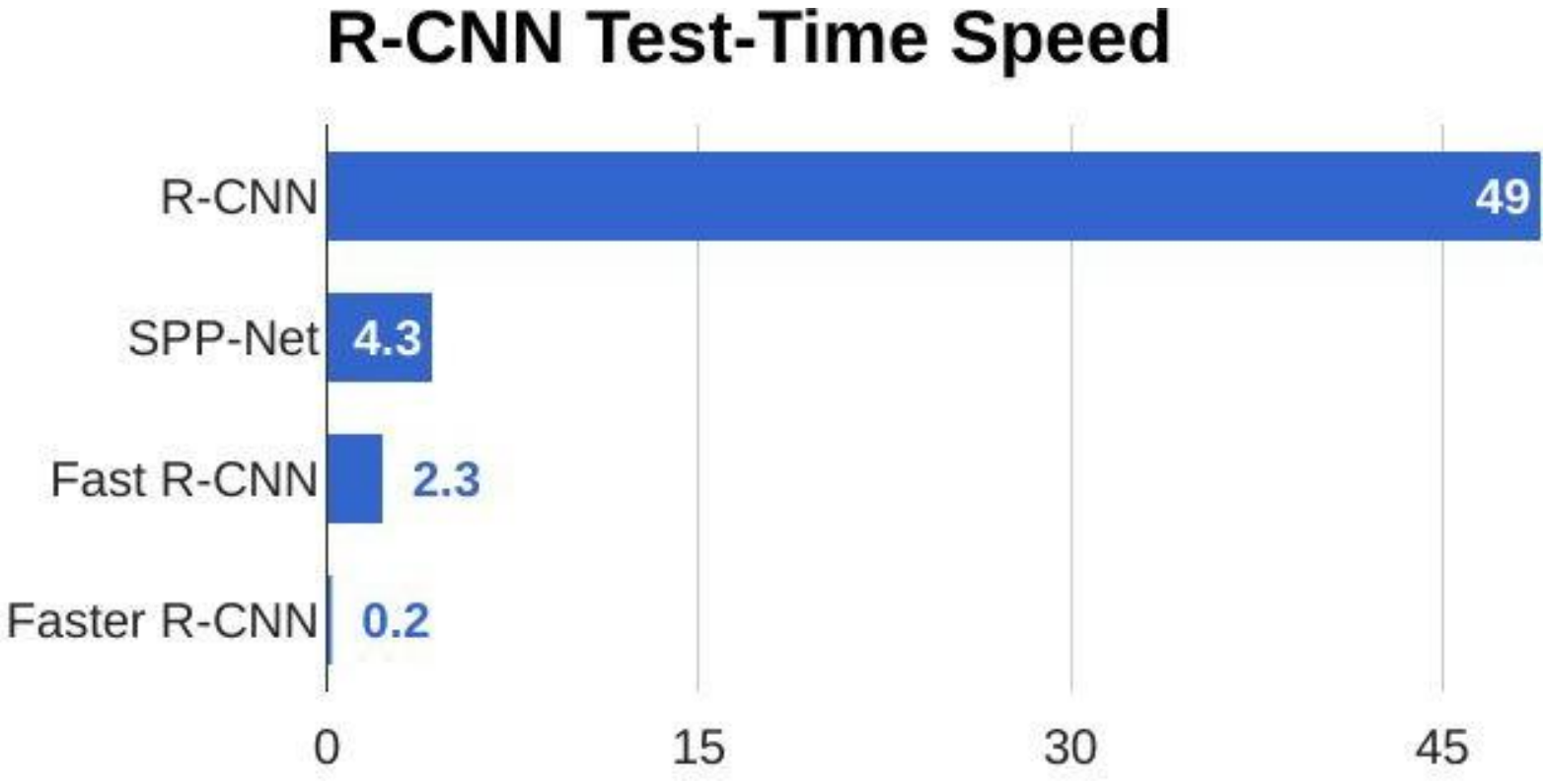


Jointly train with 4 losses:

1. RPN classify object / not object
2. RPN regress box coordinates
3. Final classification score (object classes)
4. Final box coordinates

Faster R-CNN:

Make CNN do proposals!

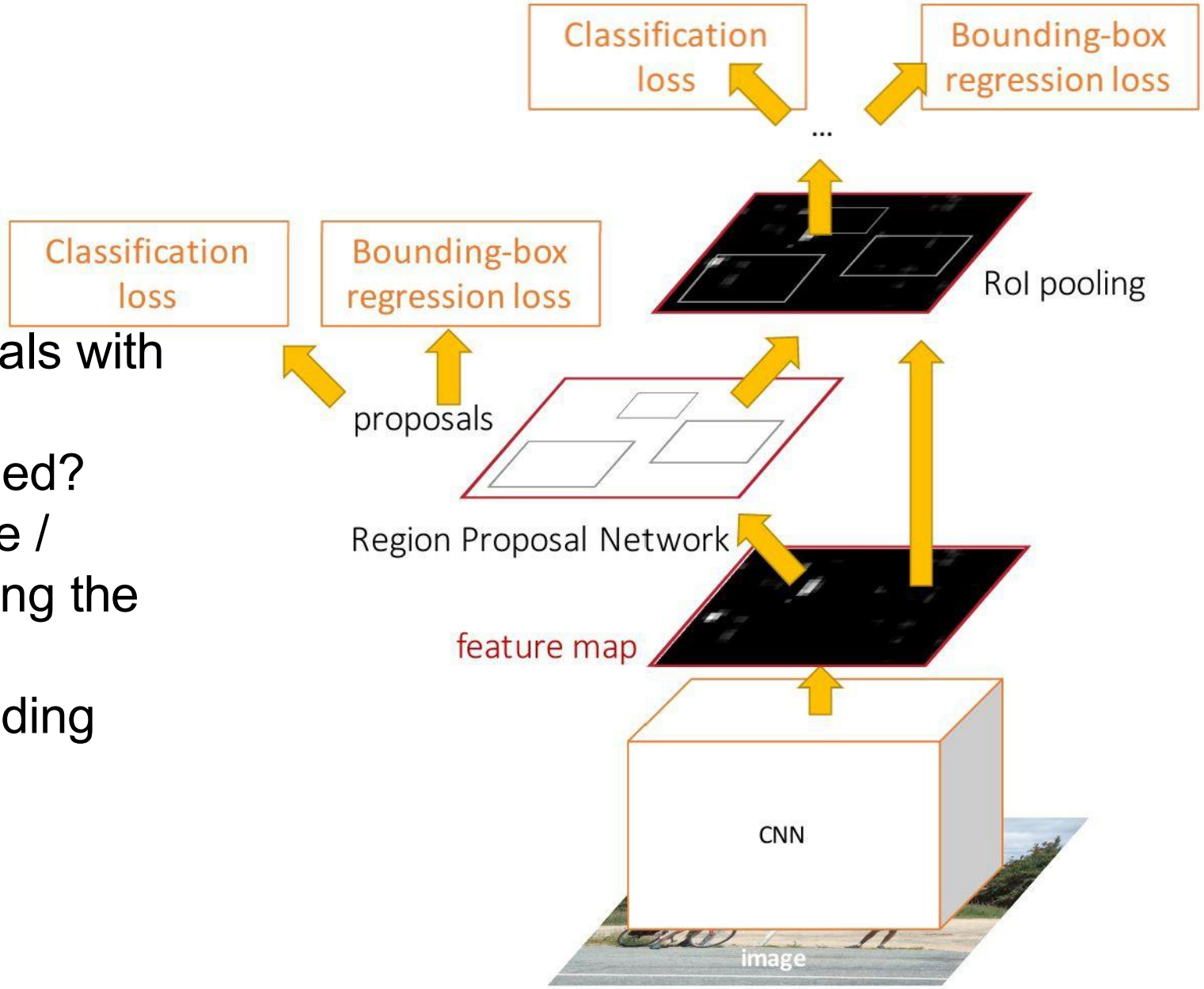


Faster R-CNN:

Make CNN do proposals!

Glossing over many details:

- Ignore overlapping proposals with **non-max suppression**
- How are anchors determined?
- How do we sample positive / negative samples for training the RPN?
- How to parameterize bounding box regression?



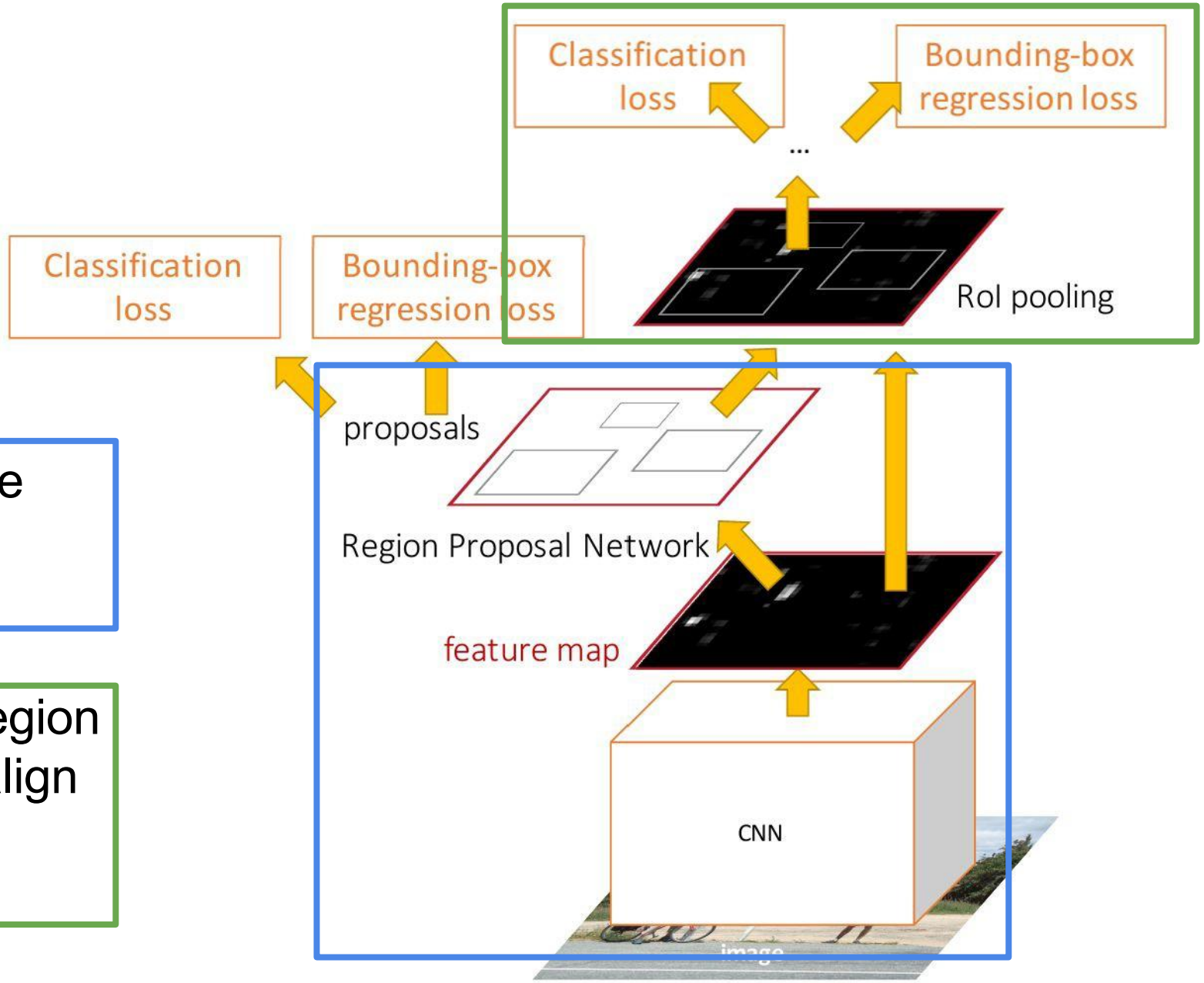
Faster R-CNN:

Make CNN do proposals!

Faster R-CNN is a
Two-stage object detector

First stage: Run once per image
- Backbone network
- Region proposal network

Second stage: Run once per region
- Crop features: RoI pool / align
- Predict object class
- Prediction bbox offset



Faster R-CNN:

Make CNN do proposals!

Do we really need the second stage?

Faster R-CNN is a
Two-stage object detector

First stage: Run once per image
- Backbone network
- Region proposal network

Second stage: Run once per region
- Crop features: RoI pool / align
- Predict object class
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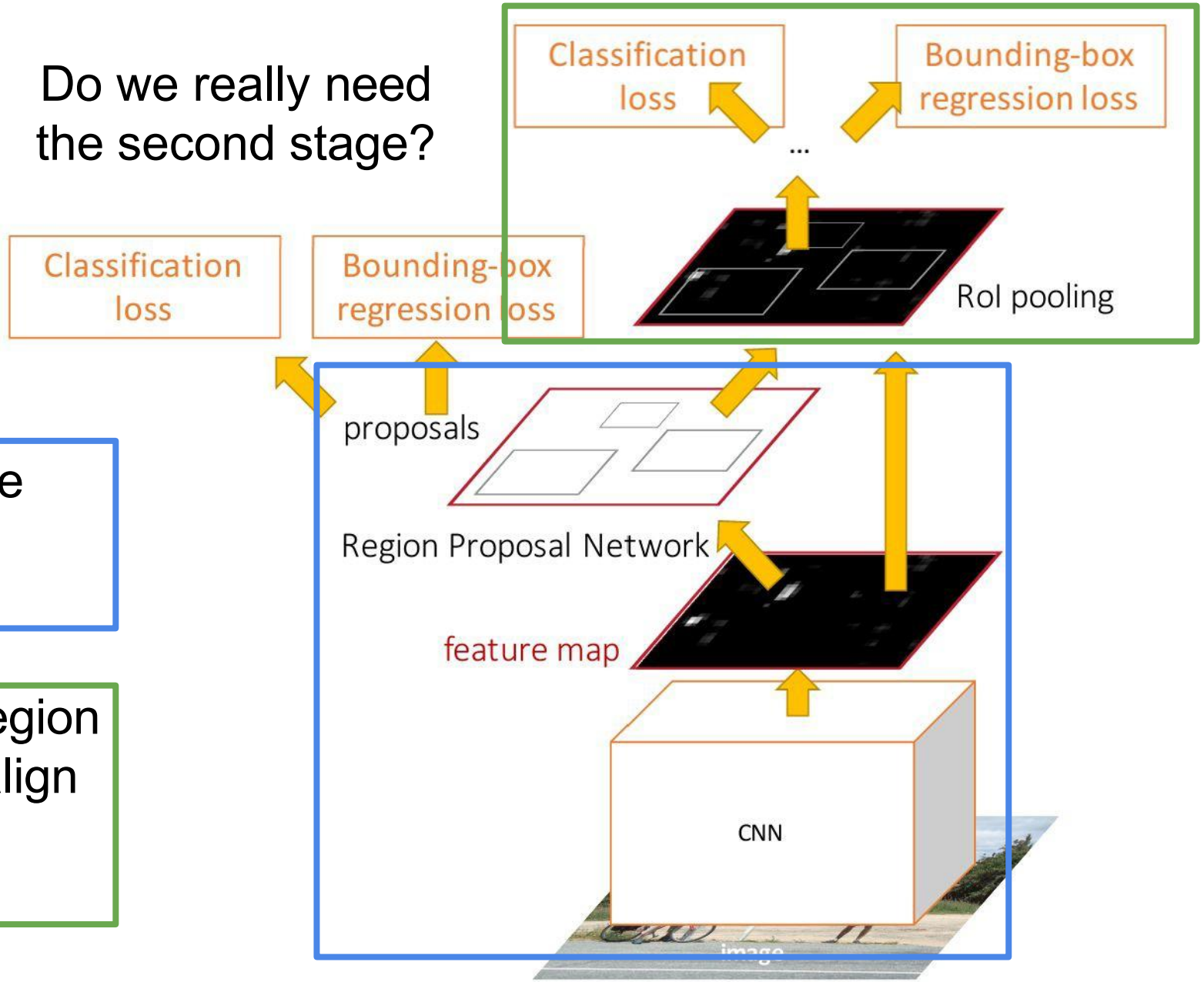
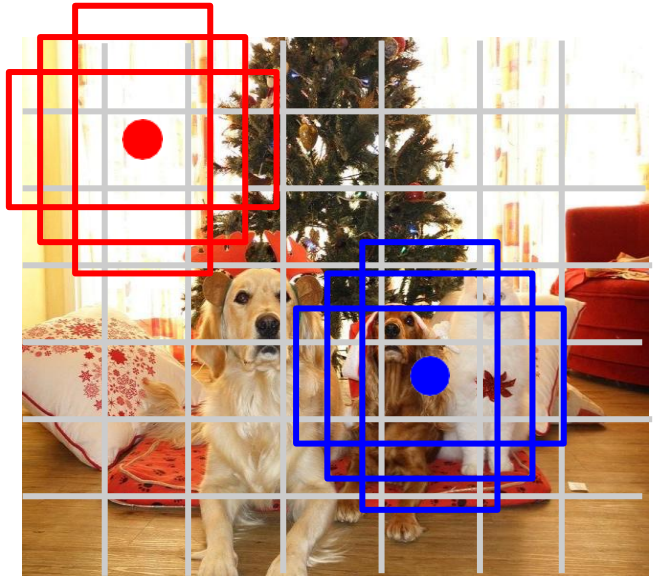


Single-stage object detection

Single-Stage Object Detectors: YOLO / SSD / RetinaNet

Input image
3 x H x W



Divide image into grid
7 x 7

Image a set of **base boxes** centered at each grid cell
Here B = 3

Within each grid cell:

- Regress from each of the B base boxes to a final box with 5 numbers: (dx, dy, dh, dw, confidence)
- Predict scores for each of C classes (including background as a class)
- Looks a lot like RPN, but category-specific!

Output:
7 x 7 x (5 * B + C)

Object Detection: Lots of variables ...

Backbone Network
VGG16
ResNet-101
Inception V2
Inception V3
Inception
ResNet
MobileNet

"Meta-Architecture"
Two-stage: Faster R-CNN
Single-stage: YOLO / SSD
Hybrid: R-FCN

Image Size
 # Region Proposals
...

Takeaways
Faster R-CNN is slower but more accurate

SSD is much faster but not as accurate

Bigger / Deeper backbones work better

Object Detection: Lots of variables ...

Backbone Network
VGG16
ResNet-101
Inception V2
Inception V3
Inception
ResNet
MobileNet

"Meta-Architecture"
Two-stage: Faster R-CNN
Single-stage: YOLO / SSD
Hybrid: R-FCN

Image Size
 # Region Proposals
...

Takeaways
Faster R-CNN is slower but more accurate

SSD is much faster but not as accurate

Bigger / Deeper backbones work better

Instance Segmentation

Classification

CAT

No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection

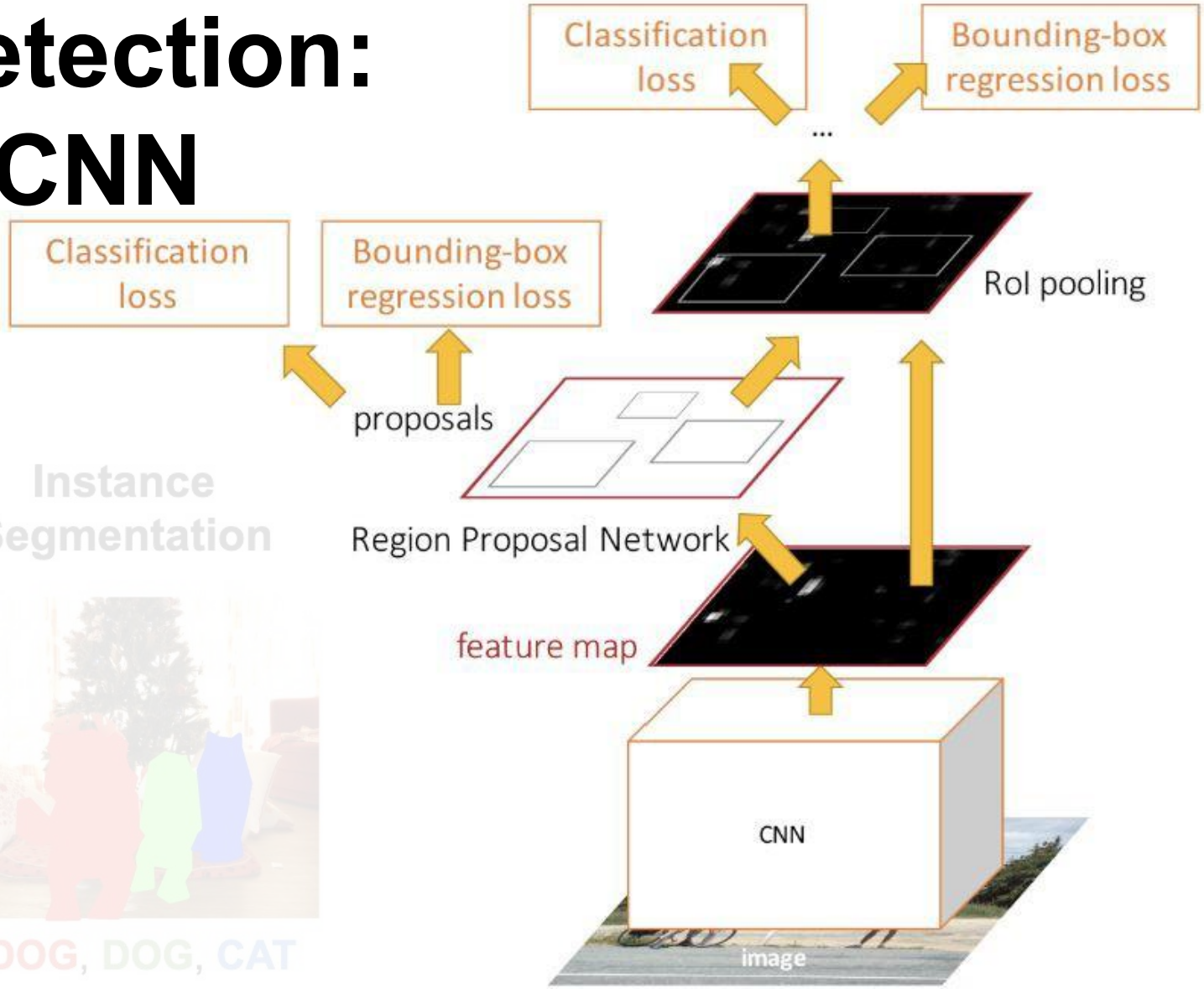
DOG, DOG, CAT

Instance Segmentation

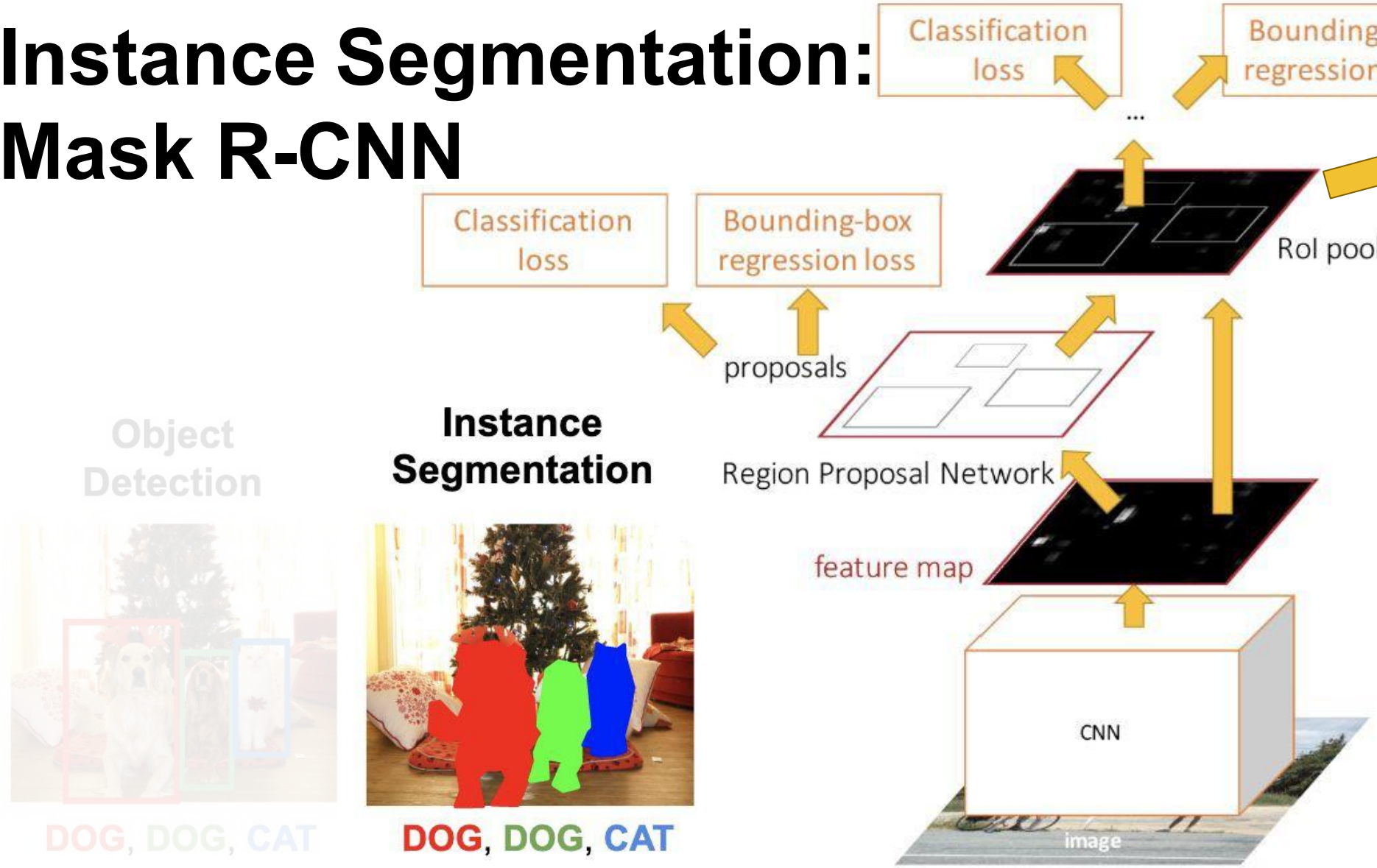
DOG, DOG, CAT

Multiple Object

Object Detection: Faster R-CNN

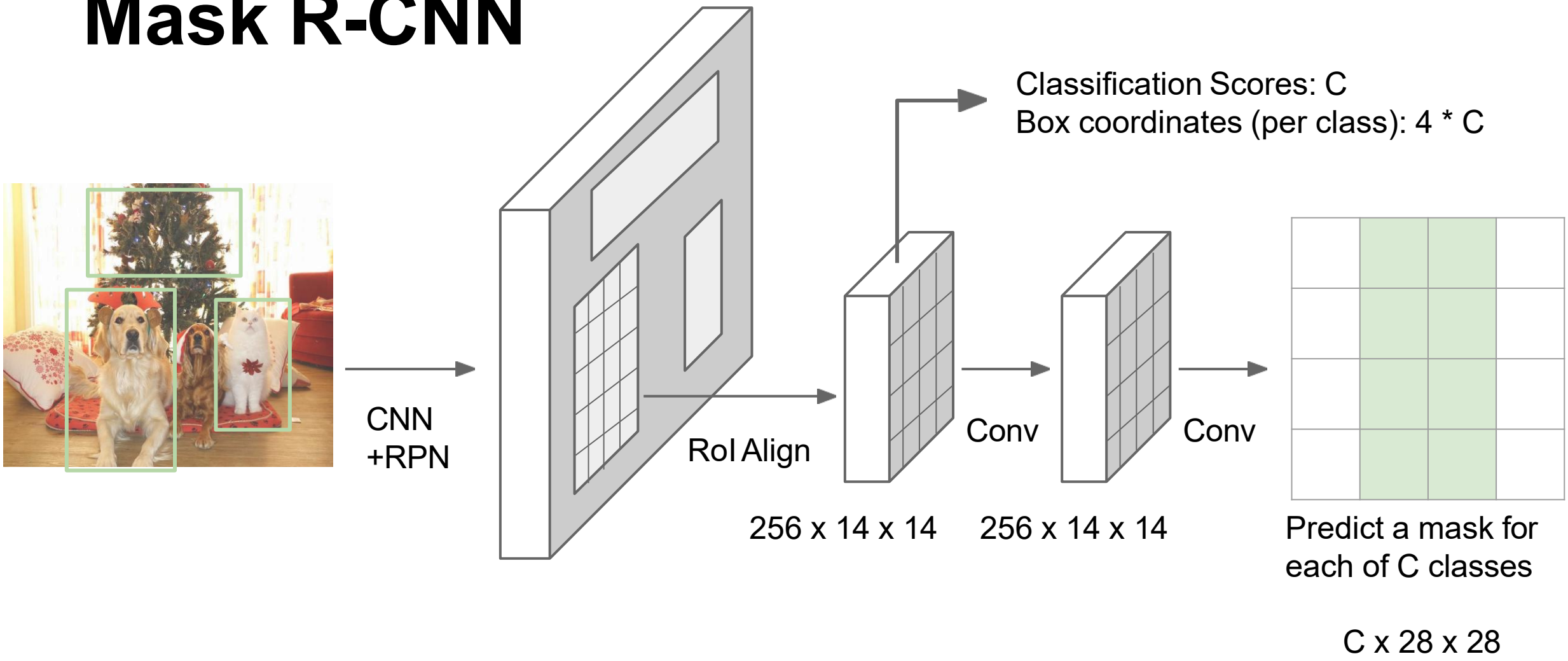


Instance Segmentation: Mask R-CNN

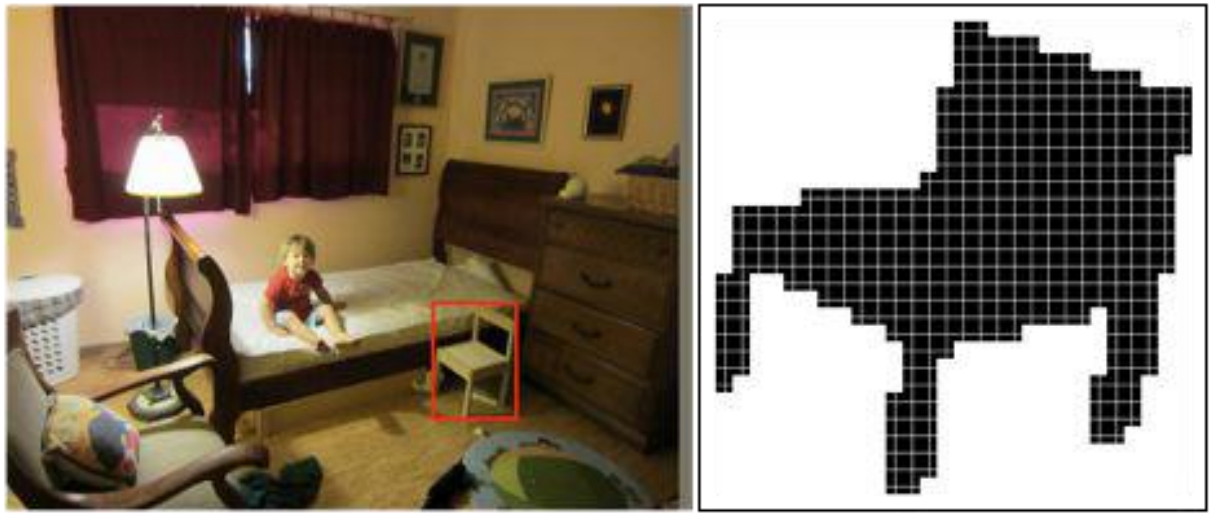


Add a small mask network that operates on each RoI and predicts a 28x28 binary mask

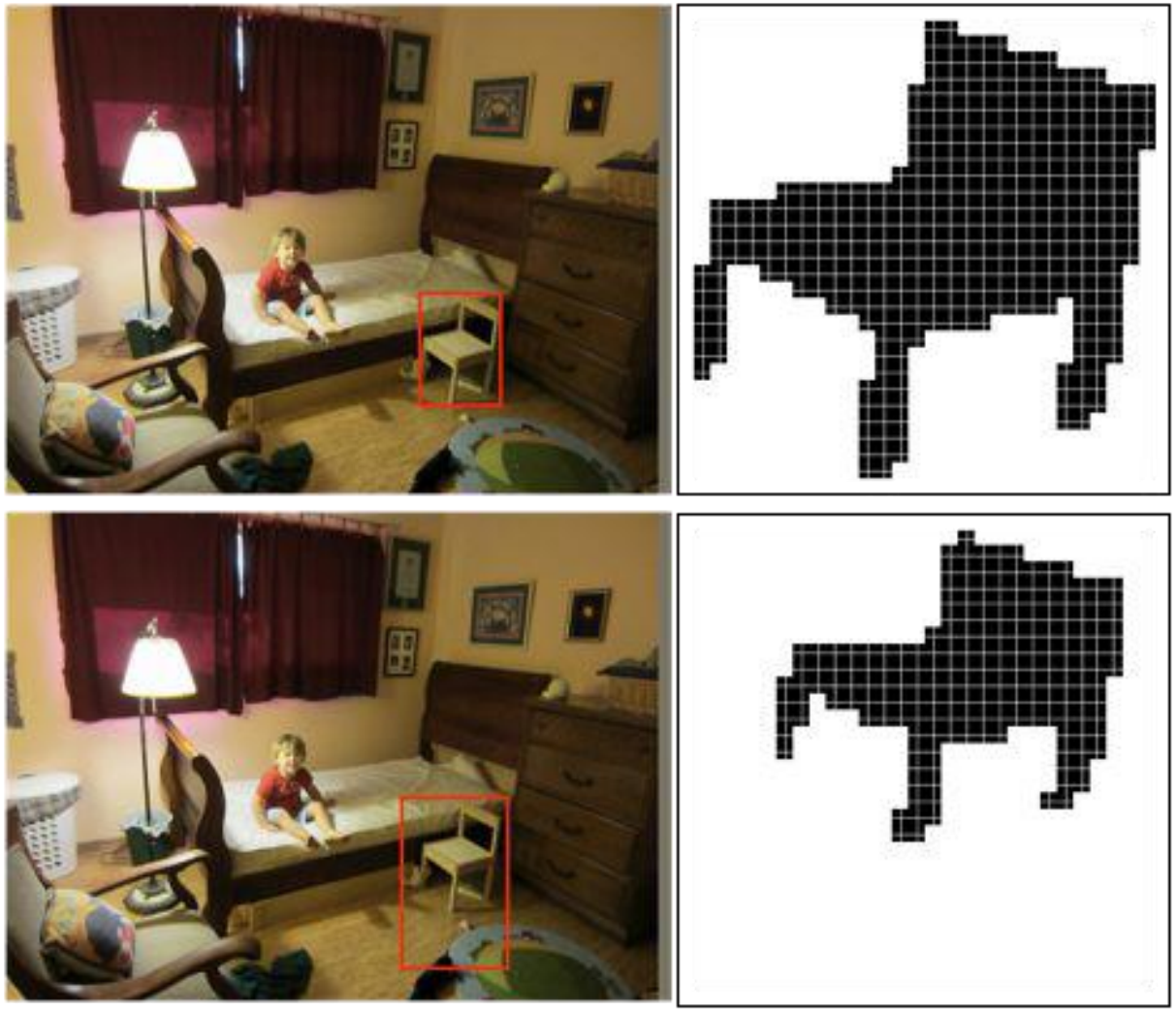
Mask R-CNN


Classification Scores: C
Box coordinates (per class): 4 * C
CNN
+RPN
RoI Align
Conv
Conv
256 x 14 x 14
256 x 14 x 14
Predict a mask for
each of C classes
C x 28 x 28

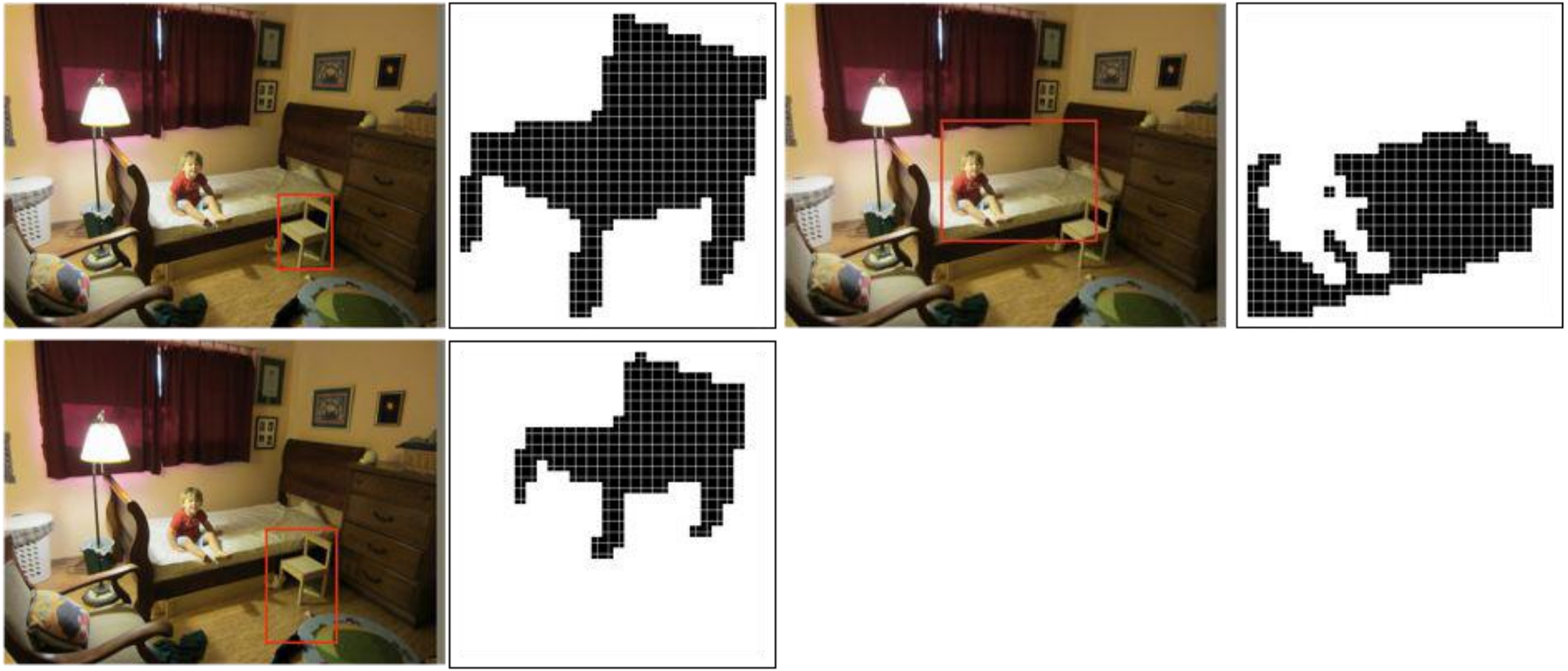
Mask R-CNN: Example Mask Training



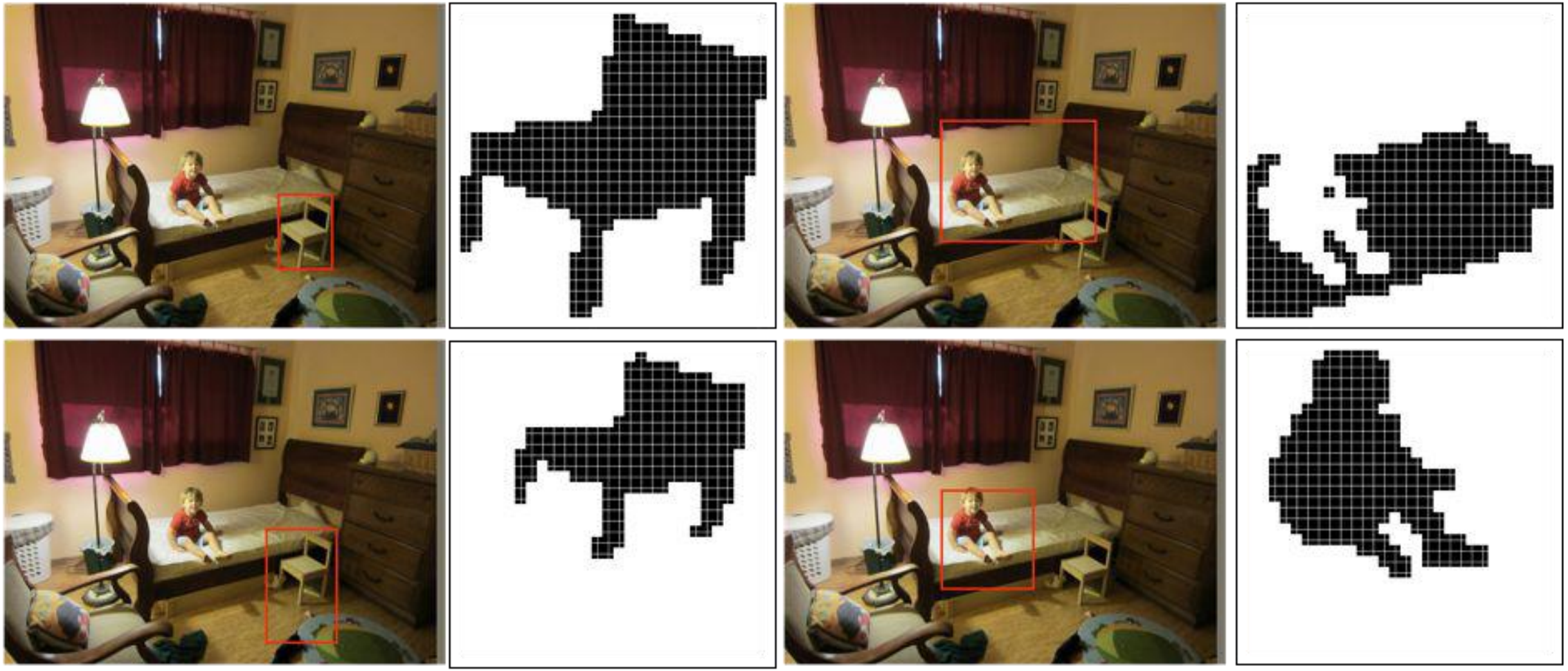
Mask R-CNN: Example Mask Training



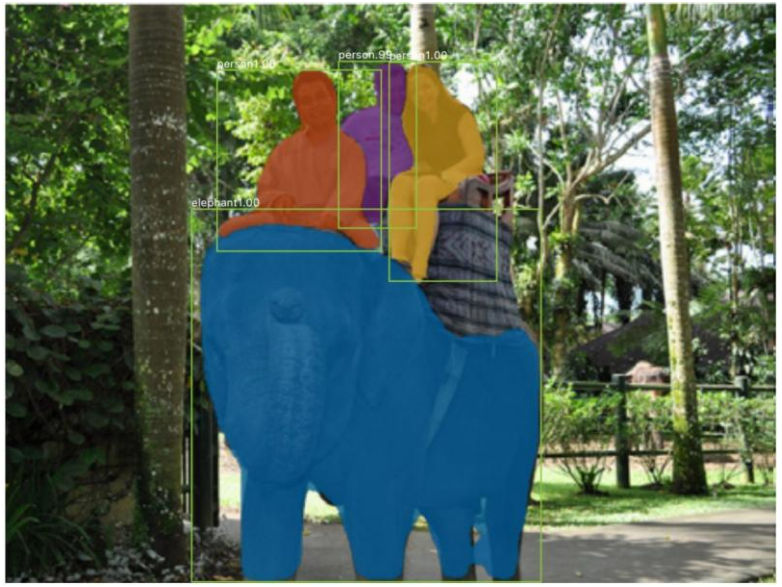
Mask R-CNN: Example Mask Training

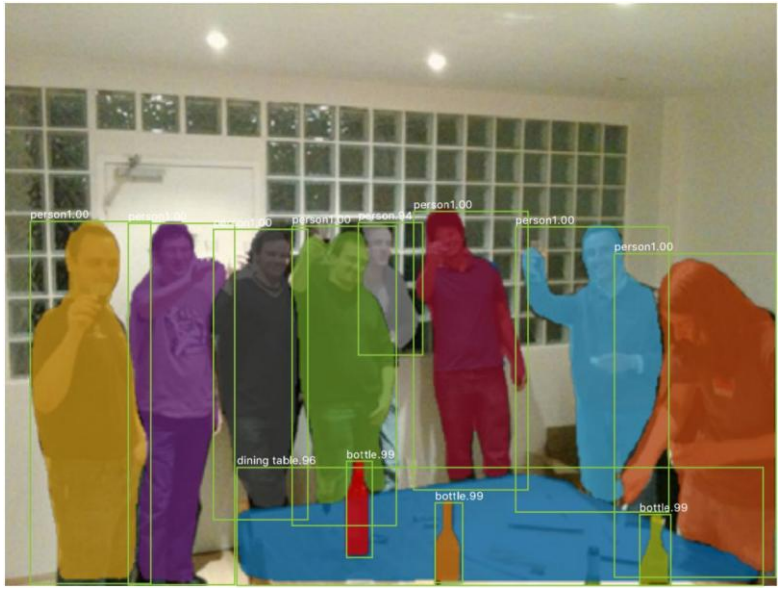


Mask R-CNN: Example Mask Training

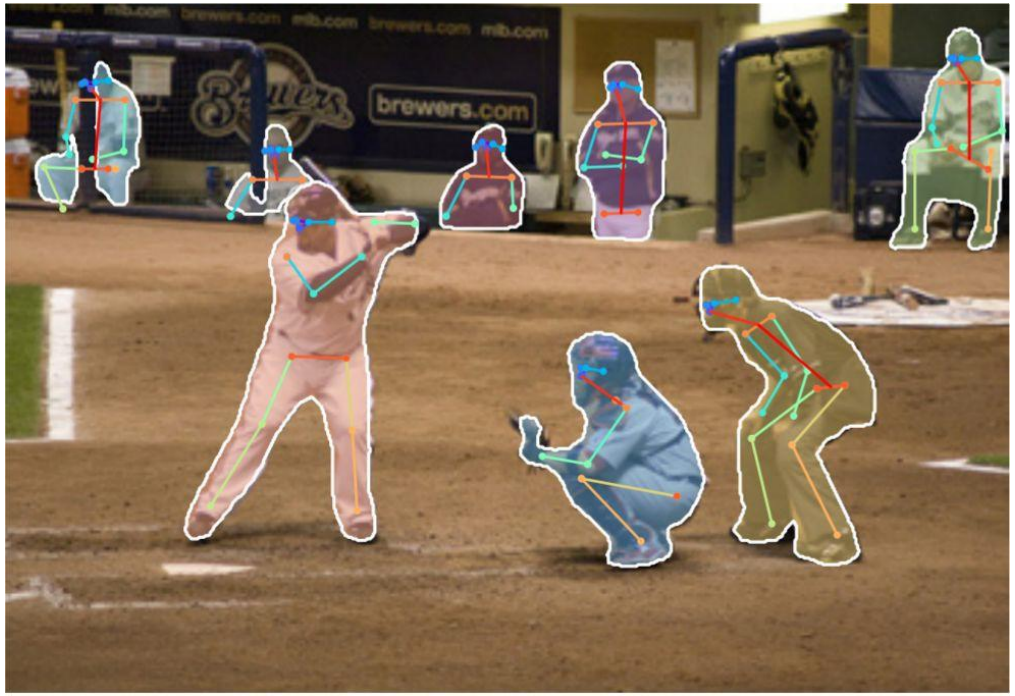


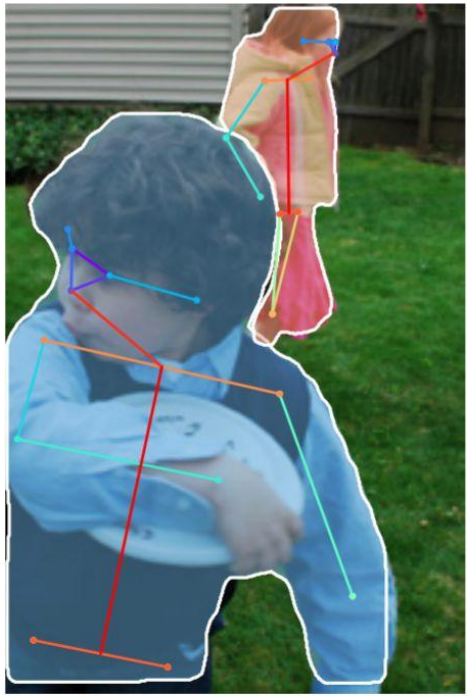
Mask R-CNN: Very Good Results!





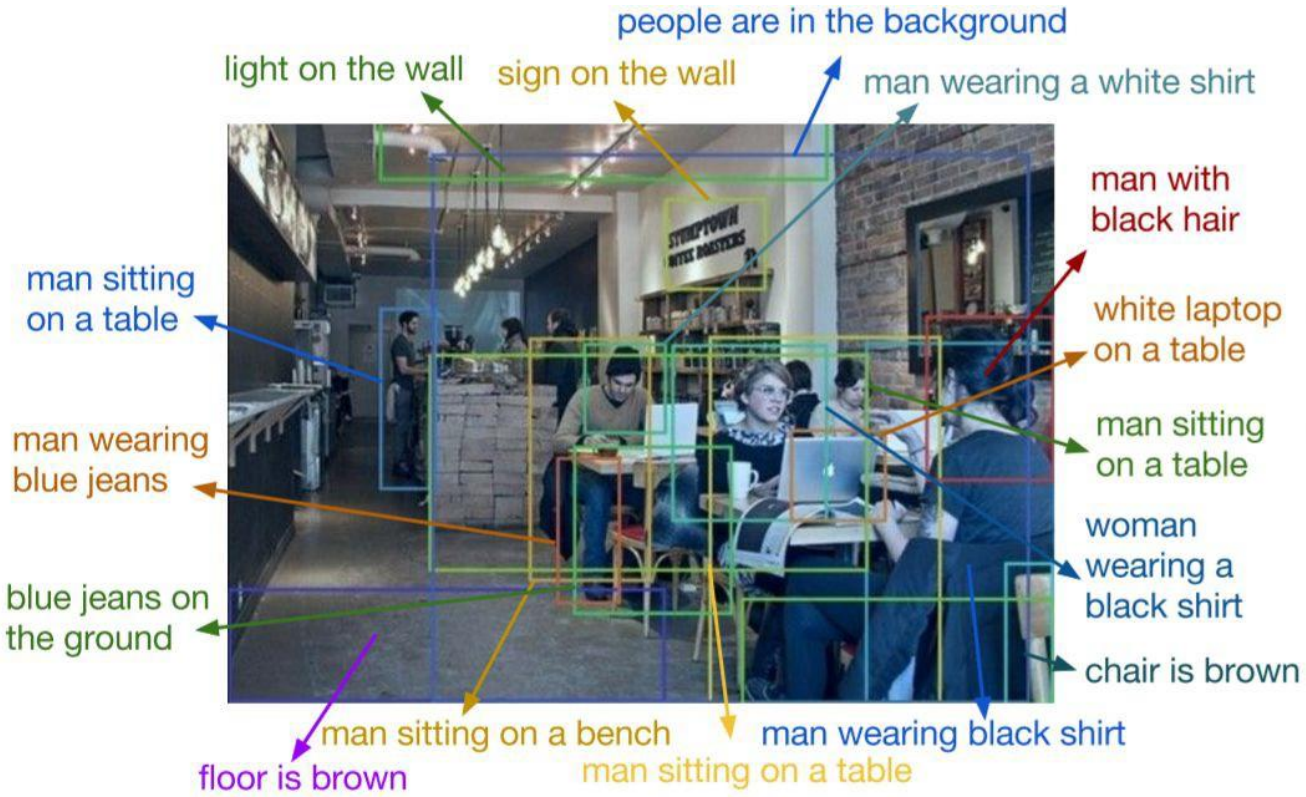
Mask R-CNN Also does pose

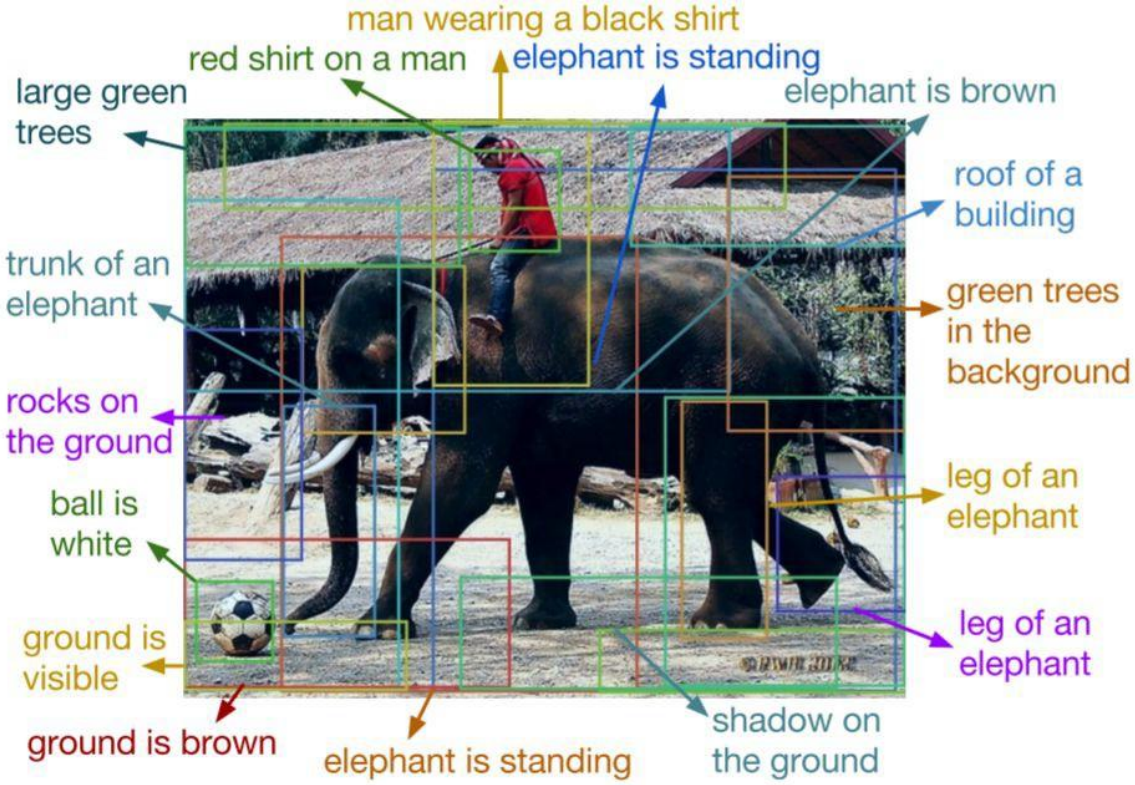




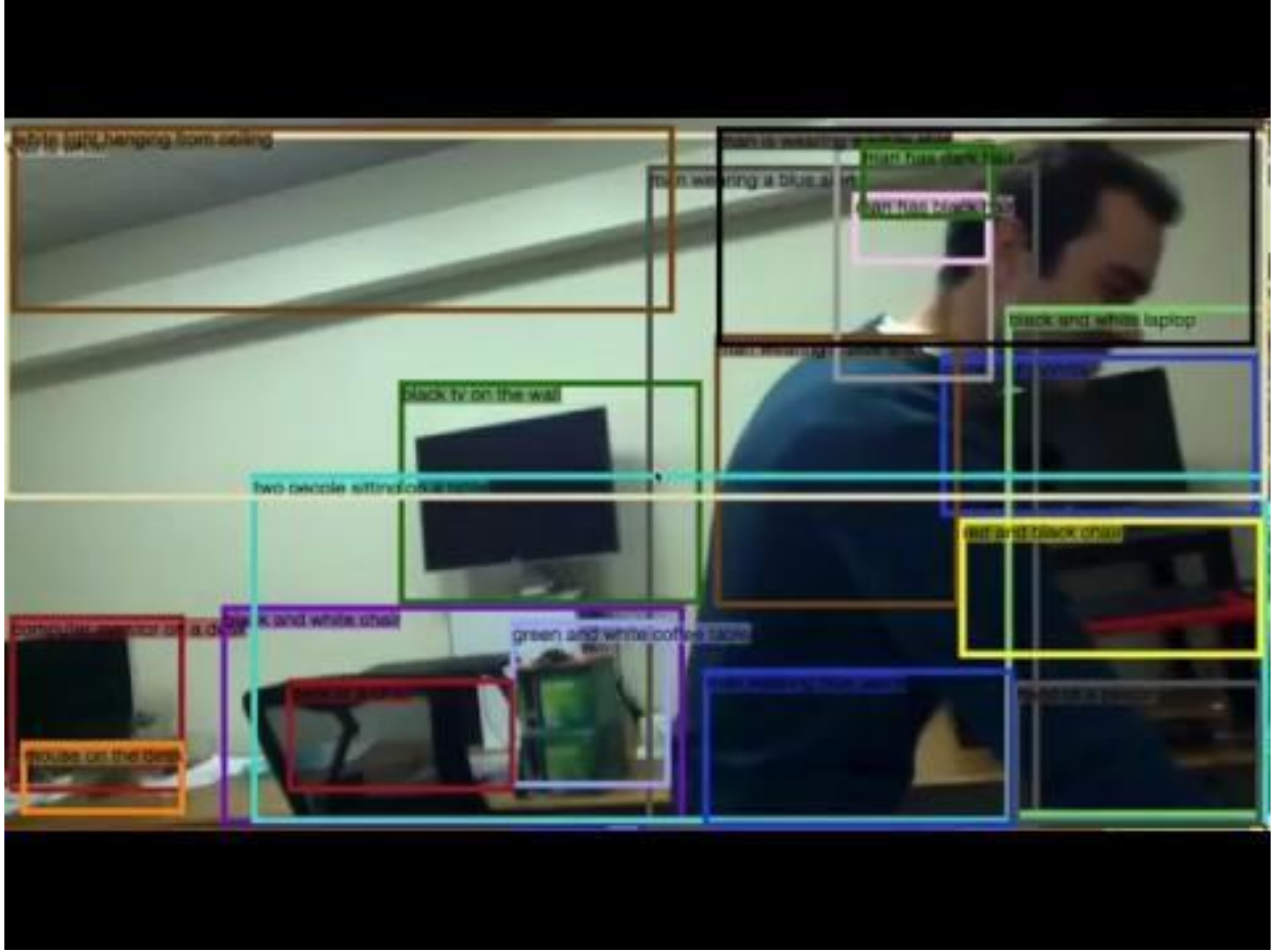
Beyond 2D Object Detection

Object Detection + Captioning
= Dense Captioning

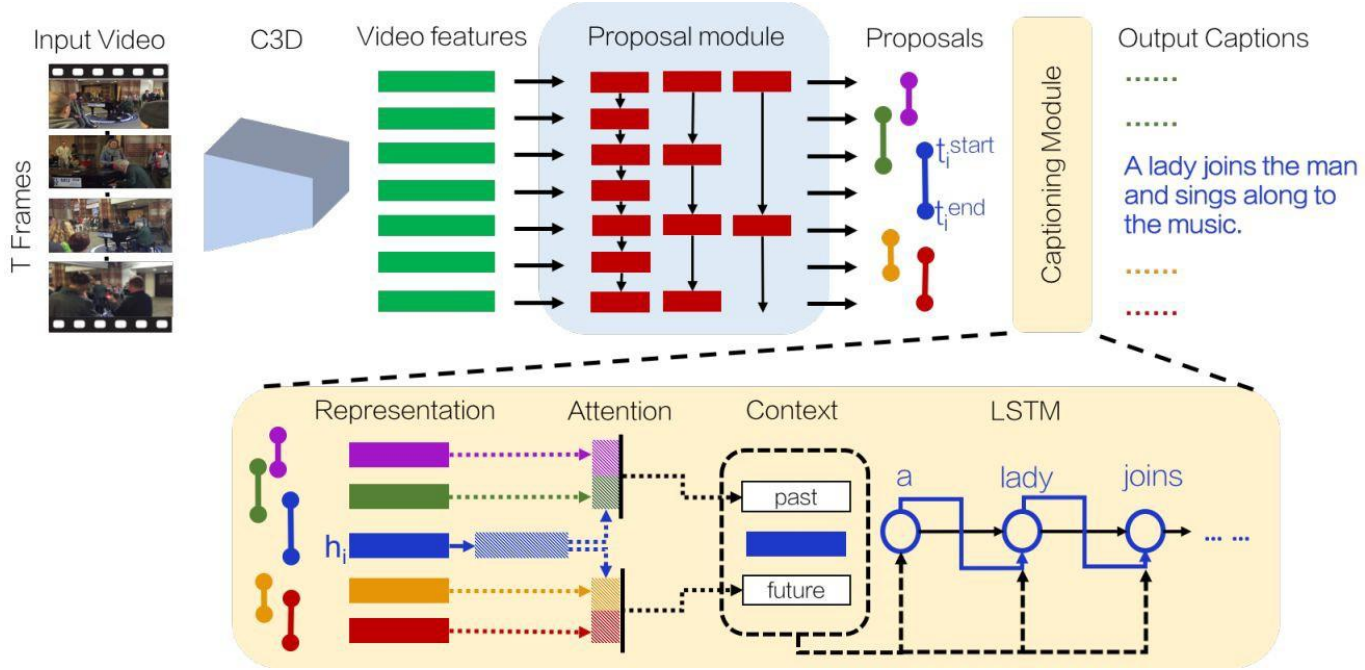


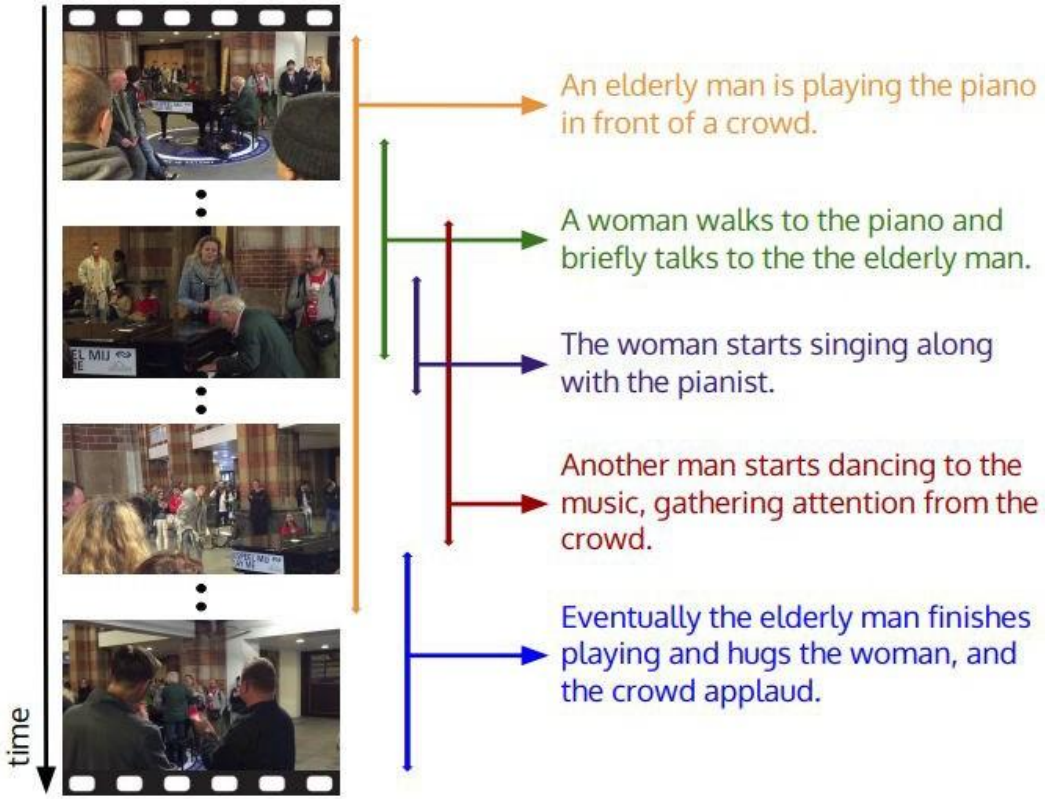


Exp

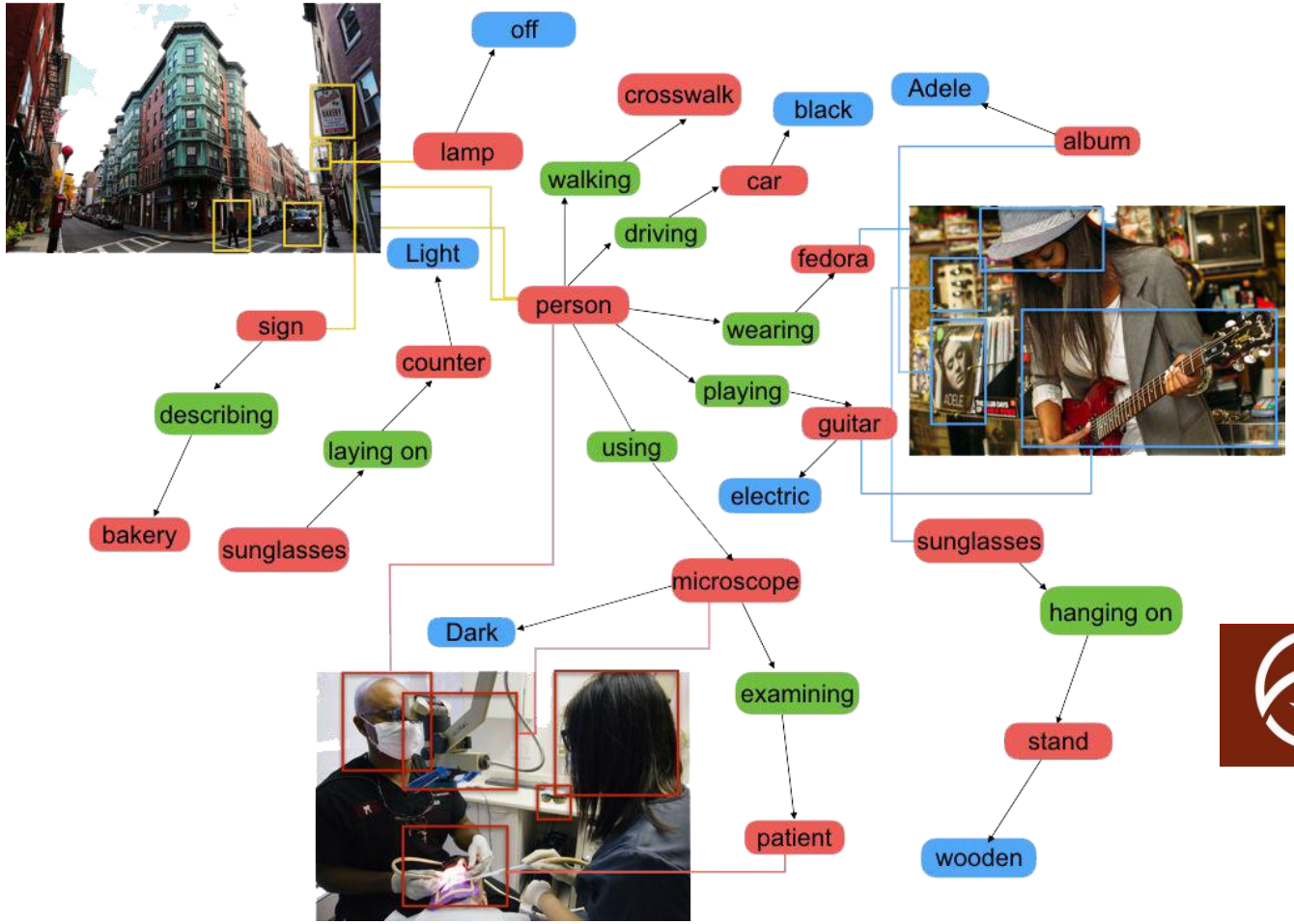

black tv on the wall
black and white laptop
green and white coffee table

Dense Video Captioning





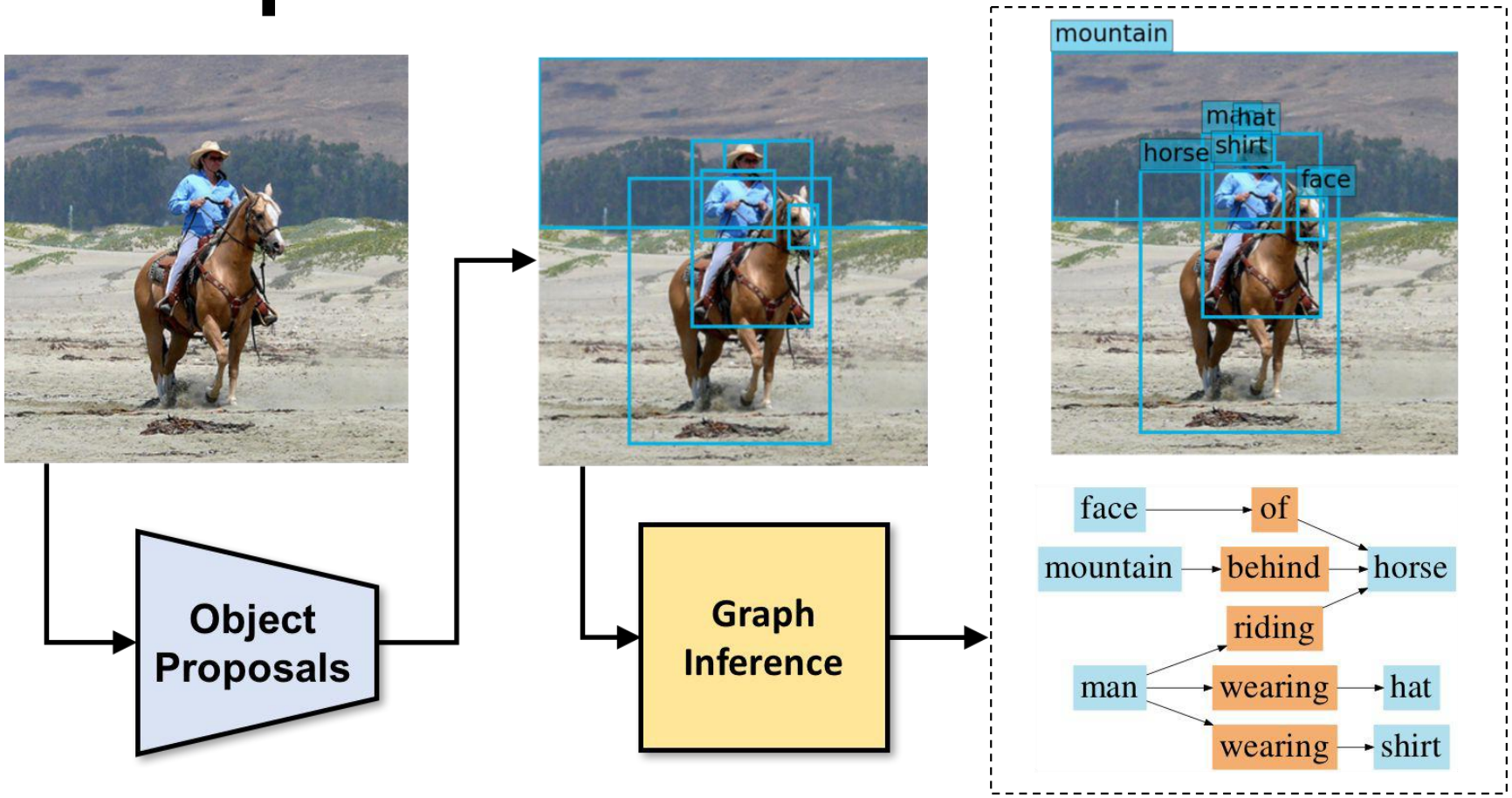
Objects + Relationships = Scene Graphs



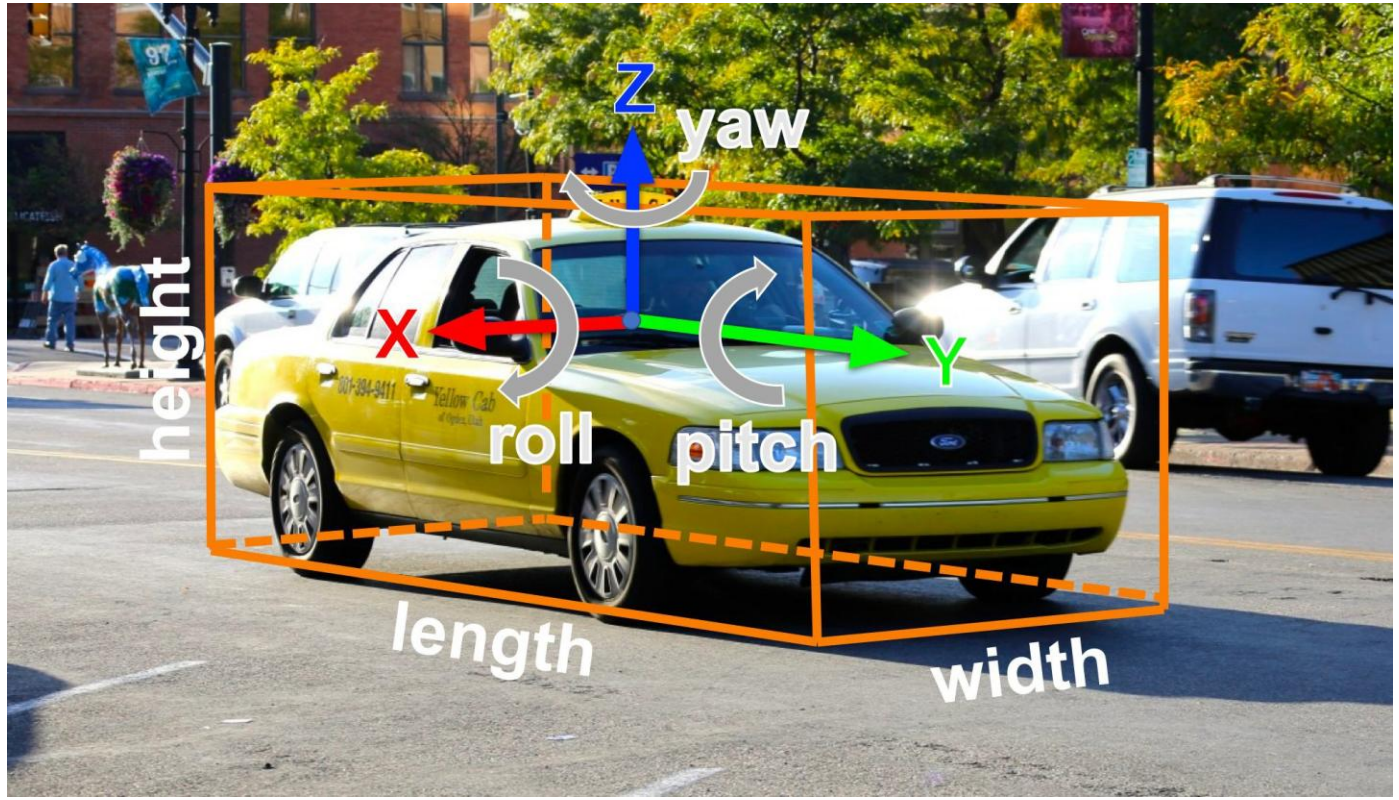
108,077 Images
5.4 Million Region Descriptions
1.7 Million Visual Question Answers
3.8 Million Object Instances
2.8 Million Attributes
2.3 Million Relationships
Everything Mapped to Wordnet Synsets

VISUALGENOME

Scene Graph Prediction



3D Object Detection

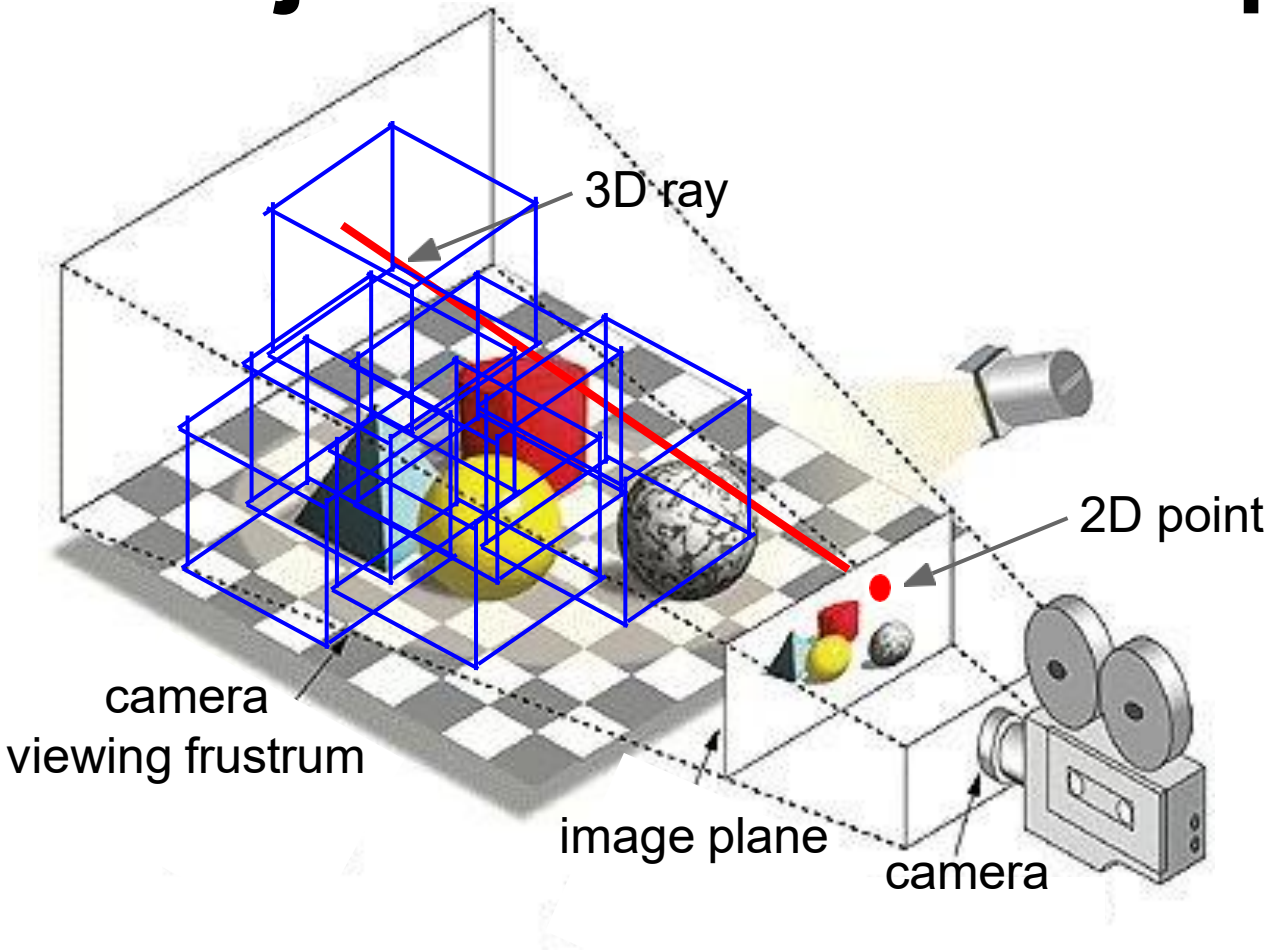


- 2D Object Detection:
- 2D bounding box
  (x, y, w, h)

- 3D Object Detection:
- 3D oriented bounding box
  (x, y, z, w, h, l, r, p, y)

- Simplified bbox: no roll & pitch

3D Object Detection: Simple Camera Model

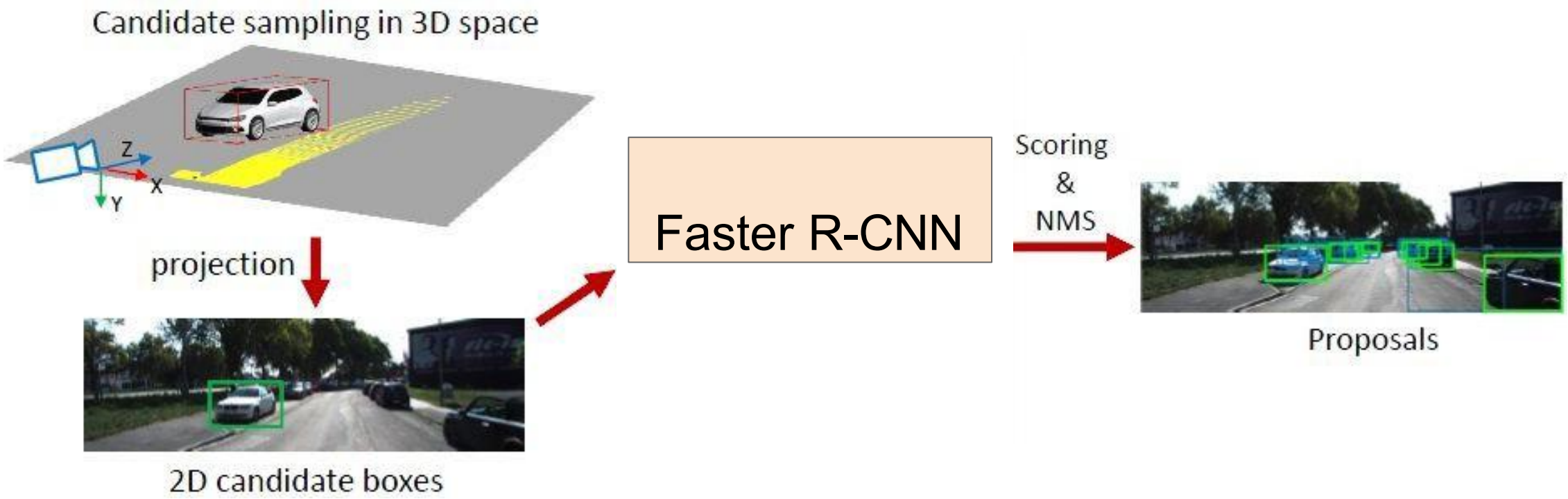


A point on the image plane corresponds to a **ray** in the 3D space

A 2D bounding box on an image is a **frustrum** in the 3D space

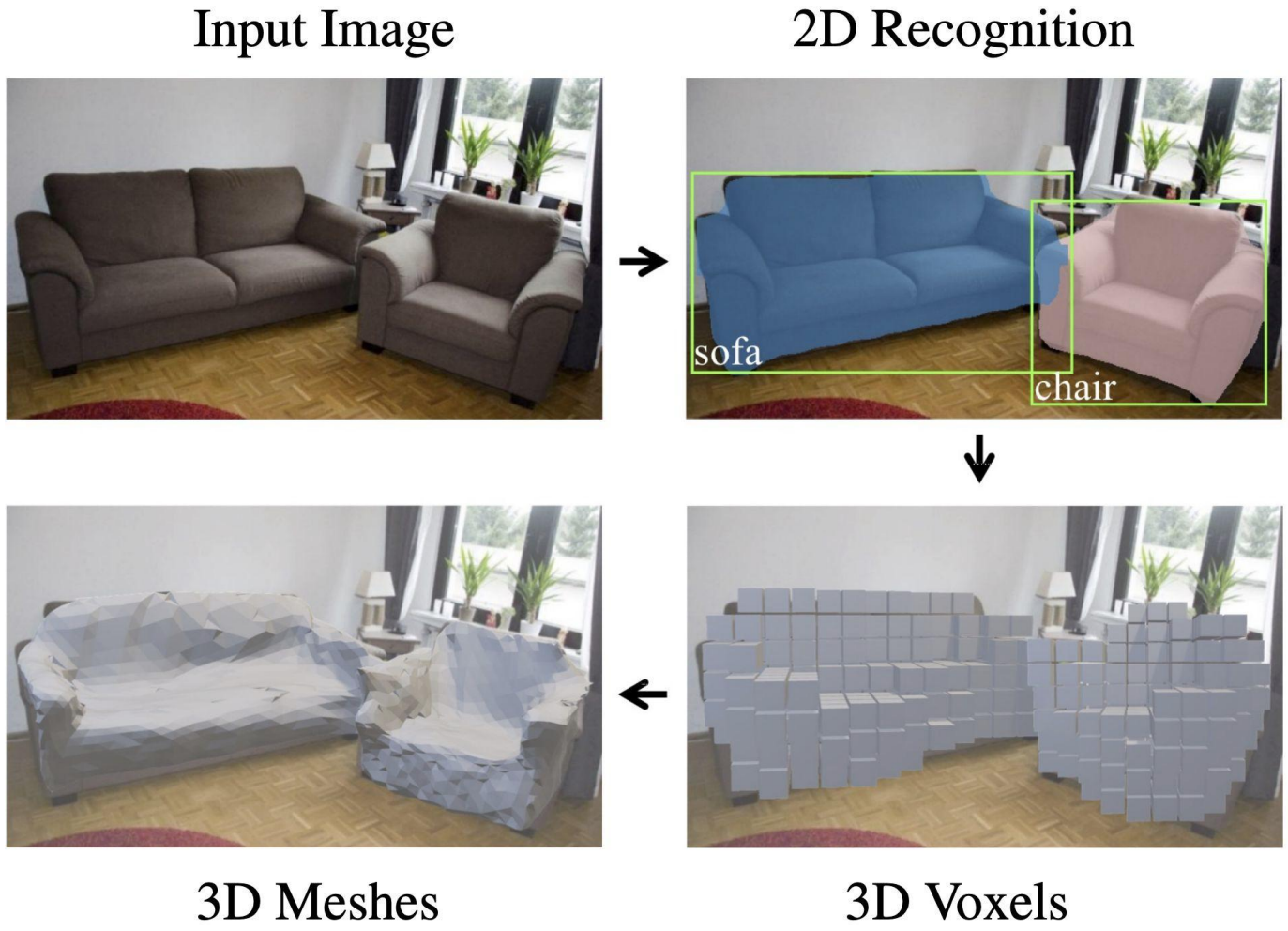
Localize an object in 3D:
The object can be anywhere in the **camera viewing frustrum**!

3D Object Detection: Monocular Camera



- Same idea as Faster RCNN, but proposals are in 3D
- 3D bounding box proposal, regress 3D box parameters + class score

3D Shape Prediction: Mesh R-CNN



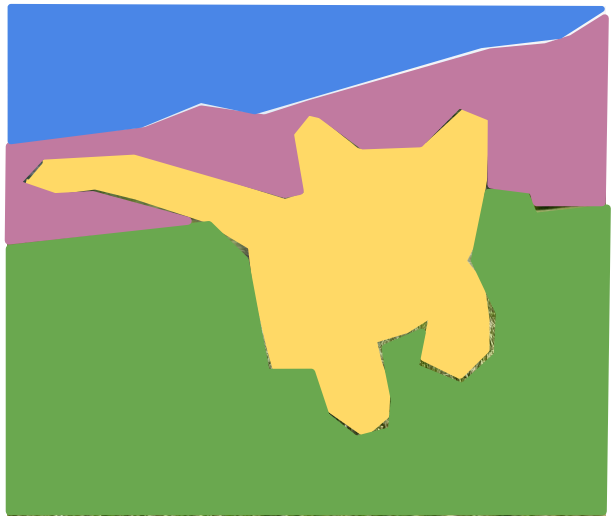
Summary: Lots of computer vision tasks!

Classification

CAT

No spatial extent

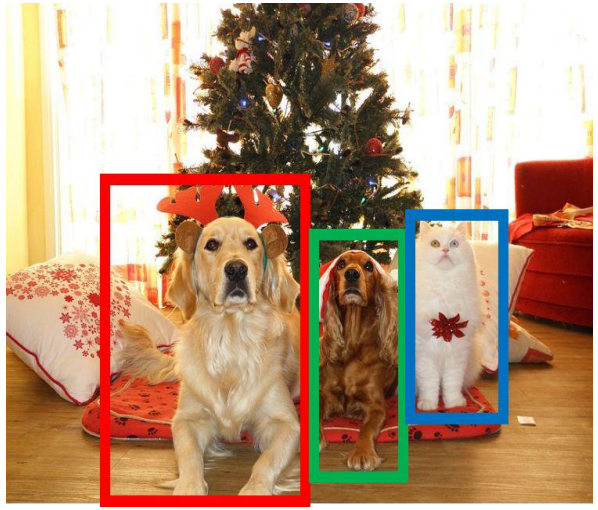
Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation

DOG, DOG, CAT

Multiple Object